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Computation of Multi-Particle Phase-Space Integrals and Their Applications in High-Energy Physics

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Resumo

Breve resumo em Português.

Abstract

Short abstract in English.

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Chapter 1

Introduction

High-energy collider experiments have, in recent years, provided ever-increasing precision in measurements of processes governed by the strong interaction. To be able to use these measurements fully, theoretical calculations must achieve similar levels of precision. At high energies, Quantum Chromodynamics (QCD) permits such predictions to be organised as a perturbative expansion in powers of the strong coupling constant α_s.

Achieving higher orders of perturbation is not simply a matter of calculating additional Feynman diagrams. Real-emission and virtual-loop contributions develop infrared singularities in kinematic configurations involving soft (low energy) or collinear (in trajectory) partons. These individual contributions are divergent and must be arranged before they can be evaluated reliably, but in sufficiently inclusive physical observables, like the hadronic R-ratio, they cancel, leaving a finite result.

The antenna subtraction formalism provides the framework for this rearrangement. The formalism describes unresolved radiation between pairs of colour-connected hard partons through universal antenna functions. These functions reproduce the relevant infrared limits locally and, after analytic integration over the unresolved phase space in dimensional regularisation, $d = 4 - 2\epsilon$, yield Laurent series whose ϵ-pole structure cancels that of the virtual contributions. Their systematic construction and integration are therefore central ingredients in higher-order QCD calculations.

This thesis develops and validates *AntCalc*, a symbolic framework for the automated construction and integration of QCD antenna functions. The work focuses on massless final-final quark-antiquark antennae through next-to-next-to-leading order (NNLO), combining symbolic amplitude generation with reduction methods for phase-space integrals. The resulting implementation is tested both through the calculation of the NNLO massless hadronic R-ratio, globally, and through the calculation of selected tests like the eikonal factor and the Altarelli-Parisi (AP) splitting function for each antenna, whenever appropriate, locally.

The remainder of the text is organised as follows. Chapter 2 introduces the theoretical foundations required throughout the work: perturbative QCD, infrared singularities, dimensional regularisation, and the antenna subtraction formalism. It defines the antenna functions, their colour structure, phase-space factorisation, and their role in fixed-order calculations. Chapter 3 describes *AntCalc*'s framework. It presents the package architecture, the construction and integration workflows, the reduction routes, and the conventions and diagnostics used to obtain the public results. Chapter 4 presents the antenna results obtained using *AntCalc*. It begins by introducing the master integrals used in the implemented reductions. Then, it develops the A_3^0 antenna from its unintegrated construction to its integrated form, together with its soft and collinear validation tests, using the eikonal factor and the AP splitting function, before presenting selected NLO and NNLO results in a shorter format, including the A_2^1, B_4^0 and C_4^0 antennae and the A_3^1,

A_2^2 and A_4^0 antenna-function sets. The chapter concludes with the exploratory massive A_3^0 extension. Chapter 5 provides a global validation through the construction of the massless hadronic R-ratio through NNLO. There, the antennae are assembled into their respective T-objects, and the cancellation of infrared poles and agreement with the known finite result are demonstrated. Chapter 6 discusses future directions for *AntCalc*, including more efficient symbolic and reduction backends, completion of the massive antenna sector, general model support, and extensions to higher perturbative orders, particularly N^{3}LO. Finally, Chapter 7 summarises the main conclusions of the thesis.

Chapter 2

Theoretical Foundations

2.1 QCD: Fields, Interactions and Perturbative Amplitudes

QCD is the quantum field theory of the strong interaction, whose elementary particles are the quarks and the gluons. Quarks are fermions that carry colour and interact through the exchange of gluons. Owing to the non-Abelian gauge structure of QCD, gluons themselves also carry colour charge and, therefore, undergo self-interaction.

At sufficiently high energy scales, the strong coupling, α_s, becomes sufficiently small for physical observables to be calculated perturbatively, as an expansion in powers of α_s. In this framework, the interaction structure of QCD determines the amplitudes for quark and gluon radiation that are studied throughout this thesis.

2.1.1 Quark and Gluon Fields

Quarks are spin-$1/2$ fields carrying flavour and a three-valued colour charge. They are QCD's matter fields. Gluons are the eight gauge fields associated with local $SU(3)_c$ colour symmetry. The QCD Lagrangian describes their propagation and interaction. It is written as follows,

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4} G^a_{\mu\nu} G^{a\,\mu\nu} + \sum_f \bar{\psi}_f \left(i\gamma^\mu D_\mu - m_f \right) \psi_f . \tag{2.1}$$

In this expression, $G^a_{\mu\nu}$ is the non-Abelian field-strength tensor, D_μ is the covariant derivative, ψ_f is the quark field and m_f is the quark mass. The flavour label f is summed over the quark species included in the theory, while the adjoint colour index a runs from one to eight. This Lagrangian is cleanly divided into two pieces, the quark and the gluon pieces.

Defining the covariant derivative as,

$$D_\mu = \partial_\mu - i g_s T^a A^a_\mu \tag{2.2}$$

where T^a represents the generators of the fundamental representation of $SU(3)_c$ and A^a_μ is the gluon gauge field, the quark Lagrangian term may be expanded to,

$$\mathcal{L}_q = \sum_f \left[\bar{\psi}_f \left(i\gamma^\mu \partial_\mu - m_f \right) \psi_f + g_s \bar{\psi}_f \gamma^\mu T^a \psi_f A^a_\mu \right] . \tag{2.3}$$

Here g_s is the strong coupling. In the expanded quark term, the first piece describes free quark propagation and the second quark-gluon interaction. This second piece is also responsible for generating the $q\bar{q}g$ vertex.

We now also define the totally antisymmetric structure constants of the $SU(3)_c$ Lie algebra f^{abc} as,

$$\left[T^a, T^b\right] = i f^{abc} T^c. \tag{2.4}$$

They may be used to define the non-Abelian field-strength tensor $G^a_{\mu\nu}$ as,

$$G^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g_s f^{abc} A^b_\mu A^c_\nu. \tag{2.5}$$

The first two terms of the field-strength tensor are Abelian-like and, in fact, they can be rewritten using the Abelian-like linear part of the non-Abelian field-strength tensor $F^a_{\mu\nu}$, as,

$$G^a_{\mu\nu} = F^a_{\mu\nu} + g_s f^{abc} A^b_\mu A^c_\nu, \quad F^a_{\mu\nu} \equiv \partial_\mu A^a_\nu - \partial_\nu A^a_\mu. \tag{2.6}$$

The second term arises because, as shown in Eq. 2.4, the $SU(3)_c$ generators do not commute. The non-Abelian field-strength tensor is used to write the gluon Lagrangian term. After expanding it becomes,

$$\mathcal{L}_g = -\frac{1}{4}\left(F^a_{\mu\nu}\right)^2 - \frac{g_s}{2} f^{abc} F^{a\,\mu\nu} A^b_\mu A^c_\nu - \frac{g_s^2}{4} f^{abc} f^{ade} A^b_\mu A^c_\nu A^{d\,\mu} A^{e\,\nu}. \tag{2.7}$$

Here, much like in $\mathcal{L}_q$, the gluon Lagrangian is also divided into three terms. The first term is quadratic in the gluon field and describes free gluon propagation. After gauge fixing, it supplies the gluon propagator. Unlike the remaining terms, it contains no non-Abelian self-interaction. The cubic and quartic terms in the gluon fields generate, respectively, the three- and four-gluon vertices, ggg and $gggg$[1].

Together, the two pieces, $\mathcal{L}_q$ and $\mathcal{L}_g$, constitute the gauge-invariant core of QCD. They generate the three elementary interaction vertices among the quark and gluon fields: $q\bar{q}g$, ggg and $gggg$ [1, 2].

2.1.2 From the QCD Lagrangian to Feynman Diagrams

The interaction terms in $\mathcal{L}_q$ and $\mathcal{L}_g$ generate Feynman rules which we can use to compute process amplitudes. Importantly, we are able to represent these processes as Feynman diagrams. These diagrams are graphical representations of terms in the perturbative expansion of a quantum amplitude. Each process may require more than one diagram to be fully represented; in that case, the complete amplitude is given as the sum of the amplitudes associated with all contributing diagrams. Fig. 2.1 depicts the two Feynman diagrams contributing to the process $\gamma^* \to q(k_1)g(k_3)\bar{q}(k_2)$, where a virtual photon splits into a quark-antiquark pair, with the gluon emitted from either the quark or antiquark line.

The final amplitude in this case would then follow,

$$\mathcal{M}^{(0)}_{\gamma^*\to qg\bar{q}} = \mathcal{M}^{(0)}_{q-\text{emission}} + \mathcal{M}^{(0)}_{\bar{q}-\text{emission}}. \tag{2.8}$$

In the diagrams in Fig. 2.1, straight, curly, and wavy lines denote fermions (quarks), gluons and photons, respectively. These diagrams represent the reduced hadronic-current process with γ^* as an off-shell and colourless source. In the full process, $e^+e^- \to \gamma^* \to qg\bar{q}$, the photon is internal between the lepton (electron-positron) and hadron (quark-antiquark) currents. All other lines are external, meaning that they represent outgoing asymptotic partons, the quarks and gluons. Each of the diagrams features

[1] A gauge-fixing term and the associated Faddeev-Popov ghost sector are required to formulate perturbative gluon propagators. Physical observables are independent of the gauge choice.

(a) Gluon emitted from the quark line. **(b)** Gluon emitted from the antiquark line.

Figure 2.1: Tree-level diagrams for the process $\gamma^* \to q(k_1)g(k_3)\bar{q}(k_2)$.

two vertices, where the lines meet. The vertices represent the interactions between the particles, which are determined by their respective Lagrangians. Each diagram contains an electromagnetic $\gamma^* q\bar{q}$ current insertion and a QCD $q\bar{q}g$ interaction vertex. The former follows from the QED sector, whereas the latter is generated by $\mathcal{L}_{\mathrm{QCD}}$. A rate or cross section often depends on the squared amplitude $|\mathcal{M}|^2$, given as,

$$|\mathcal{M}|^2 = \sum_{D,D'} \mathcal{M}_D \mathcal{M}^*_{D'}. \tag{2.9}$$

2.2 Divergences in Perturbative QCD

2.2.1 Infrared Divergences

The two QCD infrared limits, soft and collinear, can be derived explicitly as follows. We start by defining the physical system under study: consider two massless final-state partons i and j with four-momenta p_i and p_j. By momentum conservation, the four-momentum of the internal virtual propagator is $P = p_i + p_j$. The matrix element contains this propagator and in the square amplitude it appears as a $1/P^2$ factor. Expanding it and applying the massless condition ($p^2 = 0$), we get $P^2 = 2p_i \cdot p_j = 2E_iE_j(1 - \cos\theta_{ij})$ [1]. Importantly, this quantity defines the Mandelstam invariants as $s_{ij} = 2p_i \cdot p_j = P^2$, in the massless regime[2], which will serve as the natural integration variable throughout the work. The limits are then apparent: the soft limit is hit when one of the energies goes to zero and the collinear limit, when θ_{ij} does. In both cases, $1/s_{ij}^2 \to \infty$,

$$\begin{aligned} \lim_{E_i \to 0} s_{ij} &= 0 \\ \lim_{\theta_{ij} \to 0} s_{ij} &= 0. \end{aligned} \tag{2.10}$$

Note here that in terms of detection, a real emission that is either soft or collinear is experimentally indistinguishable from the corresponding n-particle state. This means that at the limits, there is nothing distinguishing the unresolved real emission ($n+1$) and the Born-level (n) cases.

Since the soft and collinear limits from real radiation are indistinguishable from the same limits in the virtual regime, cross-sections for individual degenerate states are not separately physically meaningful. The KLN theorem states that to compute a physical observable the probabilities of all degenerate states must be summed over [3, 4]. Using this theorem, we can sum all real radiation states' singularities as positive divergences and all virtual corrections' singularities as negative divergences, such that they

[2] The general definition of the Mandelstam invariant is $s_{ij} = (p_i + p_j)^2$, but as p_k^2 for any k, in this massless regime, then the massless Mandelstam invariant is defined as $s_{ij} = 2p_i \cdot p_j$.

cancel. For this cancellation to happen, however, the two parts must live within the same phase-space, which is not the case here, for these real emissions and virtual corrections, since one produces $(n+1)$ particles and the other only n. The solution here is to subtract the divergences using a local counterterm σ^S and integrate over the $n+1$ phase-space, and then add the subtracted divergences to the n-particle phase-space integrated analytically over the extra particle and then integrate the whole term over the phase-space. This is expressed schematically as,

$$\sigma = \int_{n+1} \left(d\sigma^R - d\sigma^S \right) + \int_n \left(d\sigma^V + \int_1 d\sigma^S \right) \tag{2.11}$$

where $d\sigma^R$ represents the differential real radiation cross-section, $d\sigma^V$ the differential virtual corrections cross-section and $d\sigma^S$ the differential subtraction counterterm [5]. Importantly, $d\sigma^S$ must reproduce $d\sigma^R$'s exact singular behaviour at every point in the phase space, such that the divergences cancel locally.

2.2.2 Ultraviolet Divergences

Infrared divergences are not the only kind of divergences we find in QCD. When we study one-loop diagrams new limits must be considered for the loop momentum l. The soft and collinear limits on l behave exactly as described above for the outgoing momenta. That is the reason $d\sigma^V$ is singular. But, since this momentum is not bound by the phase-space to a certain maximum, being instead integrated over the whole of momentum space, a new $l \to \infty$, so-called ultraviolet (UV) limit appears.

This limit presents two distinct but pressing problems. Firstly, the unbounded domain risks the integral diverging. This is due to, in d-dimensions, the loop measure expanding to,

$$d^d l = l^{d-1} dl \, d\Omega_{d-1}. \tag{2.12}$$

Whether the integral diverges or not depends on whether the integrand converges fast enough to counter the l^{d-1} measure term. Since each propagator contributes $\sim 1/l^2$ to the integrand, schematically, a diagram with n_p propagators in the loop and no extra powers of l in the numerator becomes,

$$\int^{\infty} l^{d-1} \cdot \frac{1}{l^{2n_p}} = \int^{\infty} l^{d-1-2n_p} dl \tag{2.13}$$

which is UV divergent whenever $d - 1 - 2n_p \geq -1 \Rightarrow d \geq 2n_p$ [2, 6].

Secondly, this limit tells us that any perturbative series constructed order by order in the bare coupling α_0, from the QCD Lagrangian [ref to the qcd lagrangian], is, beyond tree level ill-defined since its higher-order coefficients diverge [1]. This property makes the variable unsuitable as the order by order perturbative series variable we require. As such, we need to trade it for a finite renormalised coupling, α_s, for the order by order perturbative series [6].

2.3 Dimensional Regularization

The subtraction scheme from Eq. 2.11 cancels the real radiation poles directly using $d\sigma^S$, but the system as shown in the equation still is not numerically integrable and will still go to infinity when integrated due to the rightmost integral over the unresolved parton,

$$\int_1 d\sigma^S. \tag{2.14}$$

Since $d\sigma^S$ is designed explicitly to carry over all of $d\sigma^R$'s divergent structures, this integral blows up once numerically integrated over 4 dimensions. This is also true of the virtual corrections in $d\sigma^V$, as explored in Subsection 2.2.2. To solve this we integrate this term over the unresolved parton over d-dimensions and then use Dimensional Regularisation to manifest the physical divergences as explicit analytic poles in an arbitrary continuous parameter ϵ, via,

$$d = 4 - 2\epsilon \tag{2.15}$$

which leaves us with two ϵ-dependent expressions for the virtual corrections and for the now unresolved-parton-integrated subtraction cross-section,

$$\int_n \left(d\sigma^V + d\sigma_1^S \right). \tag{2.16}$$

From the KLN theorem, as described in Section 2.2.1, we know that these poles must cancel. This makes it so that the complete expression becomes finite and numerically integrable.

Though the principle is always the same, there is more than one way to implement Dimensional Regularisation. The most common methods are 't Hooft-Veltman [7], Four-Dimensional Helicity [8] and Conventional Dimensional Regularisation (CDR) [1]. This third method is the one used throughout the present work, since AndCalc's primary objective is the simplification and democratisation of the antenna framework establish in [9], which uses CDR throughout.

The use of CDR, specially the complete shift to d-dimensional space introduces some complications, however. First, the move to d-dimensional spherical coordinates implies that the phase-space integrals are now over d-dimensional solid angles, which produce known geometric artifacts like π and $(4\pi)^{-\epsilon}$ factors and Gamma functions like $\Gamma(1-\epsilon)$. To prevent these artefacts from contaminating our integrated results conform with the $\overline{\text{MS}}$ (modified minimal subtraction) renormalisation scheme [10], we must extract the $S_\epsilon = (4\pi)^{-\epsilon} e^{\epsilon \gamma_E}$ normalisation factor from the integration measure. In the standard QCD convention, the perturbative expansions are written in terms of $\alpha_s/(2\pi)$, instead of just α_s. That makes the introduction of a further normalisation parameter necessary, since the squared matrix elements do not scale just with the coupling constant α_s, but rather with the whole raw coupling constant $g^2 = 4\pi\alpha_s = 8\pi^2(\alpha_s/(2\pi))$. The scale normalisation,

$$\begin{aligned} G_k &= \left(8\pi^2\right)^k \\ &= \left(8\pi^2\right)^{n-2} \Lambda_l, \quad \Lambda_l \equiv \left(8\pi^2\right)^l \end{aligned} \tag{2.17}$$

is then introduced to normalise those expansion terms, and the loop-order-only normalisation Λ_l explicitly shown as the virtual-loop contribution. The combined normalisation here introduced is thus,

$$C(\epsilon, k) = G_k S_\epsilon = \left(8\pi^2\right)^k (4\pi)^{-\epsilon} e^{\epsilon \gamma_E} \tag{2.18}$$

where k represents the perturbative order as $k = 1$ for NLO, $k = 2$ for NNLO, ... and is, itself defined as,

$$k = (n-2) + l \tag{2.19}$$

for some antenna with n-final-state particles and l loops[3].

[3]Loops will be discussed and the k index widely used in Subsection 2.4.1.

Since multi-particle phase spaces recursively factorise, we must also normalise our phase-space integral by the d-dimensional two-body phase space volume Φ_2 [4]. This normalisation, together with the previous combined normalisation factor make the global normalisation factor $\mathcal{N}(\epsilon, k)$, used throughout the present work,

$$\mathcal{N}(\epsilon, k) = \frac{C(\epsilon, k)}{\Phi_2}. \tag{2.20}$$

As described in Subsection 2.2.2, we need to trade the bare coupling constant α_0 for a suitable order by order perturbative series variable. That is achieved as follows [9],

$$\alpha_0 \mu_0^{2\epsilon} S_\epsilon = \alpha_s \mu^{2\epsilon} \left[1 - \frac{\beta_0}{\epsilon}\frac{\alpha_s}{2\pi} + O(\alpha_s^2)\right], \quad \beta_0 = \frac{11N - 2N_f}{6} \tag{2.21}$$

where μ_0 is the bare mass parameter that keeps α_0 dimensionless, μ is the renormalisation scale (the renormalised mass parameter) that functions equally for α_s. It is more commonly scaled as $\mu^2 = q^2$ for the massless case. β_0 is the one-loop coefficient of the QCD beta function, that tracks how the renormalised coupling α_s changes with the renormalisation scale μ, and finally N is the number of colours, 3, and N_f the number of massless active quark flavours.

2.4 The Antenna Subtraction Formalism

In perturbative QCD, when a particle like a gluon, j, becomes unresolved as described above, its squared matrix element $|\mathcal{M}_{n+1}|^2$ factorises into a universal singular factor for this process, $\mathcal{S}_j$ and a lower-multiplicity squared matrix element (a level below), as follows [5, 9],

$$|\mathcal{M}_{n+1}(..., i, j, k, ...)|^2 \to \mathcal{S}_j \cdot |\mathcal{M}_n(..., I, K, ...)|^2. \tag{2.22}$$

Since this is a universal property, it is possible to compute the subtraction cross-section $d\sigma^S$ by summing over all possible unresolved pairs. Therefore,

$$d\sigma^S = \sum_{\text{unresolved pairs}} \mathcal{S}_j \cdot d\sigma^B \tag{2.23}$$

where $d\sigma^B$ denotes the Born process $\gamma^* \to q\bar{q}$, cross-section and $\mathcal{S}_j$ is the singular factor for each of the j particles being summed over.

2.4.1 Antenna Functions

Since the singular factor $\mathcal{S}_j$ must work smoothly across all unresolved limits and must also handle all spin correlations correctly, it should, optimally, be a function that preserves all the unresolved parton information without the noise coming from the hard parton process. This is easily achieved by dividing our n-level process squared matrix element by the fixed two-parton Born-level, common to all quark-antiquark antenna functions. The resulting quantity is gauge-invariant. It is the antenna, most usually notated by X_n^0 and defined as [9, 11, 12],

$$X_n^{0,\,\text{bare}} = \mathcal{N}_X \frac{|\mathcal{M}_n^0|^2}{|\mathcal{M}_2^0|^2} \tag{2.24}$$

[4] This factorisation and subsequent normalisation is explored in detail in Subsection 2.4.4

where $|\mathcal{M}_n|^2$ represents the n-level squared matrix element and $X_n^{0,\,\text{bare}}$ is the antenna that represents that same n-level. $\left|\mathcal{M}_2^0\right|^2$ is the Born-level squared matrix element. $\mathcal{N}_X$ represents the necessary normalisation over colour and coupling terms, whose general form depends on the specific antenna family under consideration. The colour-algebraic machinery required to construct it explicitly is developed in Subsection 2.4.2. It is defined, for quark-antiquark antennae [5], using the fundamental Casimir invariant C_F and the strong coupling constant $g_s = \sqrt{4\pi\alpha_s}$, as

$$\mathcal{N}_X = \frac{1}{2g_s^{2(n-2)}C_F}. \tag{2.25}$$

This definition is strictly correct and complete for real-emission, so called tree-level, antenna functions. These antenna functions represent loopless processes. QCD allows for the exchange of gluons in between quarks and the self-absorption (by the same quark) of a previously emitted gluon. These are the so-called virtual corrections. As described in Section 2.2.2, these corrections lead to UV divergences. Hence, loop antenna functions require normalisation and hence why the Eq. 2.24 is labelled as *bare*: it is the general definition without normalisation. This normalisation is achieved through the rescaling of the bare coupling constant α_0 to the renormalised coupling constant α_s through Eq. 2.21. This rescaling applies the Ward-Takahashi [13, 14] and Slavnov-Taylor identities[6] [15, 16], which guarantee the conservation of the electric and strong charges[7], respectively, through all topologies of the same process. They also require us to denote the number of loops their process supports. That number is the superscript in X_n^l, l, which up until now had been kept at zero for the loopless processes. n, on the other hand, represents the number of final-state particles in the process.

These antenna functions may now be ordered by perturbative order based on their k index. Let us define an antenna completely as $X_{n,(k)}^l$ where k is the perturbative order index, n is the number of final-state particles and l is the number of loops. Let us also define the perturbative expansion in Eq. 2.21 as,

$$\begin{aligned}\Delta\left(\frac{\alpha_s}{2\pi}\right) &= 1 - \frac{\beta_0}{\epsilon}\frac{\alpha_s}{2\pi} + O\left(\frac{\alpha_s}{2\pi}\right)\\ &= \Delta^{(0)} + \Delta^{(1)}\frac{\alpha_s}{2\pi} + O\left(\frac{\alpha_s}{2\pi}\right).\end{aligned} \tag{2.26}$$

We are now able to generalise Eq. 2.24 to all antenna functions as follows,

$$\begin{aligned}X_{n,(k)}^l &= X_{n,(k)}^{l,\,\text{bare}} + \sum_{m=1}^{\min(l,k-1)} \Delta^{(m)} X_{n,(k-m)}^{l-m,\,\text{bare}}\\ &= X_{n,(k)}^{l,\,\text{bare}} + \Delta^{(1)} X_{n,(k-1)}^{l-1,\,\text{bare}} + \Delta^{(2)} X_{n,(k-2)}^{l-2,\,\text{bare}} + \ldots\\ &= \frac{1}{\left|\mathcal{M}_2^0\right|^2}\left[\left|\mathcal{M}_n^l\right|^2 + \Delta^{(1)}\left|\mathcal{M}_n^{l-1}\right|^2 + \Delta^{(2)}\left|\mathcal{M}_n^{l-2}\right|^2 + \ldots\right].\end{aligned} \tag{2.27}$$

[5] Other antenna types are described in Subsection 2.4.3. These require different normalisations.

[6] These identities are explored further in Appendix [ref to the identities in an Appendix].

[7] In QCD terminology, "colour charge" (red, green and blue) refers to the fundamental quantum numbers carried by individual quarks and gluons, analogous to positive or negative electric charges. The "strong charge", more formally known as the strong coupling constant (g_s, where $\alpha_s = g_s^2/(4\pi)$), refers to the universal strength of the interaction between those colour-charged particles.

Lastly, the $\mathcal{N}_X$ normalisation factor will also need to be generalised. It becomes,

$$\mathcal{N}_X = \frac{1}{2g_s^{2(n-2+l)} C_F}. \tag{2.28}$$

2.4.2 Colour Algebra and the Antenna Colour Decomposition

The processes represented by the X_n^l antenna functions are composed of colour-carrying particles, namely quarks and gluons. When the amplitudes $\mathcal{M}_n^l$ are evaluated, the kinematics and colour are not explicitly separate algebraically, but since we require the antenna functions to be purely kinematic, we need to decompose them into inner colour structures. To be able to decompose, we must first clarify the Algebra of colour [9, 11].

Quarks carry a fundamental colour index $i \in \{1, 2, 3\}$ while gluons carry an adjoint colour index $a \in \{1, ..., 8\}$. The interactions between them are governed by the generators T^a. Those generators are normalised by the trace relation $\mathrm{Tr}\left(T^a T^b\right) = T_R \delta^{ab}$, where by standard convention $T_R = 1/2$ [1, 2]. This is particularly useful since when the amplitude is squared, the two colour generators must be contracted with one another, that is T^a and $(T^a)^\dagger$, and since the generators are Hermitian, $(T^a)^\dagger = T^a$. Hence, summing over all colour we get,

$$\sum_{i_1, i_2, a} (T^a)_{i_1 i_2} (T^a)_{i_2 i_1} = \sum_a \mathrm{Tr}\left(T^a T^a\right) = \sum_a T_R \delta^{aa}. \tag{2.29}$$

For a fixed a, the trace equals T_R. Summing this result over all $N^2 - 1$ adjoint indices we get,

$$\sum_a T_R \delta^{aa} = \frac{1}{2}\left(N^2 - 1\right). \tag{2.30}$$

We now also introduce the Casimir invariants: the fundamental and adjoint Casimir invariants, C_F and C_A. These represent, respectively, the total colour charge squared of the quark and of the gluon. They are defined as,

$$\begin{aligned} C_F &= \frac{N^2 - 1}{2N} \\ C_A &= N. \end{aligned} \tag{2.31}$$

The invariants are particularly useful to normalise the two hard quarks' colour from the antenna functions, by taking out a factor of $2C_F$. This factor equals,

$$2C_F = N - \frac{1}{N} \tag{2.32}$$

which will be a normalisation used throughout the present work.

In loop processes, colour can flow in different ways depending on the virtual particles inside the loops. So, instead of only simplifying using the fundamental Casimir invariant, we must also decompose the antenna structures into their distinct colour components. These are,

- The leading colour component (N): these come from planar Feynman diagrams (where virtual gluons are exchanged without crossing over each other). This component is represented by the base antenna in standard notation, e.g. A_3^1.
- The subleading colour component ($1/N$): these come from non-planar diagrams (where gluon lines do cross each other). This component is represented by the tilde antenna, e.g. $\tilde{A}_3^1$.

- The fermion loop (N_f): Instead of a virtual gluon, the loop can have a virtual quark-antiquark pair. This contribution depends strictly on the number of active flavours, N_f. This component is represented by the hat antenna, e.g. $\hat{A}_3^1$[9].

The direct expansion to Eq. 2.29 is to generalise to the four quark colour indices, instead of only two. That expansion is

$$\sum_a (T^a)_{i_1 i_2} (T^a)_{i_3 i_4} = \frac{1}{2}\left(\delta_{i_1 i_4}\delta_{i_2 i_3} - \frac{1}{N}\delta_{i_1 i_2}\delta_{i_3 i_4}\right). \tag{2.33}$$

It is the $SU(N)$ completeness relation[8] and is particularly useful for four-quark topologies like those described in Subsection 2.4.3 [1, 2].

A complete compilation of the QCD Feynman rules and the full colour trace derivations utilised by the package can be found in Appendix [ref to qcd appendix].

2.4.3 Antenna Families and T-Objects

Throughout this work the Born-level process under study is $\gamma^* \to q\bar{q}$. This process is represented at higher perturbative orders by the A, B and C antenna families. Antenna families are designations we use to identify the type of process that underlies a certain function. Consider the A_2^1 antenna: from our previous description, this antenna is a quark-antiquark antenna with a gluon loop. A_3^0 is a part of that same antenna family, the A family, those quark-antiquark antennae where gluons are exchanged or produced as final-state particles, but does not present with any loops. Instead, the gluon here is one of the now three final-state particles. Going to a higher level on the perturbative series, at NNLO, the quark-antiquark processes begin to allow for four final-state particle systems, in which the two unresolved partons may be other quark-antiquark pairs. This necessitates the creation of new antenna families since the aforementioned definition is now broken. We then designate the A_4^0 antenna for the $\gamma^* \to qgg\bar{q}$ process and the B_4^0 and C_4^0 antennae for the $\gamma^* \to q\bar{q}q\bar{q}$ [9]. We here require two different families because the process may present with either like (identical) or unlike-flavour pairs with the B antenna carrying the unlike-flavour case, often notated by $\gamma^* \to q\bar{q}q'\bar{q}'$, and the C antenna carrying the like-flavour case, denoted as $\gamma^* \to q\bar{q}q\bar{q}$. This last process, due to its topology, requires symmetrisation.

To compute these two antenna functions we start by writing the four-quark tree-level amplitude as a sum of two flavour-flow sub-amplitudes as,

$$\mathcal{M}_{q\bar{q}q\bar{q}} = \mathcal{M}_D + \mathcal{M}_E \tag{2.34}$$

where $\mathcal{M}_D$ is the direct and $\mathcal{M}_E$ is the exchanged [9] assignment amplitude.

The B_4^0 antenna, where the flavours are non-identical, so only the direct assignment amplitude is used. The antenna is built from self-interference,

$$B_4^0 = \mathcal{N}_X \frac{\mathcal{M}_D \mathcal{M}_D^*}{|\mathcal{M}_2^0|^2}. \tag{2.35}$$

On the other hand, C_4^0, where the flavours are identical, the two subamplitudes must interfere. The

[8]This expression is also called the *colour Fierz identity*.

[9]The words *direct* and *exchange* denote the way in which the like- and unlike-flavour quarks are paired amongst each other. When all flavours are identical, the pairing may be blind, since either of the quarks pairing with either of the antiquarks results in the same pair. If the flavours are not alike, however, the pairing order matters and must be considered, hence the use of the *exchanged* amplitude.

antenna is given as [1, 9],

$$C_4^0 = \mathcal{N}_X \frac{\mathcal{M}_D \mathcal{M}_E^* + \mathcal{M}_E \mathcal{M}_D^*}{\left|\mathcal{M}_2^0\right|^2}. \tag{2.36}$$

Though not deeply included in the present work, there are two other antenna types: there are quark-gluon antennae, which are organised under the D and E families, and there are gluon-gluon antennae, which are organised under the F, G and H families. As the names suggest, these types represent different Born-level processes. The first represents the boson $\to qg$ process, and the second the boson $\to gg$ process. Remarkably, these processes do not neatly fit into the Standard Model of Particle Physics (SM), as the vertices required to make a virtual photon split into either of those pairs does not exist inside the model. The implementation of those antenna functions, then, requires the use of Effective Field Theories (EFT). For the quark-gluon antennae the EFT used is SUSY and the process does not start in a virtual photon, but rather in a virtual neutralino. This neutralino splits into a gluon-gluino pair, as per SUSY and the gluino is then effectively considered an up-quark, hence describing a quark-gluon pair [17]. The gluon-gluon antennae require the use of the Higgs to gluon pair EFT framework. Here, the SM Feynman diagram describing a Higgs boson decaying into a top-quark triangle and then each of the other two vertices of that triangle emitting a gluon gets its intermediate triangle step compacted into the vertex, hence creating a $H \to gg$ process and, hence, a gluon-gluon antenna [9, 18, 19]. These processes are describing in further detail in Appendix [ref to SUSY and HiggsEFT explanation appendix].

As described in Subsection 2.4.2, antenna functions are purely kinematic. By taking all components, we are able to construct a process-wide kinematics- and colour-conscious object. These are the $\mathcal{T}$-objects. These objects are defined, before integration and hence notated by T, conceptually as [10] ,

$$T^{2(k+1)}_{\text{topology}} = (\text{colour/flavour weight}) \times T^2_{q\bar{q}} \times (\text{colour-weighted sum of antennae at } k\text{-order}). \tag{2.37}$$

This structure creates a very elegant structure for the A-type antenna relative T-objects. It goes like,

$$T^{2(k+1)}_{\text{topology}} = 2C_F T^2_{q\bar{q}} \sum_c F_c A^l_{n,(k),c} \tag{2.38}$$

where c denotes the component (leading, subleading, fermion loop, etc), F_c is the colour weight, that is N, $1/N$, N_f, etc, and $A^l_{n,(k),c}$ is the relative component antenna with n final-state particles and l loops. For this antenna type, this definition needs two extensions at NNLO that do not cleanly fit into an algorithmic formula:

- The A_3^1 antenna set represents a particularly interesting process. Unlike in lower order loop processes, here two very distinct kinds of Feynman diagram appear: those in which the loop and the emission of the final-state gluon are completely separate events, and those in which the loop involves all three external legs. The first kind of diagram maps exactly a A_2^1 antenna topology followed by a A_3^0 topology: first the two hard quarks exchange a gluon and then one of the hard quarks emits a final-state gluon[11]. Since this behaviour is not singular to the A_3^1 antenna, it must be subtracted from the bare result to retain the antenna's singular behaviour. When we then construct back the T-object, the $A_2^1 A_3^0$ term has to be added back on such that the process is considered in its totality [9].

[10] The T-objects are commonly notated by T^2, T^4 and T^6, depending on their perturbative order: LO, NLO and NNLO, respectively. That is the same as using twice our previous k-index, introduced in 2.4.1, plus one. This is the notation used in [20] and the one that will be used for the rest of the present work.

[11] This is merely a conceptual, diagrammatic interpretation, as the actual process is much less sequential than this description suggests.

- The A_2^2 antenna set, also at NNLO, is the first time we find a double-loop topology. Here the extension required is not a subtraction of some kind of behaviour, but rather an addition of a new one. This set is, in reality, composed of two different processes: the first follows the standard rule closely, as it produces a leading, a subleading and a fermion loop component. The second, however, has a singular antenna which describes the self-interference of the one-loop correction with itself, now that the system presents with two loops. This antenna is notated by $\breve{A}_2^2$. It is also the only genuinely square (not cross) term at NNLO. This means that it requires a normalisation of $(2C_F)^2$ [9].

As mentioned before Eq. 2.38, the equation is only valid for T-objects constructed from A-type antenna functions. In this work we consider all quark-antiquark antennae up to NNLO. This includes B_4^0 and C_4^0. These antennae populate the $T^6_{q\bar{q}q'\bar{q}'}$ and $T^6_{q\bar{q}q\bar{q}}$ objects, respectively.

$T^6_{q\bar{q}q'\bar{q}'}$ is built in the same way as other single-topology antenna T-objects, like $T^4_{q\bar{q}g}$, with the only difference being that here all quark flavours are summed over, requiring the inclusion of a $(N_f - 1)$ term, constructing,

$$T^6_{q\bar{q}q'\bar{q}'} = 2C_F T^2_{q\bar{q}}(N_f - 1)B_4^0. \tag{2.39}$$

The identical-flavour quark-antiquark-quark-antiquark final state is described by the sum of antennae for non-identical-flavour and for identical-flavour-only [9]. Using Eq. 2.33 in the interference term, with the identity piece vanishing by tracelessness [12], we can describe the identical-flavour T-object as,

$$T^6_{q\bar{q}q\bar{q}} = 2C_F T^2_{q\bar{q}}\left[B_4^0 - \frac{1}{N}C_4^0\right] \tag{2.40}$$

importantly without the inclusion of the quark-flavour sum term $(N_f - 1)$.

2.4.4 Phase-Space Factorisation

The previous Subsections focused on the antenna functions as objects: how they are built, combined, normalised, etc. But the antenna functions as described in Subsection 2.4.1 are the singular factor $\mathcal{S}_j$, as described in Eq. 2.23, up to some normalisation factor. The antenna functions, then, as part of $d\sigma^S$ must be integrated as, schematically, as described in Eq. 2.11,

$$\int_1 d\sigma^S \tag{2.41}$$

where $\int_1$ denotes the integral over the unresolved parton phase-space. This integral does not only denote the integral over the single unresolved parton, alone, on two counts: first the fact that the integral must cover all unresolved partons which, for three-final-particle antennae is only one, but for four-final-particle antennae it becomes two. Second, the hard partons and their interactions with the unresolved partons must also be integrated over. The integral becomes over the n-final-state-particle antenna phase space. This makes,

$$\int_1 d\sigma^S \sim \int d\Phi_{ijkl\ldots}\, X_n^l \tag{2.42}$$

up to normalisation [9].

The n-final-state-particle antenna phase space $\Phi_{ijkl\ldots}$ [13] is not the standard n-particle phase space as in

[12] This process is derived completely in Appendix [ref to the derivation].

[13] This phase space is notated by Φ_{ijk} for three-final-state-particle antennae like A_3^1 and Φ_{ijkl} for four-parton-final-particle antennae like A_4^0.

Φ_3 or Φ_4, it is, rather, the antenna's phase space, associated with the unresolved-partons emission between the two hard radiators, it is the unresolved-parton phase space of the antenna. It has no independent parametrisation and is only accessed by factorising the known n-particle phase space [9, 21]. Since, through the recursive factorisation of the Lorentz-invariant phase space property[14] [5], the phase space factorises, we are able to denote that the n-particle phase-space equals the $(n - n_u)$-particle, where n_u is the number of unresolved partons, phase-space times the antenna phase-space for n final-state particles[15]. For our case, two useful identities are produced for this formula [9],

$$d\Phi_3 = d\Phi_2 d\Phi_{ijk}$$

$$d\Phi_4 = d\Phi_2 d\Phi_{ijkl}. \tag{2.43}$$

Since $d\Phi_n$ is perfectly factorised [5], and the X_n^l antenna functions are explicitly built to be a function of the antenna's internal invariants alone and to have no dependence on how the reduced system is oriented [9], the two phase spaces are integrated separately, with the two-particle phase-space integral producing Φ_2 directly. The integrated antenna $\mathcal{X}_n^l$ is then computed as,

$$\mathcal{X}_n^l \sim \int d\Phi_{ijkl...}\, X_n^l = \frac{1}{\Phi_2} \int d\Phi_n\, X_n^l \tag{2.44}$$

which is how the phase-space prefactor appears in $\mathcal{N}(\epsilon, k)$, as described in Section 2.3.

Hence, finally, the integrated antenna $\mathcal{X}_n^l$ is given generally[16] as, taking the aforementioned overall $\mathcal{N}$ normalisation factor [9, 21],

$$\mathcal{X}_n^l = \mathcal{N}(\epsilon, k) \int d\Phi_n\, X_n^l. \tag{2.45}$$

A Note On $n_u = 0$ Cases

There are two antenna types that are exempt from this rule: A_2^1 and A_2^2, as they do not present with unresolved partons, but rather only with loops, and are completely virtual antennae. These antennae are only integrated over the loop momenta rather over the phase space, thus not requiring normalisation using the two-particle phase-space [22].

Implications on the T-objects

With this definition of the integrated antenna $\mathcal{X}_n^l$, we are now also able to introduce the often used integrated T-object as,

$$\mathcal{T}_{\text{topology}}^{2(k+1)} = 2C_F T_{q\bar{q}} \sum_c F_c \mathcal{X}_{n,(k),c}^l \tag{2.46}$$

written schematically from Eq. 2.38, though the structure may change with the antenna type used to define the relevant T-object. This equation describes that the integrated T-object, $\mathcal{T}_{\text{topology}}^{2(k+1)}$ is defined using the integrated antennae $\mathcal{X}_n^l$, integrated over their respective phase-spaces rather than blindly integrating the unintegrated T-object directly [9, 20].

[14] This property is discussed further in Appendix [ref to relevant appendix].

[15] This mechanism, as described, works at $n_u \in \{1, 2\}$. The perturbative order above, N^{3}LO, presents more complex identities for which this blueprint may not work. It also does not fit when $n_u = 0$. That case is discussed in further detail in Subsubsection 2.4.4.

[16] This definition is general for antenna functions that are already cleanly and completely given in terms of Mandelstam invariants and ϵ. This is only automatically true for antenna where $l = 0$. When $l \geq 0$, which happens for all loop antennae by definition, a prior loop-momentum integration step is necessary. This step is explored further in Section 2.5.

2.4.5 NLO and NNLO Subtraction Structure

So far, throughout the present Section 2.4, we have discussed what antenna functions are and represent, how they are normalised, what conventions are used when using them and the overall objects we use the antenna functions to compute, the T-objects. What we have not yet done, and hence not closed the section's loop is to explain why we require both the antenna functions and the T-objects. We require them, as hinted at in Subsection 2.2.1, to compute the $d\sigma^S$ differential subtraction counterterm used in the antenna subtraction scheme, as schematically described in Eq. 2.11 [5].

The main reason behind the antenna subtraction scheme is to create a finite differential prediction tool (the differential cross-section) for some F_n infrared-safe quantity, or observable, for n final-state particles, which here acts as a distribution. The terms of the subtraction formula must, then, carry their own infrared-safe quantities through the process. As a matter of example, the differential (F_n-dependent) NLO finite cross-section is given by,

$$d\sigma^{\text{NLO}} = \int_{m+1} \left(d\sigma^R F_{m+1} - d\sigma^S F_m \right) + \int_m \left(d\sigma^V + \int_1 d\sigma^S \right) F_m. \tag{2.47}$$

where $m = 2$ and is the number of final-state particles at Born-level [1].

Since the observable considered throughout this work, the hadronic R-ratio, is fully-inclusive, making no distinction between resolved and unresolved final states at any multiplicity, the measurement distribution function F_n equals one identically, for every n. The cross-section, in our case, is therefore no longer a differential prediction in F_n but rather reduces to the plain, fully-summed total. This enables us to write,

$$\sigma^{\text{NLO}} = \int_{m+1} \left(d\sigma^R - d\sigma^S \right) + \int_m \left(d\sigma^V + \int_1 d\sigma^S \right) \tag{2.48}$$

which maps directly to Eq. 2.11 and represents the scheme at NLO.

The scheme looks different at NNLO, simply due to the larger number of possible corrections. Instead of only real and virtual cases, this order's correction receives contributions where the two additional orders of α_s are realised as two extra real partons (RR), one extra real parton and one loop (RV), or two loops with no extra partons (VV). The expression then expands to meet these new terms to,

$$\sigma^{\text{NNLO}} = \int_{m+2} \left(d\sigma^{RR} - d\sigma^S \right) + \int_{m+1} \left(d\sigma^{RV} - d\sigma^T \right) + \int_m \left(d\sigma^{VV} + \int_1 d\sigma^T + \int_2 d\sigma^S \right) \tag{2.49}$$

where instead of only one, there are two distinct subtraction counterterms. These are $d\sigma^S$, which subtracts the singularities of $d\sigma^{RR}$, at both single- and double-unresolved level, and $d\sigma^T$, which subtracts the remaining single-unresolved singularities of $d\sigma^{RV}$ [9]. This inadvertently adds another level of complexity for NNLO: the KLN theorem guarantees only that the fully inclusive sum over degenerate configurations is finite, but says nothing about any particular piece of a decomposition being finite on its own. With the split including multiple subtractions at the same time, we are implying that each of these differences is separately, pointwise finite, everywhere in its own phase space.

As hinted at throughout this text but not yet stated explicitly: the subtraction terms are exact, not merely asymptotic, matches to the original cross-section terms. This follows from two facts specific to this process. First, because the antenna functions are built as exact, unique expressions (Eq. 2.24), and the sum in Eq. 2.23 runs over all unresolved pairs, of which there is only one, since the Born process $\gamma^* \to q\bar{q}$has a single colour-connected pair. Second, because the Born factor $|\mathcal{M}_2^0|^2$ depends only on the conserved quantity q^2, and therefore takes the same value whether evaluated on the exact real-emission momenta or on the reconstructed momenta used in the subtraction term. Together, these mean $d\sigma^S$ and $d\sigma^T$ coincide identically with $d\sigma^R$, $d\sigma^{RR}$, and $c\sigma^{RV}$ at every point in the phase space, for either perturbative order, not merely in the singular unresolved limit, as the general subtraction formalism only guarantees [5]. an example, σ^{NNLO}, from Eq.2.49, simplifies to,

$$\sigma^{\text{NNLO}} = \int_m \left(d\sigma^{VV} + \int_1 d\sigma^T + \int_2 d\sigma^S \right) \tag{2.50}$$

in which each of the terms is described by a different T-object, as described in Subsection 2.4.3. The objects then describe each of the cross-section terms directly, and a clear hierarchy of objects to cross-sections can be made looking at the number of unresolved partons in each of them. The expression then becomes,

$$\sigma^{\text{NNLO}} = \frac{1}{T^2_{q\bar{q}}} \left[\mathcal{T}^6_{q\bar{q}} + \mathcal{T}^6_{q\bar{q}g} + \left(\mathcal{T}^6_{q\bar{q}q\bar{q}} + \mathcal{T}^6_{q\bar{q}q'\bar{q}'} + \mathcal{T}^6_{q\bar{q}gg} \right) \right] \tag{2.51}$$

where the $1/T^2_{q\bar{q}}$ term is added to normalise the LO order to one, as $\sigma^{\text{LO}} = T^2_{q\bar{q}}$ and where $\mathcal{T}$ denotes the integrated form of a certain T-object, as described in Eq. 2.46.

The NLO case goes in the same direction exactly. Its cross-section becomes,

$$\sigma^{\text{NLO}} = \frac{1}{T^2_{q\bar{q}}} \left(\mathcal{T}^4_{q\bar{q}} + \mathcal{T}^4_{q\bar{q}g} \right). \tag{2.52}$$

Since the T-objects are described using the antenna functions, by necessity, both σ^{NLO} and σ^{NNLO} may also be given in terms of the respective antenna functions, after the normalisation procedures described throughout this text, as follows [9],

$$\begin{aligned}
\sigma^{\text{NNLO}} = \left(N - \frac{1}{N} \right) \Bigg[& N \left(\mathcal{A}^1_2 \mathcal{A}^0_3 + \mathcal{A}^1_3 + \mathcal{A}^0_4 + \mathcal{A}^2_2 \right) - \frac{1}{N} \left(\mathcal{A}^1_2 \mathcal{A}^0_3 + \tilde{\mathcal{A}}^1_3 + \tilde{\mathcal{A}}^0_4 + C^0_4 - \tilde{\mathcal{A}}^2_2 \right) \\
& + N_f \left(\hat{\mathcal{A}}^1_3 + \hat{\mathcal{A}}^2_2 \right) + \left(N - \frac{1}{N} \right) \breve{\mathcal{A}}^2_2 \Bigg] \\
\sigma^{\text{NLO}} = \left(N - \frac{1}{N} \right) & \left(\mathcal{A}^1_2 + \mathcal{A}^0_3 \right).
\end{aligned} \tag{2.53}$$

2.5 Phase-Space Integration and IBP Reduction

In the previous sections we went over what an antenna function is, why they are necessary and how we use them. In Eq. 2.45 we defined the method we use to integrate the antenna functions X^l_n, to integrated antenna functions $\mathcal{X}^l_n$. These antennae are integrated over the phase-space and also, when loops are present, their loop momenta ℓ_r, as pointed out in Footnote 16.

Though integration has been made to seem the natural endpoint of this present work, and to an extent it is, there is still use among phenomenologists for the unintegrated antenna function X^l_n, namely when using Monte Carlo (MC) integrators. These integrators evaluate the phase space numerically point-by-point and require our unintegrated antennae to perform subtractions over local divergences, as described in Subsection 2.4.5 [9]. Since the MC generator throws random phase-space points, and the subtraction happens dynamically with each throw, the unintegrated antenna functions used must be given in terms of the exact same local kinematics as the real-emission matrix elements, the Mandelstam invariants s_{ij}.

For this reason, we need to perform the loop-momentum integral before using these antennae in these integrators, if $l \geq 1$[17].

From Eq. 2.44, we have seen that through phase-space factorisation, the antennae are integrated over the antenna phase space Φ_X. If required, they are also integrated over their loop momenta. The general expression used to describe this integration setup is,

$$X_n^l \sim \int \left(\prod_{r=1}^{l} d^d \ell_r \right) d\Phi_X \, \frac{N_s(\ell_r, p_a)}{\prod_{j=1}^{N} D_j^{a_j}(\ell_r, p_a)}, \quad d = 4 - 2\epsilon \tag{2.54}$$

where ℓ_r denote the loop momenta, and the set p_a consists of a fixed mapped hard-radiator momenta $\tilde{p}_{\text{hard}}$ and the unresolved antenna momenta $p_{\text{unresolved}}$, the latter of which are integrated through Φ_X. N_s is a numerator built from scalar products, D_j are propagator or invariant denominators and a_j are integer powers. An algebraic expansion of the integrand of a single amplitude can generate multiple integrals with the same kinematic structure but different numerator and denominator powers. These integrals may then be organised into integral families, which fix the denominators and integration variables, and may then be systematically reduced. These families are denoted as the complete collection [21],

$$\mathcal{F}[\{D_j\}; \{\ell_r\}, d\Phi_X] = \{I(a_1, ..., a_N)\}, \quad a_j \in \mathbb{Z} \tag{2.55}$$

with,

$$I(\vec{a}) = \int \left[\prod_{r=1}^{l} d^d \ell_r \right] d\Phi_X \, \frac{1}{D_1^{a_1} ... D_N^{a_N}}, \quad \vec{a} = (a_1, ..., a_N). \tag{2.56}$$

To reduce these families' terms, denoted as I here, is to write them in terms of a finite basis as,

$$I(\vec{a}) = \sum_{k=1}^{N_{\text{MI}}} c_k(\vec{a}, \epsilon, \{s_{ij}\}) M_k \tag{2.57}$$

where M_k are the master integrals (MI). In essence, it takes the problem of integrating those many family terms I and separates into two parts. In I, positive a_j indices denote propagator powers while non-positive indices account for absent denominators or irreducible scalar products in the numerator. The reduction separates the calculation into determining the family-specific coefficients c_k and evaluating the master integrals M_k, so that every required family member is expressed through a substantially smaller basis.

As stated previously, MC integrators require the unintegrated antenna functions in terms of the Mandelstam invariants s_{ij}, since they must evaluate the subtractions at a sampled physical external phase-space point p. The loop-level antenna required by an MC is ultimately evaluated as a function of the external invariants. Its calculation, however, involves tensor loop integrals with loop-momentum dependence, which must be reduced before the antenna can be evaluated at a sampled phase-space point. This reduction is performed through the Passarino-Veltmann (PaVe) reduction mechanism, described in detail in Subsection 2.5.1, which reduces the tensor loop integrals to linear combinations of scalar one-loop functions, usually denoted by A_0, B_0, ..., while keeping the phase-space variables and integration untouched, as [23],

$$\int d^d \ell \, \frac{\ell^{\mu_1} ... \ell^{\mu_R}}{\prod_j D_j^{a_j}} \xrightarrow{\text{PaVe}} \sum_k c_k\big(\{s_{ij}\}\big) \, \{A_0, B_0, C_0, D_0, ...\}_k. \tag{2.58}$$

[17] Note that in standard antenna language, the term *unintegrated* already refers to phase-space integration, over Φ_X.

To finally reach the integrated antenna functions $\mathcal{X}_n^l$ we do require integrating over the phase-space. For this integral we also use a reduction method: Integration By Parts (IBP) reduction, described in detail in Subsection 2.5.2. This method generates relations among scalar integrals in a common family through vanishing total derivatives, as [24],

$$0 = \int d^d\ell \, \frac{\partial}{\partial \ell^\mu}\left(v^\mu \frac{1}{\prod_j D_j^{a_j}} \right). \tag{2.59}$$

As mentioned in previous sections, emitted final-state particles are real and on shell. That condition is enforced in the phase-space through the use of Dirac-delta functions like $\delta^+(p_i^2)$. Note that these are not ordinary Feynman propagators. To bring the phase-space integration into the same algebraic framework as the loop integrals, we introduce the method of reverse unitarity to represent these Dirac-delta functions as cut propagators, through the discontinuity relation as [21],

$$2\pi i \, \delta^+\left(p_i^2\right) = \left[\frac{1}{p_i^2 - i0} - \frac{1}{p_i^2 + i0} \right]_+ . \tag{2.60}$$

where both + signs, in δ^+ and $]_+$, represent the positive-energy prescription.

The family now contains ordinary propagators from virtual loops and cut propagators representing the unresolved phase-space. The IBP identities can then reduce the resulting combined cut-integral family to a finite set of MI.

After UV renormalisation and evaluation of the MI, the result is expanded as a Laurent series in ϵ, making the remaining explicit IR poles visible order by order [9, 7].

2.5.1 PaVe Reduction

At one-loop, amplitudes produce tensor integrals similar to those in Eq. 2.58 [23]. With general massive[18] denominator defined as,

$$D_0 = \ell^2 - m_0^2 + i0, \quad D_i = (\ell + r_i)^2 - m_i^2 + i0 \tag{2.61}$$

the tensor integral takes the form that follows,

$$I_N^{\mu_1 \dots \mu_R} = \mu^{4-d} \int \frac{d^d\ell}{i\pi^{d/2}} \frac{\ell^{\mu_1} \dots \ell^{\mu_R}}{D_0 D_1 \dots D_{N-1}} \tag{2.62}$$

where R is the tensor rank, r_i are the combinations of external momenta and m_i is the mass associated with the i-th propagator. The reduction is carried out in $d = 4 - 2\epsilon$ dimensions, so that,

$$g_{\mu\nu} g^{\mu\nu} = d. \tag{2.63}$$

PaVe reduces the Lorentz indices in the numerator $\ell^\mu \ell^\nu \ell^\rho \dots$ terms and expresses them in terms of external momenta p_i^μ, the metric tensor $g_{\mu\nu}$ and scalar one-loop integrals, through Lorentz covariance [23].

[18]Throughout this subsection, the general and massive case will be considered for completeness. The rest of the present work, however, heavily leans towards the massless regime. In that regime the internal masses vanish and the physical external partons satisfy $p_a^2 = 0$. The shifted momenta r_i, introduced in Eq. 2.61, entering propagators are combinations of external momenta and need not be lightlike.

For a rank-one N-point integral, the only available vectors are the independent external momenta, such that,

$$I_N^\mu = \sum_{i=1}^{N-1} p_i^\mu C_i \tag{2.64}$$

where C_i are scalar coefficient functions. For rank-two, the relation becomes,

$$I_N^{\mu\nu} = \sum_{i,j=1}^{N-1} p_i^\mu p_j^\nu C_{ij} + g^{\mu\nu} C_{00} \tag{2.65}$$

and for rank-three,

$$I_N^{\mu\nu\rho} = \sum_{i,j,k=1}^{N-1} p_i^\mu p_j^\nu p_k^\rho C_{ijk} + \sum_{i=1}^{N-1} \left(g^{\mu\nu} p_i^\rho + g^{\mu\rho} p_i^\nu + g^{\nu\rho} p_i^\mu\right) C_{00i} \tag{2.66}$$

and so on for higher ranks.

The reduction mechanism works because scalar products involving the loop momentum can be rewritten using the propagator denominators as,

$$2\ell \cdot r_i = D_i - D_0 - r_i^2 + m_i^2 - m_0^2 \tag{2.67}$$

which lets us cancel denominator terms, through a method like the following example set at $i = 1$,

$$\frac{\ell \cdot r_1}{D_0 D_1} = \frac{1}{2}\left(\frac{1}{D_0} - \frac{1}{D_1} + \frac{1}{D_0 D_1}\left(-m_0^2 + m_1^2 - r_1^2\right)\right) \tag{2.68}$$

thereby rewriting loop-momentum scalar products in terms of denominators and scalar integrals.

While using this reduction scheme, we might also encounter Gram matrices, defined as,

$$G_{ji} = 2 p_j \cdot p_i. \tag{2.69}$$

Contracting a general rank-one integral, as described in Eq. 2.64, with p_j gives,

$$p_j \cdot I_N = \sum_i (p_j \cdot p_i) C_i = \frac{1}{2} \sum_i G_{ji} C_i. \tag{2.70}$$

If we now define $R_j \equiv p_j \cdot I_N$, we are then able to compute the C_i coefficients in the Lorentz decomposition by taking[19],

$$G_{ji} C_i = 2R_j \Rightarrow C_i = 2\left(G^{-1}\right)_{ij} R_j. \tag{2.71}$$

Rank-One Bubble Example

To illustrate the method, let us consider the rank-one bubble,

$$B^\mu\left(p; m_0^2, m_1^2\right) = \mu^{4-d} \int \frac{d^d\ell}{i\pi^{d/2}} \frac{\ell^\mu}{D_0 D_1} \tag{2.72}$$

[19]The inversion in Eq.2.71 assumes a non-singular Gram matrix. Exceptional kinematics may require a limiting procedure or an alternative reduction method.

where the denominators follow Eq. 2.61. By Lorentz covariance we are able to write that,

$$B^{\mu} = p^{\mu} B_1 \tag{2.73}$$

which once contracted with p_{μ} makes,

$$p^2 B_1 = p_{\mu} B^{\mu} = \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{\ell \cdot p}{D_0 D_1}. \tag{2.74}$$

Using the relation in Eq. 2.67, where we consider $r_1 = p$, we get,

$$p^2 B_1 = \frac{1}{2} \left[A_0\left(m_0^2\right) - A_0\left(m_1^2\right) + \left(m_1^2 - m_0^2 - p^2\right) B_0\left(p^2; m_0^2, m_1^2\right) \right] \tag{2.75}$$

and therefore[20],

$$B^{\mu} = p^{\mu} \frac{A_0(m_0^2) - A_0(m_1^2) + (m_1^2 - m_0^2 - p^2)\, B_0}{2p^2}. \tag{2.76}$$

Here, A_0 and B_0 are the scalar one-loop integrals mentioned throughout. These two types can be defined as,

$$\begin{aligned} A_0\left(m^2\right) &= \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{1}{\ell^2 - m^2 + i0} \\ B_0\left(p^2; m_0^2, m_1^2\right) &= \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{1}{D_0 D_1}. \end{aligned} \tag{2.77}$$

2.5.2 IBP Identities and the Laporta Algorithm

IBP Identities

The integral family has already been defined in Eq. 2.56. This definition allows for the factorisation of the integral into a loop and a phase-space integral term. That factorisation results in,

$$I(\vec{a}) = \int d\Phi_X \, \mathcal{I}(\vec{a}; \{D_1, ..., D_N\}), \quad \mathcal{I}(\vec{a}; \{D_1, ..., D_N\}) = \int \prod_{r=1}^{l} d^d \ell_r \, \frac{1}{D_1^{a_1} ... D_N^{a_N}}. \tag{2.78}$$

The overall normalisation factors are irrelevant to the identities and will, therefore, be suppressed throughout the following explanation. The denominators may generally be defined as,

$$D_j = \left(\sum_{r=1}^{l} c_{jr} \ell_r + r_j \right)^2 - m_j^2 + i0 \tag{2.79}$$

where r_j is built from momenta held fixed during the loop integration, including the mapped hard partons and unresolved antenna momenta. Take one loop momentum ℓ_s, for $s = 1, ..., l$, and one vector v^{μ} from the available loop and external momenta. Dimensional regularisation allows us to set a total derivative to zero inside $\mathcal{I}$, as [21, 24],

$$0 = \int \prod_{r=1}^{l} d^d \ell_r \, \frac{\partial}{\partial \ell_s^{\mu}} \left[v^{\mu} \frac{1}{D_1^{a_1} ... D_N^{a_N}} \right] \tag{2.80}$$

[20]The result shown in Eq. 2.76 assumes generic kinematics with $p^2 \neq 0$. Exceptional cases are obtained through an appropriate limit or alternative reduction.

as previously introduced in Eq. 2.59. In dimensional regularisation, analytic continuation justifies the vanishing of the surface term at infinity, leaving an exact relation among regulated integrals[21].

Expanding the derivative gives,

$$0 = \int \prod_{r=1}^{l} d^d \ell_r \left[\frac{\partial_{\ell_s} \cdot v}{\prod_j D_j^{a_j}} - \sum_{j=1}^{N} a_j \frac{v^\mu \partial_{\ell_s^\mu} D_j}{D_1^{a_1} \ldots D_j^{a_j+1} \ldots D_N^{a_N}} \right]. \tag{2.81}$$

For these quadratic denominators D_j, differentiation with respect to ℓ_s^μ is linear in momenta,

$$\partial_{\ell_s^\mu} D_j = 2c_{js} \left(\sum_r c_{jr} \ell_r + r_j \right)_\mu \tag{2.82}$$

and therefore, $v^\mu \partial_{\ell_s^\mu} D_j$ is a scalar product. As scalar products may be expressed in terms of chosen denominators and irreducible scalar products, the latter are represented by negative family indices a_j, while zero indices denote absent denominators, we are able to write schematically,

$$\sum_{\vec{b}} \rho_{\vec{b}}(\vec{a}, \epsilon, \{s_{ij}\})\, \mathcal{I}(\vec{b}) = 0 \tag{2.83}$$

where each vector $\vec{b}$ labels a family member obtained from $\vec{a}$ through index shifts in one or more denominator powers. $\rho_{\vec{b}}$ is the corresponding coefficient in the IBP relation.

The method may be made easier to visualise with an example. Consider a one-loop bubble, akin to that studied in the previous Subsection 2.5.1. Take,

$$\mathcal{I}(a_0, a_1) = \int d^d \ell \, \frac{1}{D_0^{a_0} D_1^{a_1}} \tag{2.84}$$

where,

$$D_0 = \ell^2 - m_0^2, \quad D_1 = (\ell + p)^2 - m_1^2. \tag{2.85}$$

We now choose $v^\mu = \ell^\mu$, and build and expand the relation from Eq. 2.59 as,

$$\begin{aligned} 0 &= \int d^d \ell \, \frac{\partial}{\partial \ell^\mu} \left(\frac{\ell^\mu}{D_0^{a_0} D_1^{a_1}} \right) \\ &= d\mathcal{I}(a_0, a_1) - 2a_0 \int d^d \ell \, \frac{\ell^2}{D_0^{a_0+1} D_1^{a_1}} - 2a_1 \int d^d \ell \, \frac{\ell \cdot (\ell + p)}{D_0^{a_0} D_1^{a_1+1}}. \end{aligned} \tag{2.86}$$

We can now use,

$$\ell^2 = D_0 + m_0^2 \tag{2.87}$$

$$2\ell \cdot (\ell + p) = D_0 + D_1 + m_0^2 + m_1^2 - p^2 \tag{2.88}$$

[21] This method is explored further and completely in Appendix [ref to this].

and finally get the explicit shifted-index identity,

$$
\begin{aligned}
0 = &(d - 2a_0 - a_1)\mathcal{I}(a_0, a_1) \\
&- 2a_0 m_0^2 \mathcal{I}(a_0 + 1, a_1) \\
&- a_1 \mathcal{I}(a_0 - 1, a_1 + 1) \\
&- a_1 \left(m_0^2 + m_1^2 - p^2\right) \mathcal{I}(a_0, a_1 + 1).
\end{aligned}
\tag{2.89}
$$

Reverse Unitarity and Cut Integrals

In the previous subsubsection, we have derived the IBP identities relative to $\mathcal{I}$. As described in Eq. 2.78, the entire integrated antenna is not merely $\mathcal{I}$, but rather,

$$
I(\vec{a}) = \int d\Phi_X \, \mathcal{I}(\vec{a}). \tag{2.90}
$$

The unresolved phase-space measure is not originally an ordinary loop integral, however. Instead, it has on-shell constraints, which have to be considered and transformed before $I(\vec{a})$ is reducible via the present method.

As described in Subsection 2.4.4, phase-space factorisation isolates the antenna measure through $d\Phi_{m+r} = d\Phi_m d\Phi_X$, or in the two-hard-radiator final-final case we are considering throughout the present work, $d\Phi_X = d\Phi_n / d\Phi_2$, for n final-state particles. That n-particle phase space is given generally as,

$$
d\Phi_n(q : p_1, ..., p_n) = (2\pi)^d \delta^{(d)}\left(q - \sum_{i=1}^{n} p_i\right) \prod_{i=1}^{n} \frac{d^d p_i}{(2\pi)^{d-1}} \, \delta^+\left(p_i^2 - m_i^2\right) \tag{2.91}
$$

where, as described in the start of the current section, the δ-function imposes the mass-shell condition for the i-th final-state particle, while the superscript + selects its positive-energy solution. This may be written as,

$$
\delta^+\left(p_i^2 - m_i^2\right) = \theta\left(p_i^0\right) \, \delta\left(p_i^2 - m_i^2\right) \tag{2.92}
$$

This δ^+-function may be reexpressed as a discontinuity as follows [21, 25],

$$
2\pi i \, \delta^+\left(p_i^2 - m_i^2\right) = \left[\frac{1}{p_i^2 - m_i^2 - i0} - \frac{1}{p_i^2 - m_i^2 + i0}\right]_+ \tag{2.93}
$$

where the $\pm i0$ terms specify the propagator prescriptions. The subscript + represents the positive-energy cut selection.

These objects are called *cut propagators*, because they come about from cutting a propagator line, thus putting the represented particle on-shell, as a real final-state particle. Algebraically, it also allows phase-space constraints to be written in the same denominator language as virtual propagators. A cut propagator associated with a required final-state particle must be retained during the reduction and sectors in which such a cut is absent do not represent the physical phase-space integral and thus, vanish [21]. Using the discontinuity relation to refactor the δ-functions, we are able to rewrite the integrated antenna expression schematically as,

$$
\mathcal{X}_n^l \sim \int \left[\prod_{r=1}^{l} d^d \ell_r\right] \left[\prod_{u} d^d p_u\right] \frac{N_s(\ell_r, p_u; \tilde{p}_{\text{hard}})}{\prod_\alpha D_{v,\alpha}^{a_\alpha} \prod_\beta D_{c,\beta}^{a_\beta}} \tag{2.94}
$$

where $D_{v,\alpha}$ are ordinary virtual propagators, $D_{c,\beta}$ are cut propagators, $\tilde{p}_{\text{hard}}$ remains fixed as the hard-radiator combined momenta and p_u are integrated unresolved momenta.

After rewriting the integral in this way, the total derivative introduced in the previous Subsection 2.5.2 may now be taken with respect to any loop momentum ℓ_r^μ or integrated unresolved momentum p_u^μ. This process produces relations involving both virtual and cut denominators. This then allows us to reduce loop and phase-space integrals simultaneously using IBP reduction [21, 25].

As an example, let us now consider the massless three-particle phase-space Φ_3. It may be described as,

$$d\Phi_3(q;p_1,p_2,p_3) \propto d^dp_1d^dp_2d^dp_3\,\delta^+\left(p_1^2\right)\,\delta^+\left(p_2^2\right)\,\delta^+\left(p_3^2\right)\,\delta^{(d)}(q-p_1-p_2-p_3). \tag{2.95}$$

We start by using momentum conservation to eliminate p_3,

$$p_3 = q - p_1 - p_2. \tag{2.96}$$

This first step eliminates the rightmost δ-function. Then, using reverse unitarity, each of the remaining δ-functions may be rewritten as cut propagators. Schematically, it gives,

$$d\Phi_3 \sim d^dp_1d^dp_2\frac{1}{[p_1^2]_c[p_2^2]_c[(q-p_1-p_2)^2]_c} \tag{2.97}$$

with

$$\left[\frac{1}{p^2}\right]_c \equiv \frac{1}{2\pi i}\left[\frac{1}{p^2-i0}-\frac{1}{p^2+i0}\right]_+. \tag{2.98}$$

Laporta Algorithm

At NLO and NNLO these reduction systems become very complex and their solution ever more time-intensive. The Laporta algorithm is a systematic-elimination method that uses IBP relations to express more complicated integrals in terms of a smaller set of unreduced ones (master integrals) [26].

The algorithm first classifies the family members into sectors. These sectors are defined as,

$$\theta_j \equiv \theta(a_j) = \begin{cases} 1, & a_j > 0, \\ 0, & a_j \le 0, \end{cases} \qquad \vec{\theta} = (\theta_1, ..., \theta_N). \tag{2.99}$$

After fixing an integral family and identifying its admissible sectors, the Laporta algorithm assigns an ordering to the resulting integrals according to their complexity. This hierarchy is based on the number of propagators present t, the excess denominator power r, and the numerator complexity encoded through negative indices s[22]. These quantities are defined using each integral's a-index, taken from the family's

[22]The present t, r and s metrics are not to be taken as unique Laporta orderings, as these vary by implementation. Instead, they are illustrative objects.

$\vec{a}$ vector, as previously defined. They are, generally,

$$t = \sum_{j=1}^{N} \theta(a_j), \tag{2.100}$$

$$r = \sum_{j=1}^{N} \max(a_j - 1, 0), \tag{2.101}$$

$$s = \sum_{j=1}^{N} \max(-a_j, 0). \tag{2.102}$$

Here, $\theta(a_j)$ records whether denominator D_j is present. The hierarchy is defined by comparing t first, then r and finally s. The algorithm uses this chosen hierarchy to select, within each IBP relation, the most complex integral to eliminate first.

As an example, consider an arbitrary family defined as $\mathcal{F}[\{D_1, D_2, D_3\}; d\Phi]$, with the denominators chosen as $D_1 = s_{12}$, $D_2 = s_{13}$ and $D_3 = s_{23}$. We may take the integrals below, all members of this family, and organise them using their respective sector vectors as follows,

$$I_1 = \int d\Phi \frac{s_{13}}{s_{12}} \Rightarrow I_1 = I(1, -1, 0), \tag{2.103}$$

$$I_2 = \int d\Phi \frac{2}{s_{12}s_{13}s_{23}} \Rightarrow I_2 = 2I(1, 1, 1), \tag{2.104}$$

$$I_3 = \int d\Phi \frac{1}{s_{12}^3} \Rightarrow I_3 = I(3, 0, 0) \tag{2.105}$$

where $d\Phi$ is the common phase-space measure. Importantly, two integrals may share a sector if they contain the same denominators, independently of those denominators' powers. Also, only positive indices denote denominator presence, meaning that integrals I_1 and I_3 in the example above are members of the same sector, $I_1, I_3 \in (1, 0, 0)$. In our example, only two sectors appear: $(1, 0, 0)$ and $(1, 1, 1)$. These are our two target sectors. The three integrals are organised in terms of the t, r and s as follows,

	t	r	s
$I(1, -1, 0)$	1	0	1
$I(1, 1, 1)$	3	0	0
$I(3, 0, 0)$	1	2	0

(2.106)

then, by order of complexity, the algorithm orders the integrals as,

$$I(1, 1, 1) \succ I(3, 0, 0) \succ I(1, -1, 0). \tag{2.107}$$

After determining the integral families and sectors present, we apply the IBP identities to the seed integrals. These seed integrals are a part of a finite, bounded set of integrals chosen from each relevant sector, whose IBP identities must be sufficiently numerous to reduce every target integral and every non-zero subtopology reached along the way.

After generating IBP identities for the selected seed integrals, any relation containing these representatives is first solved for $I(1, 1, 1)$, the highest-ranked member, and the resulting expression is substituted recursively into the remaining system [26].

IBP reduction may produce lower sectors, but may also generate new integrals from the initially relevant sectors. We may define the relevant sectors as the target sectors plus all lower sectors produced by IBP reduction. Lastly, some sectors may be cancelled due to symmetry, if one denominator is shifted to another under loop-momentum shift or a relabelling of external momenta. If two sectors become equivalent, one is discarded from the independent reduction and the other is retained as its representative. In short, a sector is deemed relevant if it is an original target sector, or a lower sector reached from another via IBP reduction. The relevant sector must also be non-scaleless in a chosen kinematics, not redundant under a family symmetry and, for a cut family, it must retain all required cut denominators throughout.

It is important to note that cut families add a physical restriction: if an IBP reduction step cancels a cut propagator, the sector it produces is missing one of the required on-shell final-state particles and it no longer represents the intended phase-space. If that happens, the integral is set to zero [21].

The reduction is complete when the remaining non-zero family members cannot be eliminated by the available identities. These survivors form a master-integral basis.

2.5.3 Master Integrals

The IBP reduction method reduces the integral to a linear combination of master integrals as,

$$I(\vec{a}) = \sum_{k=1}^{N_{\mathrm{MI}}} c_k(\vec{a}, \epsilon, \{s_{ij}\}) M_k \tag{2.108}$$

where c_k are the reduction coefficients and represent what comes from the reduction algebra and M_k are the MI. Our remaining task is evaluating the finite set M_k [26].

Master integrals are not uniquely defined: an integral is a master only relative to the chosen family, kinematics, IBP system, and basis. This means that a different basis with different masters may be used to reduce the same integrals and span the exact same integral space [26].

Master integrals are scalar integrals with a fixed denominator/cut structure. They are generically represented as,

$$M_k = \int \left[\prod_r d^d \ell_r\right] \left[\prod_u d^d p_u\right] \frac{1}{\prod_\alpha D_{v,\alpha}^{a_\alpha} \prod_\beta D_{c,\beta}^{a_\beta}}. \tag{2.109}$$

For integrated antennae, the cut denominators in these master integrals encode the real unresolved phase space. Hence, a master may be a pure phase-space volume, a phase-space integral with invariant denominators, or a loop-plus-phase-space cut integral. The MI are also not necessarily finite, in fact, in dimensional regularisation they commonly contain ϵ poles [21].

For a massless fully inclusive antenna system, q^2 is often the only independent kinematic scale once the dimensional-regularisation normalisation factors have been separated. Through dimensional analysis, the MI are strongly constrained to take the form,

$$M_k(q^2, \epsilon) = \left(q^2\right)^{\lambda_k + \kappa_k \epsilon} f_k(\epsilon) \tag{2.110}$$

where λ_k is fixed by mass dimension and f_k is dimensionless, depending only on ϵ. For this reason, phase-space MI can often be written in terms of Gamma functions, as [21],

$$f_k(\epsilon) \sim \frac{\Gamma^m(1 - a\epsilon)}{\Gamma^n(1 - b\epsilon)}. \tag{2.111}$$

When evaluating master integrals, different types often call for different approaches. Simple one-scale phase-space masters often integrate directly after the phase-space is parametrised. Angular and energy-fraction integrations typically reduce to Beta functions. When a master depends on a non-trivial kinematic variable, such as $x = m^2/q^2$, the most common approach is to differentiate the master with respect to that variable. The derivative produces integrals in the same family, which IBP is then able to reduce to the master basis, as [27],

$$\frac{d}{dx}\vec{M}(x,\epsilon) = A(x,\epsilon)\vec{M}(x,\epsilon). \tag{2.112}$$

where $\vec{M} = (M_1, ..., M_{N_{\rm MI}})^{\rm T}$ is the vector of masters and A is a matrix of coefficient functions. Boundary conditions can then be used to determine the solution. These methods are discussed in further detail in Appendix [ref to appropriate appendix].

Once the MI are evaluated, we expand them around $\epsilon = 0$, as a Laurent series [21],

$$M_k = \sum_{n=n_{\rm min}}^{\infty} m_{k,n}\epsilon^n \tag{2.113}$$

to expose their pole and finite structure. The lowest point $n_{\rm min}$ can, and often is, negative, resulting in an expansion similar to,

$$M_k = \frac{m_{k,-2}}{\epsilon^2} + \frac{m_{k,-1}}{\epsilon} + m_{k,0} + O(\epsilon). \tag{2.114}$$

The required expansion depth is determined by the reduction coefficient. If $c_k \sim \epsilon^{-p}$, the M_k must be expanded through to ϵ^p to obtain the finite contribution to $c_k M_k$.

2.6 The Massive Extension

Up until now every part of the theory has been explained generally, but with a clear preference for the massless regime. That is because the vast majority of the present work's ensuing chapters will focus on that regime. But a still significant part of it will explore an extension of the discussed methods into the massive regime.

The fundamental change is the massive on-shell condition; the remainder of this section traces its consequences for the antenna construction and integration. Since final-state particles now have mass, their squared momenta are not set to zero, they are set to their squared mass.

$$p_q^2 = p_{\bar{q}}^2 = 0 \xrightarrow{\text{Massive Regime}} p_q^2 = p_{\bar{q}}^2 = m_q^2. \tag{2.115}$$

where p_i represents the momentum and m_i represents the mass of particle i.

The antenna subtraction formalism and the reduction architecture that was developed and described throughout this chapter are retained, however. What changes is the intermediate steps and kinematics: the massive on-shell condition changes the kinematics themselves and the unresolved limits, with soft radiation remaining singular but heavy-quark collinear behaviour becoming quasi-collinear. Massive integration is also different as the massive phase-scale is not equal to its massless counterpart. A new scale of m_q^2/q^2 is also introduced and leads to distinct cut-integral families and master integrals [28].

2.6.1 Massive Kinematics and Infrared Structure

In Subsection 2.2.1, the Mandelstam invariants were introduced for the massless regime. The general definition is,

$$\begin{aligned} s_{ij} = (p_i + p_j)^2 &= p_i^2 + 2p_i \cdot p_j + p_j^2 \\ &= m_i^2 + m_j^2 + 2p_i \cdot p_j. \end{aligned} \tag{2.116}$$

Through momentum conservation, which states that $q = \sum_i p_i$, we also have to reconsider the q^2 expression, as,

$$q^2 = \sum_i m_i^2 + 2\sum_{i<j} p_i \cdot p_j. \tag{2.117}$$

Soft radiation remains singular in the massive regime and therefore continues to generate explicit infrared poles after integration. A non-zero mass regulates collinear limits involving a massive parton, whereas collinear limits between two massless partons remain singular. If the two final-state collinear partons are massless, then the limit follows the massless collinear limit from the massless regime. If the two partons are massive, no strict collinear singularity exists. But if one of the collinear partons is massive, consider that particle i of mass m_i and energy E_i, and the other is massless, consider that particle j, such that $m_j = 0$, then,

$$p_i \cdot p_j = E_i E_j (1 - \beta_i \cos\theta), \quad \beta_i = \sqrt{1 - \frac{m_i^2}{E_i^2}} < 1. \tag{2.118}$$

As the angle between the partons θ tends towards zero, the mass prevents the invariant from reaching the collinear singular surface,

$$\lim_{\theta \to 0} p_i \cdot p_j = E_i E_j (1 - \beta_i) \neq 0. \tag{2.119}$$

This regulated collinear region still carries physically important information, particularly in the joint high-energy and small-angle region, $E_i >> m_i$ and $\theta << 1$,

$$p_i \cdot p_j \sim \frac{E_i E_j}{2}\left(\theta^2 + \frac{m_i^2}{E_i^2}\right) \tag{2.120}$$

and consequently since,

$$\int_0^{\theta^2_{\max}} d\theta^2 \, \frac{1}{p_i \cdot p_j} \sim \int_0^{\theta^2_{\max}} d\theta^2 \left(\theta^2 + \frac{m_i^2}{E_i^2}\right)^{-1} \sim \ln\left(\frac{E_i^2}{m_i^2}\right) \tag{2.121}$$

the massive-massless quasi-collinear region replaces the strict collinear pole by a mass logarithm. The quasi-collinear limit, then, is the limit in which both the mass m_i and the opening angle θ become small together. It is through this limit that the massive antennae recover the massless splitting behaviour, as $m_i \to 0$ [28].

Mass does not regulate UV divergences, as $\ell^2 >> m_i^2, p_i^2$. It is noteworthy, however, that in a theory with massive quarks, the quark mass and quark field must be renormalised as,

$$g_{s,0} = \mu^\epsilon Z_g g_s, \quad m_{q,0} = Z_m m_q, \quad \psi_{q,0} = \sqrt{Z_2^q}\,\psi_q. \tag{2.122}$$

2.6.2 Massive Phase-Space Integration and Reduction Methods

For final-state particles of momenta p_i and masses m_i, the phase-space becomes,

$$d\Phi_n(q; p_1, ..., p_n) = (2\pi)^d \delta^{(d)}\left(q - \sum_{i=1}^{n} p_i\right) \prod_{i=1}^{n} \frac{d^d p_i}{(2\pi)^{d-1}} \delta^+\left(p_i^2 - m_i^2\right). \tag{2.123}$$

At the level of the phase-space measure, the formal change from the massless case, as described in Subsubsection 2.5.2, is,

$$\delta^+(p_i^2) \rightarrow \delta^+(p_i^2 - m_i^2) \tag{2.124}$$

which physically imposes that $p_i^2 = m_i^2$ and $p_i^0 > 0$.

Phase-space factorisation still holds for the massive regime, but the momentum mapping must preserve the relevant mass-shell conditions. So,

$$d\Phi_{m+r}(\{p\}; q) = d\Phi_m(\{\tilde{p}\}; q) d\Phi_X. \tag{2.125}$$

For a final-final antenna with hard radiator i, k and unresolved particles between them, this is schematically,

$$d\Phi_{m+r} = d\Phi_m(..., \tilde{p}_I, \tilde{p}_K, ...; q) d\Phi_{X_{ijk}} \tag{2.126}$$

while $\tilde{p}_I^2 = m_i^2$, $\tilde{p}_K^2 = m_k^2$ and momentum conservation is satisfied,

$$\tilde{p}_I + \tilde{p}_K = p_i + \left(\sum_{\text{unresolved } j} p_j\right) + p_k. \tag{2.127}$$

Antenna integration is computed in the same manner as in the massless regime,

$$\mathcal{X}_n^l \sim \int d\Phi_X \left(\prod_{r=1}^{l} d^d \ell_r\right) X_n^l(\{p\}, \{\ell\}) \sim \left(q^2\right)^{\lambda + \kappa\epsilon} f\left(\left\{\frac{m^2}{q^2}\right\}, \epsilon\right). \tag{2.128}$$

PaVe and IBP reduction remain applicable in the massive regime, following the same principles introduced for the massless case. For IBP reduction, however, the massive case is significantly harder analytically: the phase-space in non-trivial and has mass-dependent thresholds; the master integrals are generally multi-scale, and can involve logarithms, polylogarithms and more complicated transcendental functions; and lastly, boundary conditions are less direct, although common choices are the massless limit at $m \rightarrow 0$ and the threshold limit at $\beta \rightarrow 0$ [28].

Chapter 3

Package Framework

AntCalc is an active research *Wolfram Language* package developed to offer direct access to QCD antenna functions, both before and after integration, by automating the symbolic steps required to build and integrate the supported antenna functions.

To complete its function, *AntCalc* makes use of the *FeynCalc* [29, 30, 31, 32] environment for symbolic-amplitude work, and *LiteRed2* [33] for IBP-backed routes. It uses a profile-based system to enable modularity. Its internal architecture is organised as a profile-driven route system: route-specific information is centralised in profiles, which guide the selection and configuration of the appropriate build, reduction, and integration workflows. This establishes a common contract between the package components, reducing the amount of route-specific logic that must be duplicated when existing functionality is extended or new routes are introduced.

The package supports public and experimental routes. The public antenna building and integration routes, which for the current version compute all massless quark-antiquark final-final antenna functions up to NNLO, have been validated against the literature. Experimental routes are unfinished or unchecked and should not be considered validated.

AntCalc's inner working structure can be described using the diagram in Figure 3.1.

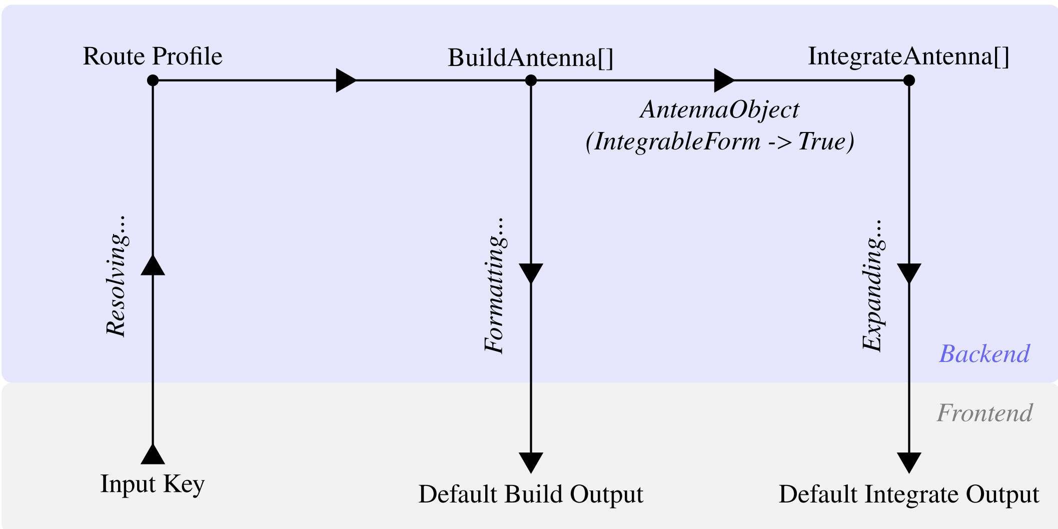

Figure 3.1: Profile-driven architecture of *AntCalc*. Route profiles configure the build and integration workflows, while the build-side route context carries the evolving calculation data. The `AntennaObject` provides the integration-ready handoff between the two stages.

3.1 Design Philosophy and Scope

AntCalc was built to combine the symbolic amplitude capabilities of the FeynCalc environment with the IBP-reduction capabilities of LiteRed2, adapting and orchestrating the parts of those systems required for the construction and integration of QCD antenna functions.

The central design objective of the package is to return results in the conventional forms used in antenna calculations. Though it supports a wide variety of options to inspect and configure the backend operations, the default outputs of its two central functions are designed to reflect this objective. For example, for loop antennae, the default `BuildAntenna[...]` result is given in PaVe-reduced form, as is customary in the literature. The default `IntegrateAntenna[...]` result is also a Laurent ϵ-series up to ϵ^2 for the NLO-order antennae $\mathcal{A}_2^1$ and $\mathcal{A}_3^0$, and up to ϵ^0 for the NNLO-order ones $\mathcal{A}_4^0$, $\mathcal{A}_3^1$ and $\mathcal{A}_2^2$ sets and $\mathcal{B}_4^0$ and C_4^0, matching the conventional truncation order, though higher order terms are accessible through the use of the aforementioned options.

Its structure was also designed to be modular and extensible. Each of the two central functions (`BuildAntenna` and `IntegrateAntenna`) is implemented as an encapsulated public module, which communicates with the rest of the codebase using the profile system: the route profile selects and configures the appropriate workflows and the route context carries the evolving results and diagnostics between stages. Using route profiles and route contexts, as described in Section 3.2, the addition of further modules is made easier, reducing duplicated logic and direct dependencies. These new modules need not be compatible with previous implemented functions through function-specific couplings. They need only conform with the profile system, being selected and configured by route profiles, and accepting inputs from and delivering outputs to the route contexts.

Currently, *AntCalc* is organised into two tiers: public and experimental. It already has all massless quark-antiquark final-final antenna functions up to NNLO implemented as part of the public tier, along with every other object required to compute the massless R-ratio up to that perturbative order. These are:

- A_2^0;
- A_2^1;
- A_2^2, $\tilde{A}_2^2$, $\hat{A}_2^2$, $\check{A}_2^2$;
- A_3^0;
- A_3^1, $\tilde{A}_3^1$, $\hat{A}_3^1$;
- A_4^0, $\tilde{A}_4^0$;
- B_4^0;
- C_4^0.

The massive A_3^0 route is also accessible, but remains experimental.

3.2 Profile-Driven Architecture

As described in Section 3.1, *AntCalc* relies on a profile-driven routing system. This system maps the route key to its route-specific profile and workflow.

A public antenna route key is a three-element list which describes a canonical route whose execution may be refined by options. The route key takes the form,

$$\{\text{Antenna Type}, \text{Final-State Multiplicity}, \text{Loop Order}\}. \tag{3.1}$$

As an example, the route key for the one-loop three-parton [1] antenna set is $\{\mathrm{A}, 3, 1\}$, and represents not only the leading A_3^1 antenna, but also its subleading and quark-loop counterparts, and its default public output is the ordered list $\{A_3^1, \tilde{A}_3^1, \hat{A}_3^1\}$, in this specific order.

The package links this key to the profile it identifies and follows that profile's directions on how the route should be constructed and/or integrated. The data produced while the route executes is also stored in the route context association. The profile system does not consist of a single profile, but rather of three main and some conventional auxiliary profiles which are routinely read while the route executes. The principal build-side profile is `AntennaProfile[key]`, which specifies information such as antenna type, multiplicity and loop order, colour normalisation, extraction rules, components, etc. `AntennaReductionProfile[key]` is separate, but still build-side. It specifies the PaVe-reduction preferences for the appropriate routes. In this case, up to NNLO, these are the $\{\mathrm{A}, 2, 1\}$ and $\{\mathrm{A}, 3, 1\}$ loop routes. `AntennaIntegrationProfile[key]` is the integration-side profile, which specifies its side's backend, IBP basis families, expansion depth, etc. The specifics of each stage's profiles and options will be discussed in their respective sections, 3.3 and 3.5.

While profiles select and configure computation paths, the workflows themselves perform the calculations. The route dispatcher and workflow code consult the profiles to select and configure each stage. As an example, which will be further explained and explored in the two upcoming sections, let us consider the call `BuildAntenna[A,2,1]`. Its full route profile is,

```
AntennaProfile[{A, 2, 1}] :=
    <|
        "Key" -> {A, 2, 1},
        "Name" -> "A21",
        "AntennaType" -> A,
        "NumFinalParticles" -> 2,
        "LoopOrder" -> 1,
        "TreeAmplitude" -> AntennaAmplitude[{A, 2, 0}],
        "BornInterference" -> BornInterference[],
        "Extraction" -> "LoopScalar",
        "ColourNorm" -> colourNorm,
        "ExcludedParticles" ->
            {S[_], V[1], V[2], V[3], U[1], U[2], U[3], U[4]},
        "DropSumOver" -> True,
        "ReductionProfile" -> AntennaReductionProfile[{A, 2, 1}],
        "ConventionProfile" -> BuildAntennaConventionProfile[{A, 2, 1}]
    |>
```

In this profile, the key, name, antenna type, number of final particles (multiplicity), and loop order have been discussed before. The remaining entries configure the construction of the route: `TreeAmplitude` supplies the tree-level A_2^0 amplitude with which the one-loop amplitude is interfered, `BornInterference`

[1] It is understood that throughout the rest of the present work, unless explicitly mentioned, all antenna functions considered are massless quark-antiquark final-final antennae.

selects the Born-level self-interference used to normalise the loop interference. `Extraction` selects the scalar one-loop extraction appropriate for this antenna, while `ColourNorm` provides the colour normalisation used. `ExcludedParticles` and `DropSumOver` are settings passed to the *FeynCalc*/*FeynArts* part of the build workflow, selecting what particles are not to be considered in the Feynman diagram generation step and requesting the removal of external `SumOver` structures. Lastly, `ReductionProfile` commands the workflow to complete a lookup on the selected route's antenna, A_2^1 in this case, and determines whether it requires the PaVe reduction backend, from within its `AntennaReductionProfile` association. Since it is a one-loop antenna, it does. `ConventionProfile` selects the convention bridge so the public results follow *AntCalc*'s stated normalisation and dimensional-regularisation conventions, through its separately defined `BuildAntennaConventionProfile` association.

3.3 The Build Stage

This first stage is tasked with building the unintegrated antenna functions from their respective Feynman diagrams. Its central function is `BuildAntenna[type, multiplicity, loop order]`. The route through this stage begins by taking the route key from some public call, like `BuildAntenna[A, 3, 1]`. It then resolves the corresponding build workflow from the route key, and implements it. Finally, it produces a rich build-side, backend, association which is then reformatted into the public unintegrated antenna expression or component list. This workflow is structured as follows,

```
BuildAntenna[type, multiplicity, loop order]
    -> route key;
    -> loop-order dispatcher;
    -> route-specific build workflow;
    -> build-data association / route context;
    -> public-output boundary;
    -> output: expression, component list, AntennaObject, diagnostics,
       or run record.
```

The loop-order dispatcher is the most important element of this workflow. It dispatches the route, using its key, to its loop order's path, as different loop orders necessitate different mechanisms, the most important of which being the introduction of the reduction of the loop momenta ℓ^μ for one- and two-loop antennae.

Functionally, the route-context for this stage usually contains the initial profile itself (the declarative `AntennaProfile`), the amplitude generated from it, the sectors, computed interferences, components after separation, diagnostics, and its normalised interference. Depending on the route, the context might also retain prototype components, full colour components, reference square components or loop-specific amplitudes.

A diagram in the spirit of the one in Fig. 3.1, but only relative to the Build stage is shown below, in Fig. 3.2.

3.3.1 Tree-Level Routes

As mentioned, the workflow handles distinct loop-order routes differently, though the differences only appear after the dispatcher is called. Tree-level routes like those relative to the A_3^0 antenna, start by retrieving the profile and the interference type (`BornInterference` in this case), and computing the

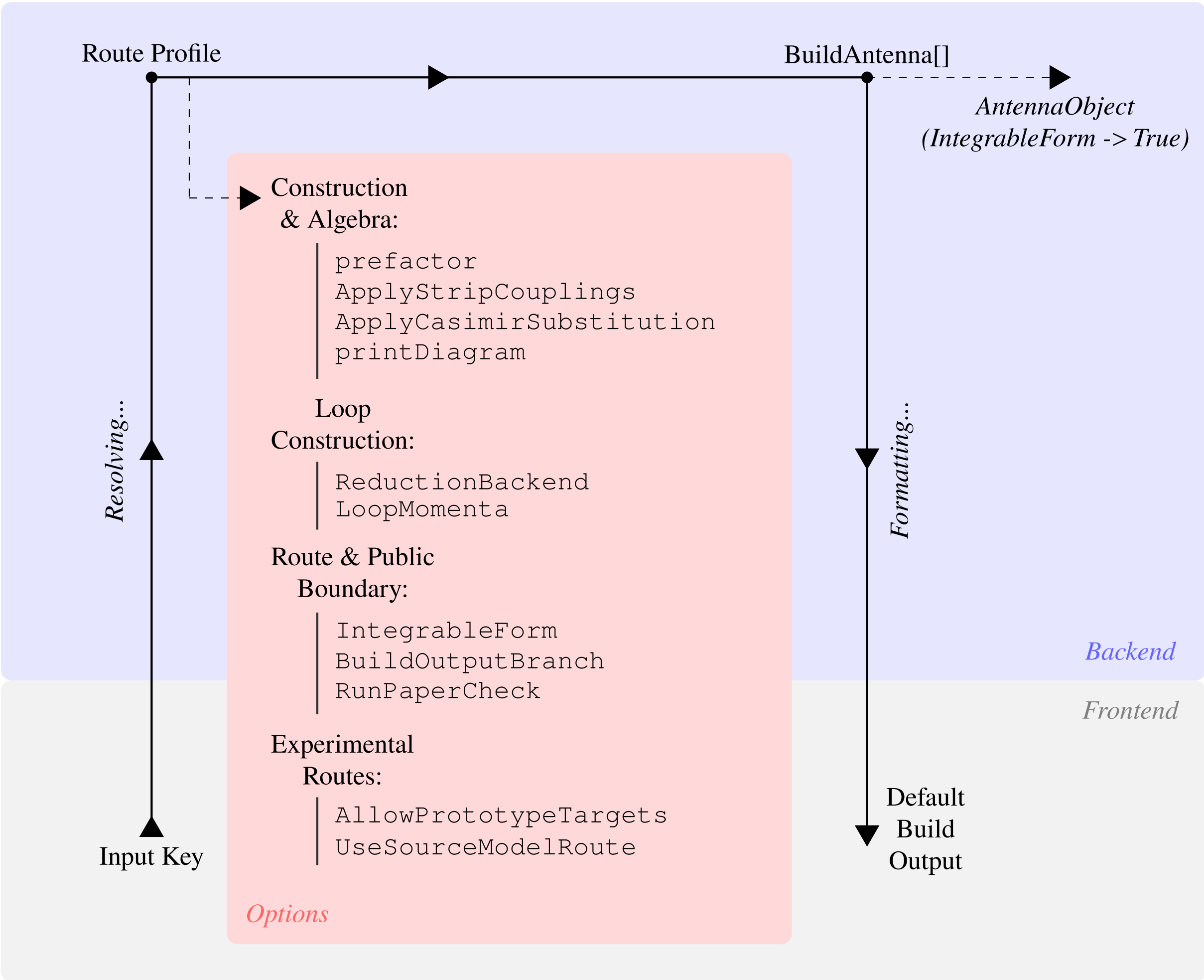

Figure 3.2: Profile-driven architecture of *AntCalc*'s Build stage. The principal build-specific options to the `BuildAntenna` function are displayed on the red card.

relevant process's amplitude. It uses the profile's `Production` field to select the physical construction path, and the `Extraction` and `ColourNorm` fields to select the extraction mode and the colour normalisation used for each antenna. The colour normalisation used by default has been described previously in Eq.2.32, as $c_N \equiv N - \frac{1}{N}$. The `Extraction` mode also introduces the Born normalisation as described in Eq. 2.27 as $B_0 \equiv |\mathcal{M}_2^0|^2$. The routes selected in the `Production` and `Extraction` fields depend on the methodology required to compute the antenna functions and retrieve the components, if required. The public tree-level antenna functions are constructed using different production and extraction modes, as described in Table 3.1.

Production Mode	**Extraction Mode**	**Current Route**
`SelfInterference`	`BornScalar`	A_2^0
`SelfInterference`	`TreeColourCoefficients`	A_3^0
`ColourOrderedAntenna`	`TreeColourCoefficients`	A_4^0, $\tilde{A}_4^0$
`SectorSelfInterference`	`ColourNormScalar`	B_4^0
`SectorSymmetrisedInterference`	`MinusTwoColourNormOverSUNN`	C_4^0

Table 3.1: Overview of the tree-level production and extraction modes.

These production paths all lead to the extraction of the required antenna functions, but the way in which they do it changes with the antenna's multiplicity. `SelfInterference` starts by generating the amplitude, computing its self-interference, and then extracting the antenna. `ColourOrderedAntenna` generates the full-process amplitude and computes the full-colour interference. It then applies the colour-ordered reconstruction as specified by `ColourOrderedSpecs`, and then extracts the components, in this case the Leading and Subleading components. `SectorSelfInterference` generates the full-process amplitude and splits it into profile-defined sectors. It selects the relevant sector, computes its self-interference and extracts the antenna function. Finally, `SectorSymmetrisedInterference` starts by generating the full-process amplitude, and splitting it into direct and exchange sectors. It then computes the required symmetrised interference and extracts the antenna. A reference square is also retained for diagnostics and comparison. The extraction paths lead to the separation of the whole-process results into colour-independent antenna functions. The `BornScalar` mode is used solely in A_2^0, and it does not algebraically divide the computed interference by its Born form, establishing the antenna as the reference normalisation. It returns $A_2^0 = 1$. `TreeColourCoefficients`, as used for A_3^0, A_4^0 and $\tilde{A}_4^0$, starts by normalising the computed amplitude by the colour and the Born normalisation. It then simplifies and expands the normalised interference and searches for the N, $1/N$ and N_f coefficients, the `Leading`, `SubLead` and `QuarkLoop` coefficients in the code. If the terms are present, as described in the previous text, the interference $\mathcal{J}$ becomes,

$$\frac{\mathcal{J}}{c_N B_0} = N X_{\text{Lead}} + \frac{1}{N} X_{\text{SubLead}} + N_f X_{\text{QuarkLoop}} \tag{3.2}$$

and the X terms extracted as the antenna functions. If there are no terms of that shape, like in the A_3^0 case, then $X = \mathcal{J}/(c_N B_0)$, colour- and Born-normalised. It is important to note that, though present, no quark loop antenna should ever appear in the tree-level route. The dispatcher still considers it nonetheless since it is written to be reusable. For B_4^0, the `ColourNormScalar` mode is used, after the primary-current sector has been self-interfered. The extraction then returns the colour- and Born-normalised sector interference, as,

$$B_4^0 = \frac{\mathcal{J}_{\text{sector}}}{c_N B_0}. \tag{3.3}$$

Lastly, `MinusTwoColourNormOverSUNN` is used for C_4^0, and it takes the symmetrised interference of the direct and exchange sectors and normalises it as,

$$C_4^0 = -\frac{N \mathcal{J}_{\text{sym}}}{2 c_N B_0}. \tag{3.4}$$

The C_4^0 antenna requires a further $-N/2$ normalisation factor due to its origin as the interference of the direct and exchange flavour-flow amplitudes for identical quarks. The minus sign reflects the exchange of identical fermions, the $1/2$ factor is required when the symmetrised source contains both cross terms, as discussed in Eq. 2.36. The remaining N factor, along with the usual c_N in the denominator adjusts the interference's colour coefficient into the colour-free C_4^0 antenna.

3.3.2 One-Loop Routes

One-loop routes are dispatched to `BuildLoopAntennaData` where the primary backend reduction method is selected through the nested `ReductionProfile`, which by default is the reduction using PaVe, as described in Subsection 2.5.1.

The production method consists of generating the tree-level and one-loop amplitudes, using `TreeAmplitude` and `MAmpOneLoop`, respectively, and then forming the tree-loop interference. The interference is colour- and one-loop-normalised, using c_N and $\Lambda_l = (8\pi^2)^l$, for loop order l, as described in Eq. 2.17. To this normalised interference is then applied the reduction and the components extracted. These antenna functions also differ in production and extraction method, however. Their production and extraction modes are shown in Table 3.2, below.

Production Mode	**Extraction Mode**	**Current Route**
`TreeAmplitude + MAmpOneLoop`	`LoopScalar`	A_2^1
`TreeAmplitude + MAmpOneLoop`	`LoopColourCoefficients`	$A_3^1, \tilde{A}_3^1, \hat{A}_3^1$

Table 3.2: Overview of the one-loop production and extraction modes.

Both antenna sets are produced in the same manner, as discussed. The relevant tree-level antenna amplitude, $\mathcal{M}_2^0$ and $\mathcal{M}_3^0$ respectively for A_2^1 and the A_3^1 set, is first computed. Then it is interfered with the loop amplitudes, $\mathcal{M}_2^1$ and $\mathcal{M}_3^1$. The interference is computed, before being normalised as,

$$\mathcal{J}_n^l = 2\,\mathrm{Re}\left(\left(\mathcal{M}_n^0\right)^\dagger \mathcal{M}_n^l\right) \tag{3.5}$$

where n represents the number of final-state particles, as usual.

Since A_2^1's interference does not decompose into colour components, $A_2^1 = \Lambda_1 \mathcal{J}_2^{1\prime} = \Lambda_1 \mathcal{J}_2^1/(c_N B_0)$. The A_3^1 set, on the other hand, appears closely like the A_4^0 set, but with three, rather than two components, the leading, subleading and quark loop antenna functions. The interference is given as in Eq. 3.2. Lastly, as discussed in Eq. 2.21, together with Eq. 2.27, the A_3^1 set needs to be UV renormalised. Consider the notation $X_n^{l,\text{bare}}$ here to denote a bare antenna, such that, for example, $A_2^{1,\text{bare}} = A_2^1$. For the A_3^1 set, the interference expands to,

$$\frac{\Lambda_1}{c_N B_0}\mathcal{J}_3^1 = N T_{\text{Lead}} + \frac{1}{N} T_{\text{Sublead}} + N_f T_{\text{QL}}. \tag{3.6}$$

Differently from other processes at this level, the A_3^1 route necessitates a further subtraction, as the interference components come in terms of a genuine one-loop three-parton antenna and an iterated lower-order contribution. That makes it so that the bare (before UV renormalisation) A_3^1 set antenna functions become,

$$\begin{aligned} A_3^{1,\text{bare}} &= T_{\text{Lead}} - A_2^1 A_3^0 \\ \tilde{A}_3^{1,\text{bare}} &= -\left(T_{\text{Sublead}} + A_2^1 A_3^0\right) \\ \hat{A}_3^{1,\text{bare}} &= T_{\text{QL}}. \end{aligned} \tag{3.7}$$

The components are then, finally, renormalised as discussed,

$$\begin{aligned} A_3^1 &= A_3^{1,\text{bare}} - \frac{11}{6\epsilon} A_3^0 \\ \tilde{A}_3^1 &= \tilde{A}_3^{1,\text{bare}} \\ \hat{A}_3^1 &= \hat{A}_3^{1,\text{bare}} + \frac{1}{3\epsilon} A_3^0. \end{aligned} \tag{3.8}$$

Each of these components may also be extracted alone as a single antenna, using the `Component` option. If, while using *AntCalc*, one ran `BuildAntenna[A,3,1,` `Component->Subleading]`, for example, only the unintegrated $\tilde{A}_3^1$ antenna would be returned.

3.3.3 Two-Loop Routes

Loop order two routes are dispatched to `BuildTwoLoopAntennaData`. There is only, formally, one set of this kind, A_2^2. It is remarkably different from the rest, however. First, these objects, alongside A_2^0 and A_2^1, are not formally antenna functions as they fail to meet the definition of a structure that captures the singular behaviour of an unresolved parton radiated between two hard, colour-connected partons [9]. They are, instead, form factors required by the antenna framework. The A_2^2 set, at two loops, is obtained from the two-loop quark form factor [22]. Second, this four-component set is not generated like A_4^0 and A_3^1 were, rather these four components come from two distinct source contributions. Their production and extraction modes are similar and handled as a bundle inside *AntCalc*, but, as shown in Table 3.3, their sources differ,

Contribution	**Extraction Mode**	**Current Route**
`TwoLoopInterf`	`TwoLoopTTermComponents`	A_2^2, $\tilde{A}_2^2$, $\hat{A}_2^2$
`OneLoopSelf`	`TwoLoopTTermComponents`	$\breve{A}_2^2$

Table 3.3: Overview of the two-loop contributions and extraction modes.

The first three antenna objects are formed by interfering the LO tree-level amplitude with the two-loop amplitude. This interference follows the blueprint defined in Eq. 3.5, as the interference becomes,

$$\mathcal{J}_2^{2,[2\times 0]} = 2\,\mathrm{Re}\left(\left(\mathcal{M}_2^0\right)^\dagger \mathcal{M}_2^2\right). \tag{3.9}$$

The superscript $[2 \times 0]$ denotes the nature of the operation: interfering a two-loop with a zero-loop amplitude. It is also usual to split the T-object relative to these four components, $T_{q\bar{q}}^6$, into two, based on the two contributions.

The last object, $\breve{A}_2^2$, comes from self-interfering the one-loop amplitude as,

$$\mathcal{J}_2^{2,[1\times 1]} = \left(\mathcal{M}_2^1\right)^\dagger \mathcal{M}_2^1. \tag{3.10}$$

Both of these interferences are equally Born- and loop-normalised, using B_0 and Λ_2. They are also colour-normalised, but while $\mathcal{J}_2^{2,[2\times 0]}$ requires only c_N, $\mathcal{J}_2^{2,[1\times 1]}$ requires c_N^2 since it is computed using two one-loop amplitudes, each of which requires a c_N normalisation factor, as described in Subsection 2.4.2. They, then, become,

$$\begin{aligned} \frac{\Lambda_2}{c_N B_0}\mathcal{J}_2^{2,[2\times 0]} &= N A_2^{2,\mathrm{bare}} + \frac{1}{N}\tilde{A}_2^{2,\mathrm{bare}} + N_f \hat{A}_2^{2,\mathrm{bare}} \\ \frac{\Lambda_2}{c_N^2 B_0}\mathcal{J}_2^{2,[1\times 1]} &= \breve{A}_2^{2,\mathrm{bare}}. \end{aligned} \tag{3.11}$$

The antenna objects have to then be UV renormalised as follows,

$$
\begin{aligned}
A_2^2 &= A_2^{2,\text{bare}} - \frac{11}{6\epsilon} A_2^1 \\
\tilde{A}_2^2 &= \tilde{A}_2^{2,\text{bare}} \\
\hat{A}_2^2 &= \hat{A}_2^{2,\text{bare}} + \frac{1}{3\epsilon} A_2^1 \\
\breve{A}_2^2 &= \breve{A}_2^{2,\text{bare}}.
\end{aligned}
\tag{3.12}
$$

This route may be schematically represented as shown in Fig. 3.3, below.

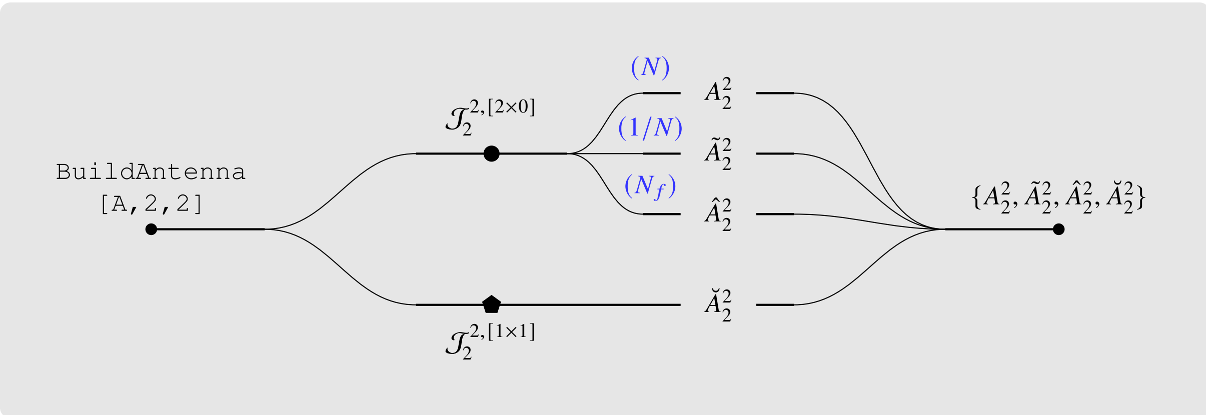

Figure 3.3: Diagram showing the build-side workflow leading to the generation of the A_2^2 antenna set. The two subsets are separated by the interferences used in their generation.

These renormalised results are now used to form the overall $T_{q\bar{q}}^6$ object. It is defined as the sum of the two contributions' sub-T-objects, $T_{q\bar{q}}^{6,[2\times0]}$ for the first, and $T_{q\bar{q}}^{6,[1\times1]}$ for the second contribution, following the interferences $\mathcal{J}$ notation as,

$$
\begin{aligned}
T_{q\bar{q}}^6 &= \left(N - \frac{1}{N}\right) T_{q\bar{q}}^2 \left[T_{q\bar{q}}^{6,[2\times0]} + T_{q\bar{q}}^{6,[1\times1]}\right] \\
&= \left(N - \frac{1}{N}\right) T_{q\bar{q}}^2 \left[\frac{\Lambda_2}{c_N B_0} \mathcal{J}_2^{2,[2\times0]} + \frac{\Lambda_2}{c_N^2 B_0} \mathcal{J}_2^{2,[1\times1]}\right] \\
&= \left(N - \frac{1}{N}\right) T_{q\bar{q}}^2 \left[N A_2^2 + \frac{1}{N} \tilde{A}_2^2 + N_f \hat{A}_2^2 + \left(N - \frac{1}{N}\right) \breve{A}_2^2\right].
\end{aligned}
\tag{3.13}
$$

3.4 Public Boundary and Integration handoff

As previously discussed, each `BuildAntenna` workflow produces a route context. That context contains the initial profile alongside some selected pieces of data acquired during the building process. The ordinary, default, `BuildAntenna[...]` call formats this contexts into the conventional public output, which consists of a single expression or ordered list of components. This output is not compatible with a following `IntegrateAntenna[...]` call, as this second call requires the route profile information. To make the two functions compatible we must use the handoff option `IntegrableForm`. Calling `BuildAntenna[..., IntegrableForm -> True]` returns an `AntennaObject` instead of the default public output. It retains the route key, the selected output, the profile, the complete build data and the source-contribution provenance. This `AntennaObject` is the intended input of the `IntegrateAntenna[...]` function.

3.5 The Integration Stage

The integration stage is tasked with taking the complete `AntennaObject`, as produced by a `BuildAntenna[..., IntegrableForm -> True]`, resolving the integration route from the key, performing all intermediate calculations and applying all necessary conventions, and finally returning a Laurent-expanded integrated antenna or component list. The workflow's structure is,

```
IntegrateAntenna[AntennaObject]
    -> recover route key and resolve profile;
    -> choose integration backend;
    -> complete IBP/PaVe reduction;
    -> substitute master integrals when applicable;
    -> extract public integrated antennae;
    -> output: selected component, list, diagnostics, run record, or
       master-combination view.
```

It is relevant to note that `IntegrateAntenna` requires the profile to complete its function like `BuildAntenna`, since it also encodes the relevant routing information for the reduction and integration process.

Much like for `BuildAntenna`, there is no single route context object, rather the relevant information is stored around multiple objects that are accessed when required. These objects store initiation-crucial elements, like the key and the profile, the selected component and the input (unintegrated) antenna, some IBP-crucial elements, like the bases required. the master integrals to use, and the series result, along with others like the route story, or the diagnostics.

An integration-stage-specific diagram in the spirit of the one in Fig. 3.1 can also be drawn in further detail. That diagram is shown in Fig. 3.4.

3.5.1 Integration Profiles and Selection

The integration-specific profile for each route selects the kind of backend process (and external package workflow used), and the basis family required for the process. The antennae are divided by topology, that is, number of final-state particles and number of loops, and all of those that share the same topology in the current implementation, share the same basis-family set. The exception to this rule is the A_2^2 set, in which different sets are used depending on configuration. The A_2^1 antenna is also an exception to the use of basis families, as it is reduced using PaVe reduction and then the PaVe MI are substituted by their analytical results in terms of ϵ. An overview of these is shown in the following Table 3.4.

Default Backend	**Bases-Family Set**	**Current Route**
PaVe / *FeynHelpers*	massless two-parton vertex	A_2^1
IBP / *LiteRed2*	`X30`	A_3^0
IBP / *LiteRed2*	`X40`	$A_4^0, \tilde{A}_4^0, B_4^0, C_4^0$
IBP / *LiteRed2*	`A31`	$A_3^1, \tilde{A}_3^1, \hat{A}_3^1$
IBP / *LiteRed2*	`A22TwoLoopTree`	$A_2^2, \tilde{A}_2^2, \hat{A}_2^2$
IBP / *LiteRed2*	`A22OneLoopSelf`	$\check{A}_2^2$

Table 3.4: Overview of the integration-side route-profiles.

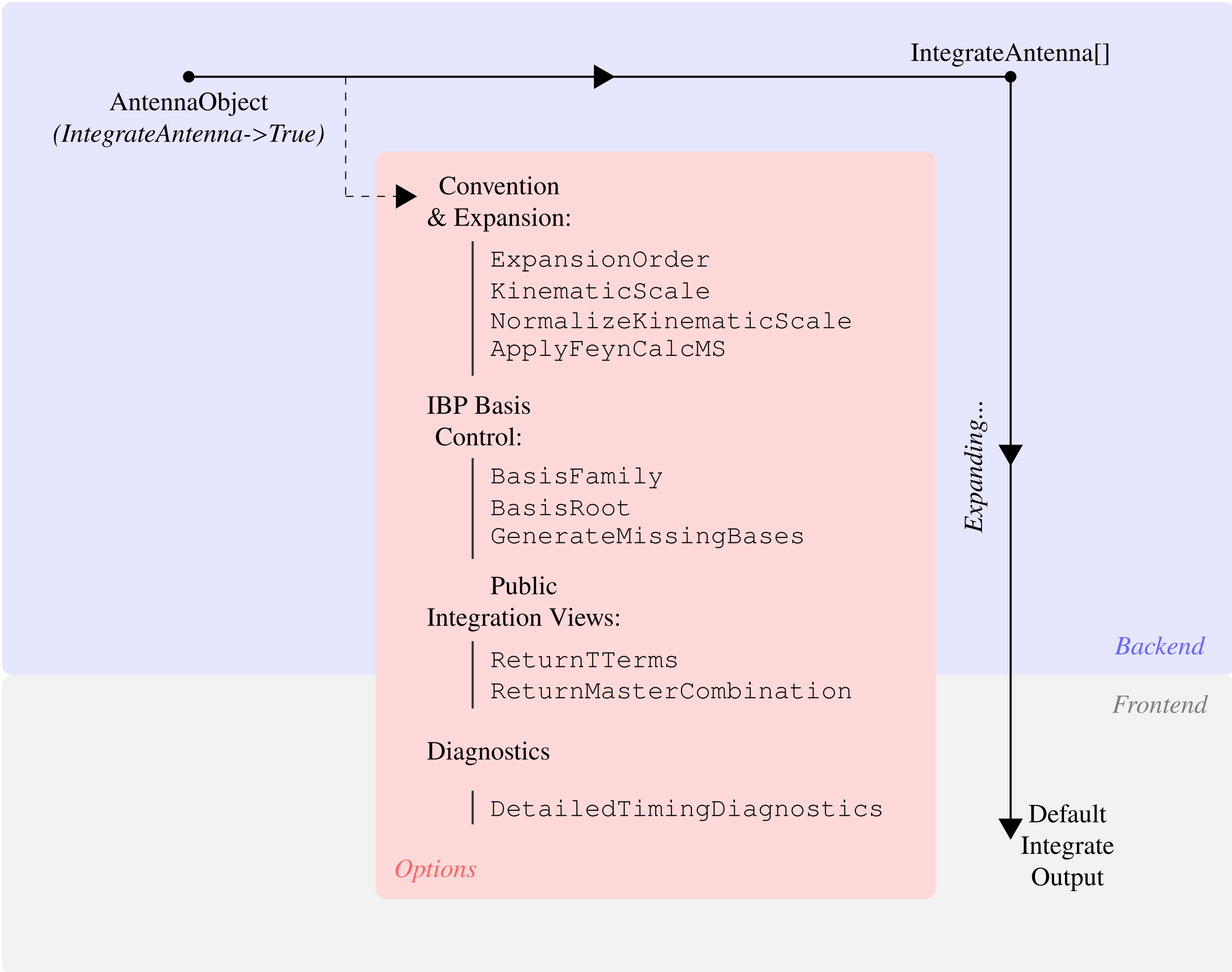

Figure 3.4: Profile-driven architecture of *AntCalc*'s Integration stage. The principal integration-only options are shown inside the red card.

As discussed previously for the build-side, the profile here also does not complete any calculations itself. It simply provides the workflow with the required information for it to work. For example, consider the integration of the B_4^0 antenna: it reaches `IntegrateAntenna` as an `AntennaObject`. For the key, the workflow is able to resolve the integration-side route-profile and then it reduces the input unintegrated antenna from `AntennaObject` with the IBP method, using *LiteRed2*, and using the bases and MI from the `X40` basis-family set.

3.5.2 Integration Routes

PaVe Route

The A_2^1 PaVe route is the distinct low-multiplicity virtual case. It is the only route that does not involve IBP. Due to being the two-parton one-loop form factor, this A_2^1 antenna's integration workflow requires only the reduction and integration of the single loop momentum, making PaVe reduction the best choice. The workflow then uses *Package-X* through *FeynHelpers*, to evaluate the PaVe scalar one-loop functions and, after applying the required branch and scale conventions, the kinematic-scale normalisation, and Laurent expansion in ϵ, it gives the integrated $\mathcal{A}_2^1$ antenna. This process is shown schematically in Fig. 3.5, below.

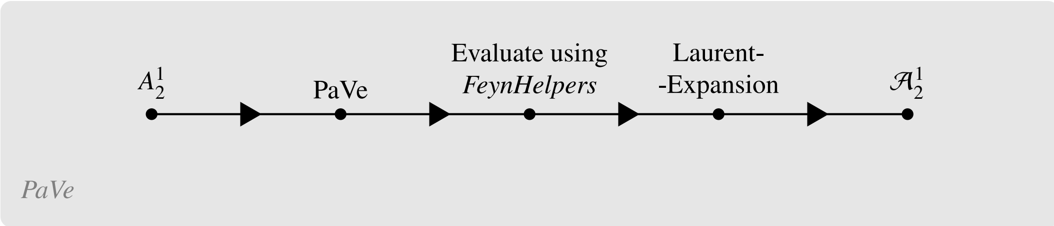

Figure 3.5: PaVe reduction for the A_2^1 antenna integration.

IBP Route

All other antenna functions are integrated using IBP reduction using *LiteRed2* and as described in Sections 2.5. The workflow rewrites the momenta, invariants, and denominators in *LiteRed2*'s terms. It then adds the phase-space cut structure for the selected family, loads its bases and after expanding the unintegrated antenna expression, going term-by-term, matches each one to its relevant family. After each match, the term is IBP reduced relative to its matched family-basis MI. This last step is handled completely by *LiteRed2*'s internal engine. After all terms have been reduced, they are summed to form the final reduced antenna function. For the public routes, the MI have already been hand-matched with known MI. These known MI are then substituted by their values after integration, which is also hand-calculated and finally, after applying all required normalisation and conventions, the integrated antenna functions are Laurent-expanded in ϵ. The IBP route process is shown schematically in Fig. 3.6.

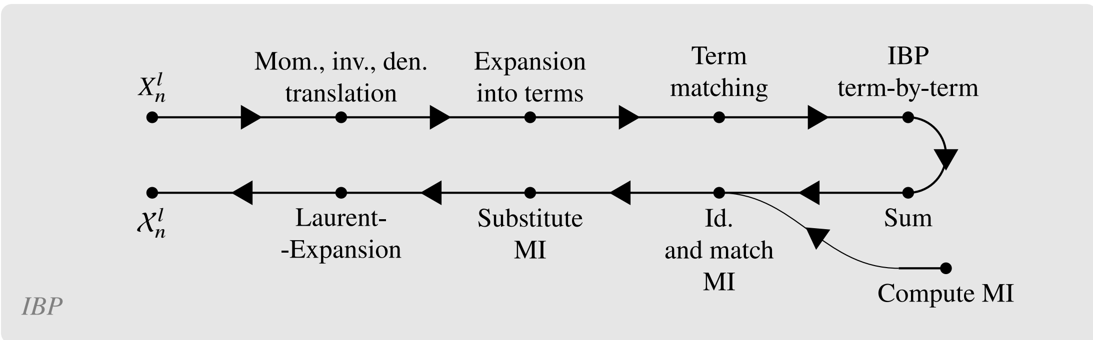

Figure 3.6: IBP reduction workflow for the remaining antennae integrations. The MI computation is external to *AntCalc*'s internal operation, and hence is represented by an external incoming arrow.

3.5.3 Normalisations and Convention Bridge

As discussed in Section 2.3, before integration, the antenna functions must be normalised using the $\mathcal{N}(\epsilon, k)$ factor, introduced in Eq. 2.20. Since in Section 3.3, where we discussed the Build stage, the unintegrated antenna functions were normalised using the loop-order factor, Λ_l, in this stage, the rest of the normalisations and conventions must be applied. That begins with the application of what remains from the G_k factor and then $C(\epsilon, k)$. Finally, we reach the full $\mathcal{N}(\epsilon, k)$ by dividing by the two-particle phase-space measure. This is, schematically,

$$
\begin{aligned}
\mathcal{X}_n^l &= \mathcal{N}(\epsilon, k) \int d\Phi_{X_{ijk}}\, X_n^l \\
&= \frac{1}{\Phi_2} \left(8\pi^2\right)^{n-2} S_\epsilon \int d\Phi_{X_{ijk}} \left(\Lambda_l X_n^l\right).
\end{aligned}
\tag{3.14}
$$

Note, however, that not all routes require that the rest of the G_k be applied, based on their n. There are also routes that did not require the application of the Λ_l factor in the Build stage. The following Table 3.5 shows how each antenna or antenna set interacts with G_k.

Route	n, l	G_k **relative to build** Λ_l
A_2^1	$2, 1$	$G_1 = \Lambda_1$ (no extra multiplicity factors needed)
A_3^0	$3, 0$	$G_1 = 8\pi^2\Lambda_0 = 8\pi^2$
A_3^1 set	$3, 1$	$G_2 = 8\pi^2\Lambda_1$
A_4^0 set, B_4^0, C_4^0	$4, 0$	$G_2 = (8\pi^2)^2\Lambda_0 = (8\pi^2)^2$
A_2^2 set	$2, 2$	$G_2 = \Lambda_2$ (no extra multiplicity factors needed)

Table 3.5: Overview of the interactions of the antenna routes with the normalisation factor G_k.

Expansion Order

The last step before reaching the integrated antenna function $\mathcal{X}_n^l$, or the ordered list of component integrated antennae, is to expand the working expression as a Laurent series. Following the literature, the maximum Laurent-expansion order is route-dependent. Lower-order antenna functions ($\mathcal{A}_2^1$ and $\mathcal{A}_3^0$) are expanded to ϵ^2, while higher-order functions are expanded to ϵ^0. This latter restriction comes from the used and verified master-integral representations, of which some are only encoded through ϵ^0. There might be need to expand to a lesser or to a greater term, however. That can be set using the `ExpansionOrder` option, as,

$$\begin{aligned}
\texttt{IntegrateAntenna[A30]} &\xrightarrow{\text{Default Result}} \frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + \frac{19}{4} - \frac{7\pi^2}{12} + \epsilon\left(\frac{109}{8} + ...\right) + \epsilon^2\left(\frac{639}{16} + ...\right) \\
\texttt{IntegrateAntenna[A30]} &\xrightarrow{\texttt{ExpansionOrder->0}} \frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + \frac{19}{4} - \frac{7\pi^2}{12}
\end{aligned} \tag{3.15}$$

where `A30=BuildAntenna[A,3,0,IntegrableForm->True]`, is the A_3^0 set's `AntennaObject`. Though the example in Eq.3.15 above shows the truncation of a series whose default expansion is deeper than the one resulting from the option, using `ExpansionOrder` for a higher-than-default order is also possible and supported. That call, however, is not validated against the literature, as the MI *AntCalc* uses are, some, truncated.

3.5.4 Component Extraction and Integrated $\mathcal{T}$-Objects

The last part of the Build stage, before it generates the output, is component extraction. In that part, the full result of the calculations the workflow went over is separated into colour-dependent terms and the coefficients relative to each of those terms is labelled as one of the components (leading, subleading, etc.). All multi-component antenna routes are reduced and integrated using IBP reduction, which is independent of colour. That makes it sensible to completely separate the components before reducing. This way, not only does the process become much cheaper if only one component is selected, but the reductions are also made cleaner, since different components may require different bases and MI. The following diagram in Fig.3.7, describes schematically this process, using the A_3^1 antenna set as an example.

It is also important to note here that the result of the A_3^1 set's integration is given, much like it was for

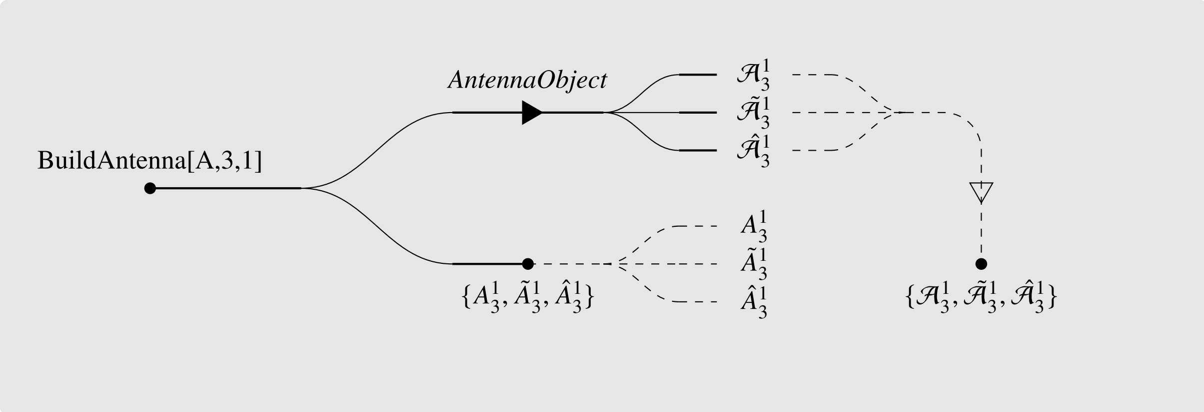

Figure 3.7: Diagram showing the component extraction steps and paths for the build-only and build-and-integrate routes. The combined route is shown above and the build-only route below it. The dashed lines represent the options to either separate the ordered lists into single components on the build-only route, or to merge the components into a single ordered list for the combined route. The A_3^1 antenna set is used as an example.

the Build stage in 3.7, as,

$$\begin{aligned}
\mathcal{A}_3^1 &= \mathcal{T}_{\text{Lead}} - \mathcal{A}_2^1 \mathcal{A}_3^0 \\
\tilde{\mathcal{A}}_3^1 &= -\left(\mathcal{T}_{\text{Sublead}} + \mathcal{A}_2^1 \mathcal{A}_3^0\right) \\
\hat{\mathcal{A}}_3^1 &= \mathcal{T}_{\text{QL}}.
\end{aligned} \tag{3.16}$$

These resulting $\mathcal{A}_3^1$ and $\tilde{\mathcal{A}}_3^1$ are already UV renormalised by this stage.

Those $\mathcal{T}$-object may still be useful, however. They can be extracted instead of the antenna functions, essentially by breaking the process before that last operation, in Eq. 3.16 and returning an analogous ordered list made up of the $\mathcal{T}$-objects, as,

$$\begin{aligned}
\texttt{IntegrateAntenna[A31]} &\xrightarrow{\text{Default Result}} \{\mathcal{A}_3^1, \tilde{\mathcal{A}}_3^1, \hat{\mathcal{A}}_3^1\} \\
\texttt{IntegrateAntenna[A31]} &\xrightarrow{\texttt{ReturnTTerms->True}} \{\mathcal{T}_{\text{Lead}}, \mathcal{T}_{\text{Sublead}}, \mathcal{T}_{\text{QL}}\}
\end{aligned} \tag{3.17}$$

where `A31=BuildAntenna[A,3,1,IntegrableForm->True]`, is, once again, the A_3^1 set's `AntennaObject`.

3.5.5 A_2^2 Set Integration Routes

As described in this section, the A_2^2 set is composed of two subsets of antennae with different provenance and IBP basis families. Both the build- and the integrate-side routes are completely separate based on subset. Those routes may be schematically drawn in the spirit of the diagram in Fig. 3.7, as in the diagram in Fig. 3.8, below. The workflow interprets the overall `[A,2,2]`, then separates the process into different build-side production modes as shown in the diagram in Fig. 3.3. It then merges the four components into an ordered list. That list is sent to the integrate-side workflow. This workflow starts by separating them into the subsets and then, considering that all components are selected for integration, it reduces them using the subset-specific basis families. Finally, once all four are integrated, they are once again merged onto an ordered list, which is delivered as the default output.

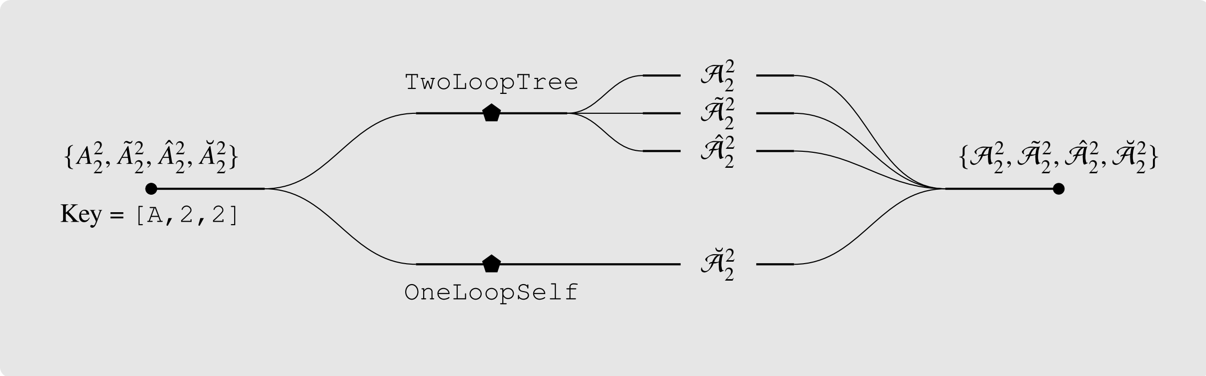

Figure 3.8: Diagram showing the component extraction steps and integration-basis families used in the integration-side workflow of the A_2^2 antenna set.

3.6 Higher-Level Functions

Besides the two canonical functions, *AntCalc* also supports a number of higher-level functions. This higher-level layer does not introduce new antenna-construction or integration methodology; instead it orchestrates the use of the two canonical functions to compute observables like the R-ratio, retrieve T-objects or to streamline repeated Build or Integrate evaluation.

There are a total of five higher-level functions. They are,

- `BuildAndIntegrateAntenna`
- `BuildAllAntennae`
- `BuildAndIntegrateAllAntennae`
- `TObject`
- `BuildRRatio`.

Each of these functions completes a useful higher-level task and their use can be schematically seen in Fig. 6.1. The first one streamlines the retrieval of the integrated antenna result to a single function, instead of having to build the unintegrated antenna using the `IntegrableForm` option and then to plug the intermediate result into the integration workflow. This function essentially follows the top path on the diagram in Fig. 3.7, for the A_3^1 route: it starts by calling `BuildAntenna` using the appropriate option, `IntegrableForm->True`. It then, internally, plugs the result into `IntegrateAntenna`. It forwards the controls needed by the combined workflow to their respective stages, including component selection and Laurent-expansion order. That means that one can build and integrate only the subleading component, expanded only up to ϵ^{-1}, for example. In the diagram, this function goes all the way from its namesake node, down to where $\mathcal{X}_n^l$ leaves the line, being its schematic solution, the integrated antenna function.

There are also two bulk wrappers: `BuildAllAntennae` and `BuildAndIntegrateAll Antennae`. These two functions complete exactly what their names imply: the first one builds all public antenna functions and returns an ordered list of all antennae (and components); the second builds and integrates every public antenna function and returns an ordered list of all antennae (and components). In the diagram, the first function is separate from the rest of the workflow due to its own nature. The second is shown at the bottom, since it calls `BuildAndIntegrateAntenna` for every public, massless,

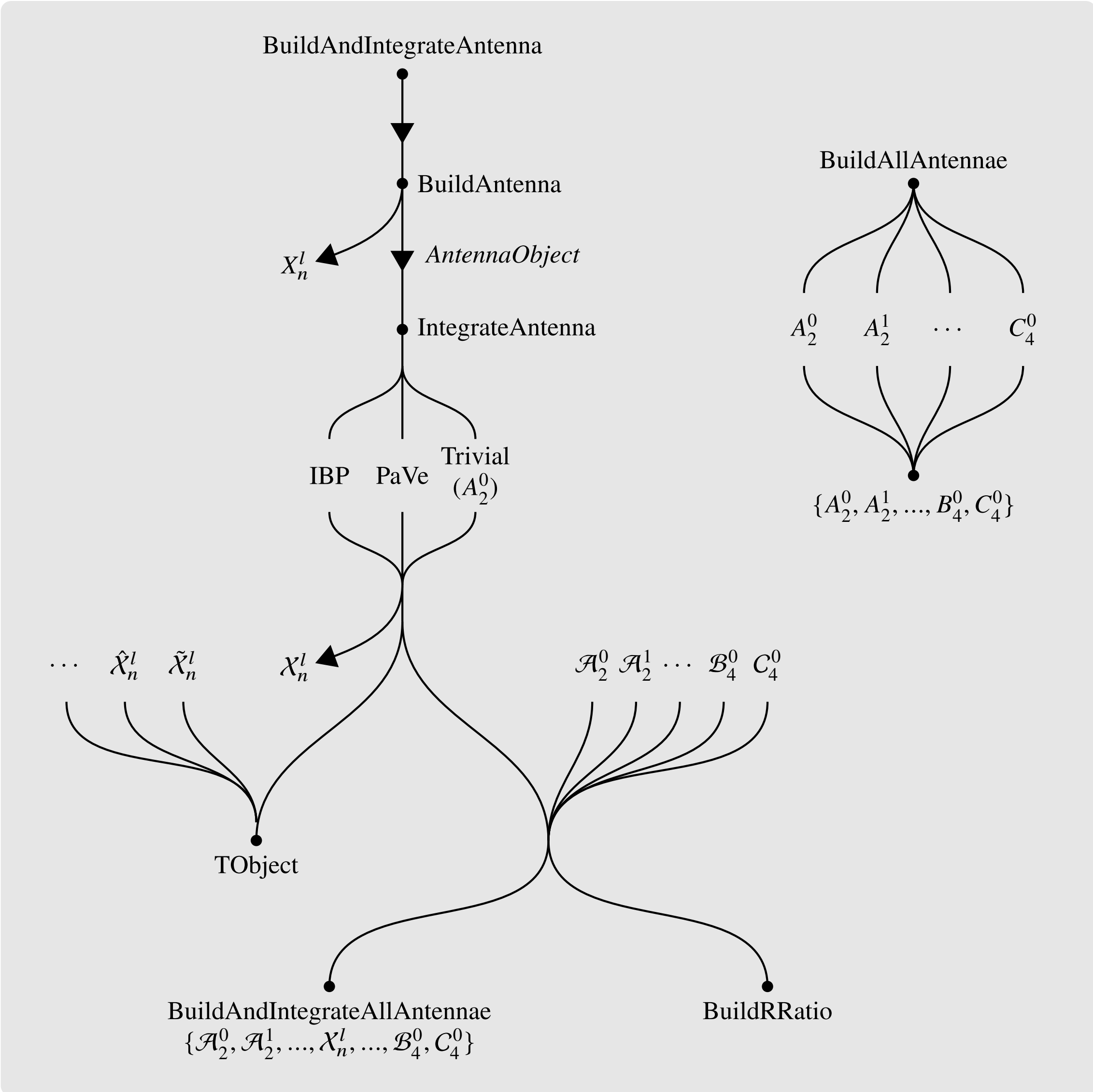

Figure 3.9: Diagram showing the complete *AntCalc* workflow including higher-level functions.

route. Their outputs can be modulated based on the requested order. That is, we can limit the wrappers to generate only antenna functions, built or integrated, up to, for example, NLO. In that case, only the list $\{A_2^0, A_2^1, A_3^0\}$ (or the same but integrated) would be returned. These functions are able to produce (build or build and integrate) all publicly available antenna functions, that is, all massless quark-antiquark final-final (SMQCD[2]) antennae, up to NNLO.

We may also retrieve a specific $\mathcal{T}$-object directly, using `TObject`. The public supported $\mathcal{T}$-objects are described in Table 3.6, below. These computation routes follow closely the methods discussed in Subsection 2.4.3. The function calls the one-shot integrated routes, using `BuildAndIntegrateAntenna` for the necessary ingredient antennae, then assembles them into the $\mathcal{T}$-object structure.

As an example, let us consider $\mathcal{T}^6_{q\bar{q}g}$. As shown in Eq. 3.7 and Table 3.6, this $\mathcal{T}$-object requires the $\mathcal{A}_3^1$ set, $\mathcal{A}_2^1$, and $\mathcal{A}_3^0$. Upon calling `BuildAndIntegrateAntenna` for those antennae, `TObject`

[2]The label SMQCD is not standard in the literature. It is used here to denote the massless QCD sector of the Standard Model.

$\mathcal{T}$-Object	Ingredients	Call
$\mathcal{T}^2_{q\bar{q}}$	Born contribution	`TObject[2,"qqbar"]`
$\mathcal{T}^4_{q\bar{q}}$	$\mathcal{A}^1_2$	`TObject[4,"qqbar"]`
$\mathcal{T}^4_{q\bar{q}g}$	$\mathcal{A}^0_3$	`TObject[4,"qqbarg"]`
$\mathcal{T}^6_{q\bar{q}}$	$\mathcal{A}^2_2, \tilde{\mathcal{A}}^2_2, \hat{\mathcal{A}}^2_2, \breve{\mathcal{A}}^2_2$	`TObject[6,"qqbar"]`
$\mathcal{T}^6_{q\bar{q}g}$	$\mathcal{A}^1_2, \mathcal{A}^0_3, \mathcal{A}^1_3, \tilde{\mathcal{A}}^1_3, \hat{\mathcal{A}}^1_3$	`TObject[6,"qqbarg"]`
$\mathcal{T}^6_{q\bar{q}q'\bar{q}'}$	$\mathcal{B}^0_4$	`TObject[6,"qqbarqprimeqprimebar"]`
$\mathcal{T}^6_{q\bar{q}q\bar{q}}$	$\mathcal{B}^0_4, \mathcal{C}^0_4$	`TObject[6,"qqbarqqbar"]`
$\mathcal{T}^6_{q\bar{q}gg}$	$\mathcal{A}^0_4, \tilde{\mathcal{A}}^0_4$	`TObject[6,"qqbargg"]`

Table 3.6: Overview of the TObject routes.

assembles them as follows,

$$\mathcal{T}^6_{q\bar{q}g} = \left(N - \frac{1}{N}\right) T^2_{q\bar{q}} \left[N\left(\mathcal{A}^1_3 + \mathcal{A}^1_2\mathcal{A}^0_3\right) - \frac{1}{N}\left(\tilde{\mathcal{A}}^1_3 + \mathcal{A}^1_2\mathcal{A}^0_3\right) + N_f\hat{\mathcal{A}}^1_3\right]. \tag{3.18}$$

Lastly, *AntCalc* is also able to generate the massless SMQCD hadronic R-ratio, using the integrated antenna ingredients required at the selected perturbative order. The function is called based on the requested perturbative order, that is, to get the complete NNLO massless hadronic R-ratio, we use `BuildRRatio[SMQCD, maxOrder->NNLO]`, but we may also only compute the R-ratio up to NLO using `BuildRRatio[SMQCD, maxOrder->NLO]`. By default, `BuildRRatio` returns the computed finite coefficient after application of the observable convention bridge; this result has been validated against a literature reference target.

3.7 Validation and Development Diagnostics

The supported non-trivial massless public release antenna routes are validated against literature results, specifically against those in [9] and [20], by comparing their Laurent-expanded results. The same integrated antennae are used as ingredients to reproduce the known $\mathcal{T}$-objects and, with them, assemble the massless hadronic R-ratio for their antenna type. These checks test the package at increasing levels: individual integrated antenna, exclusive final-state contributions, and the inclusive observable.

As discussed in Section 3.2, the route system can retain selected diagnostic information from an evaluation, during its calculation. That makes it easy to locate any inconsistency during the route calculation that may then be inspected using the built-in diagnostics tool. One example of its use could be checking how the poles of each R-ratio order's antennae cancel each other to make a finite result. As an example, let us consider the SMQCD R-ratio up to NLO. We can run it and then, in the diagnostics association, see what antennae it uses as ingredients. We may then simply compare those against one another (in this case there are only two, $\mathcal{A}^1_2$ and $\mathcal{A}^0_3$), and see how their poles cancel. The check may go as follows. In the following example, the common colour and perturbative prefactors are omitted, since they do not affect the cancellation of the poles in $\mathcal{A}^1_2 + \mathcal{A}^0_3$.

```
In[1]:= <<AntCalc`
In[2]:= diagNLORRatio = BuildRRatio[SMQCD,maxOrder->NLO,
```

```
ReturnDiagnostics->True][[2]];
In[3]:= ingredients = diagNLORRatio["Ingredients"];
In[4]:= A21 = ingredients["intA21"]; A30 = ingredients["intA30"];
In[5]:= resList = {};
In[6]:=
Do[
coefA21 = Coefficient[A21,Epsilon,i];
coefA30 = Coefficient[A30,Epsilon,i];
AppendTo[resList, coefA21+coefA30]
,
{i,-2,2}
];
In[7]:= resList
Out[7]:=
```

$$\left\{0, 0, \frac{3}{4}, \frac{45}{8} - 6\zeta_3, \frac{383}{16} - \frac{7\pi^2}{16} - \frac{\pi^4}{10} - 9\zeta_3\right\}$$

The first two entries are the coefficients of ϵ^{-2} and ϵ^{-1}, respectively, and vanish as required. The remaining entries are the finite and positive-power coefficients, which do not cancel. This diagnostic inspection makes the origin of the NLO pole cancellation explicit. The release-level observable validation additionally compares the raw NNLO R-ratio series with the known pole-free and finite result.

The `Ingredients` entry used above is only one part of the diagnostic information. For a complete retained history of a calculation, *AntCalc* can instead return an `AntennaRunRecord`, which records the relevant route stages and metadata. This information was used to trace the source of discrepancies during development; the resulting validation comparisons are presented in Chapter 5.

Chapter 4

Antenna Construction and Integration: Selected Case Studies

4.1 Master Integrals Used in *AntCalc*

As discussed in Chapter 3, *AntCalc* makes use of two reduction methods, Passarino-Veltmann reduction and integration-by-parts reduction. Both of those methods reduce a larger loop-momentum or phase-space integral into a linear combination of smaller and often less complex master integrals, as discussed in Subsection 2.5.3[1] [34]. The MI required by each route depend on the antenna being integrated, though some antennae share the same MI, especially those of the same type and in the same perturbative order, like all tree-level antenna functions at NNLO [21]. *AntCalc* also applies integral reduction at two different stages: loop antennae are reduced before producing the public build-stage output, but only over the loop momenta, and all antennae are reduced over the whole phase space during the integration stage. This means that loop antenna functions like A_3^1, for example, may require more than one set of MI throughout a complete build-and-integrate run, and the ones required for one stage may not be required for the other. Table 4.1 shows the MI used in each stage for each antenna [9].

Antenna type(s)	**Build-Side MI**	**Integration-Side MI**
A_2^0	–	–
A_3^0	–	R_3
A_4^0, $\tilde{A}_4^0$, B_4^0, C_4^0	–	R_4, R_6, $R_{8,a}$, $R_{8,b}$
A_2^1	–	B_0, C_0
A_3^1, $\tilde{A}_3^1$, $\hat{A}_3^1$	B_0, C_0, D_0	$V_{5,a}$, $V_{5,b}$, V_8
A_2^2	$A_{22,\mathrm{LO}}$, A_3, A_4, B_0, C_0	$A_{22,\mathrm{LO}}$, A_3, A_4
$\tilde{A}_2^2$	$A_{22,\mathrm{LO}}$, A_3, A_4, A_6	$A_{22,\mathrm{LO}}$, A_3, A_4, A_6
$\hat{A}_2^2$	A_4, B_0, C_0	A_4
$\check{A}_2^2$	$A_{22,\mathrm{LO}}$	$A_{22,\mathrm{LO}}$

Table 4.1: Overview of the master integrals used in the build- and integration-stage by each antenna type [9, 21].

Both PaVe and IBP MI are shown in the table, but they are quite different in structure and integration

[1] The PaVe MI are most commonly referred to as *scalar functions*, instead of *master integrals*. They are, in effect, still master integrals, however, and the moniker will be kept throughout the rest of the present text [34].

processes and will be discussed separately. The PaVe are B_0, C_0, and D_0. All others are IBP MI.

4.1.1 PaVe Scalar Functions

As mentioned, only three of the MI considered are PaVe scalar functions. Those are B_0, C_0 and D_0 and are the PaVe two-, three- and four-point scalar integrals, respectively. The PaVe method also features the A_0 (one-point) scalar function, that does not feature in *AntCalc*. It also accounts for scalar integrals with more than four external legs, as those can be further reduced in four dimensions to combinations of D_0 functions [7, 23].

The B_0 Function

The B_0 function represents a scalar one-loop diagram with two external lines and two internal propagators. It is represented, in d dimensions as,

$$B_0\left(k^2, m_0, m_1\right) = \frac{(2\pi\mu)^{4-d}}{i\pi^2} \int d^d p \frac{1}{\left(p^2 - m_0^2 + i\varepsilon\right)\left((p+k)^2 - m_1^2 + i\varepsilon\right)} \tag{4.1}$$

where k is the external momentum, and m_0 and m_1 are the internal masses. μ is the t'Hooft normalisation scale.

It is also noteworthy that this integral, unlike higher-point ones, is UV divergent and generates the pole term Δ_B,

$$\Delta_B = \frac{2}{4-d} - \gamma_E + \log(4\pi) \tag{4.2}$$

where γ_E is Euler's gamma constant. This function may also be written as an integral over the Feynman parameter x. That yields the explicit analytical form [23],

$$B_0\left(k^2, m_0, m_1\right) = \Delta_B - \int_0^1 dx \, \log\left[\frac{k^2x^2 - x\left(k^2 - m_0^2 + m_1^2\right) + m_1^2 - i\varepsilon}{\mu^2}\right] + O(d-4). \tag{4.3}$$

The C_0 Function

The C_0 function represents a scalar loop with three external lines and three internal propagators. It is a common feature in the vertex corrections of electroweak physics. It is written in d dimensions as,

$$\begin{aligned} C_0(k_1, k_2, m_0, m_1, m_2) = &\frac{(2\pi\mu)^{4-d}}{i\pi^2} \\ &\int d^d p \, \frac{1}{\left(p^2 - m_0^2 + i\varepsilon\right)\left((p+k_1)^2 - m_1^2 + i\varepsilon\right)\left((p+k_1+k_2)^2 - m_2^2 + i\varepsilon\right)}. \end{aligned} \tag{4.4}$$

This integral is UV finite and may be rewritten as a double integral over the Feynman parameters x and y as,

$$C_0 \propto \int_0^1 dx \int_0^{1-x} dy \, \left[ax^2 + by^2 + cxy + dx + ey + f\right]^{-1} \tag{4.5}$$

where the coefficients a, b, c, d, e and f are linear combinations of the external momenta and internal mass parameters [7].

The D_0 Function

The D_0 function describes a scalar loop with four external lines and four internal propagators. It is often called the scalar box integral. It is written as,

$$D_0(k_1, k_2, k_3, m_0, m_1, m_2, m_3) = \frac{(2\pi\mu)^{4-d}}{i\pi^2} \int d^d p \, \frac{1}{D_0 D_1 D_2 D_3} \tag{4.6}$$

with denominator factors,

$$\begin{aligned} D_0 &= p^2 - m_0^2 + i\varepsilon \\ D_i &= \left(p + \sum_{a=1}^{i} k_a \right)^2 - m_i^2 + i\varepsilon. \end{aligned} \tag{4.7}$$

Evaluating D_0 is highly non-trivial as it features numerous physical cuts and branch points. The resulting analytical expression is expressed in terms of Spence functions (dilogarithms) [7].

4.1.2 IBP Master Integrals

The IBP MI employed in *AntCalc* can be divided into four groups, based on the antenna functions they are used with. This division goes as,

- NLO tree-level: R_3;
- NNLO tree-level: R_4, R_6, $R_{8,a}$, $R_{8,b}$;
- NNLO one-loop: $V_{5,a}$, $V_{5,b}$, V_8;
- NNLO two-loop: $A_{22,\mathrm{LO}}$, A_3, A_4, A_6.

Phase-Space Measures and Normalisations

These MI are computed over different phase-space measures, depending on the number of final-state particles. These phase-space integrals are defined over all n-particle Mandelstam invariants for each specific phase-space. For that reason, we introduce the shorthands,

$$\begin{aligned} &\mathcal{S}_2 = s_{12}, \quad d\mathcal{S}_2 = ds_{12} \\ &\mathcal{S}_3 = s_{12} + s_{13} + s_{23}, \quad d\mathcal{S}_3 = ds_{12} ds_{13} ds_{23} \\ &\mathcal{S}_4 = s_{12} + s_{13} + s_{14} + s_{23} + s_{24} + s_{34}, \quad d\mathcal{S}_4 = ds_{12} ds_{13} ds_{14} ds_{23} ds_{24} ds_{34}. \end{aligned} \tag{4.8}$$

Using the shorthands, the differential phase-spaces are defined in d dimensions as [21, 25],

$$\begin{aligned} d\Phi_2 &= (2\pi)^{2-d} s_{12}^{\frac{d-4}{2}} 2^{1-d} d\Omega_{d-1} d\mathcal{S}_2 \delta\Big(q^2 - \mathcal{S}_2\Big) \\ d\Phi_3 &= (2\pi)^{3-2d} 2^{-1-d} \Big(q^2\Big)^{2-\frac{d}{2}} (s_{12}s_{13}s_{23})^{\frac{d-4}{2}} d\Omega_{d-1} d\Omega_{d-2} d\mathcal{S}_3 \delta\Big(q^2 - \mathcal{S}_3\Big) \\ d\Phi_4 &= (2\pi)^{4-3d} \Big(q^2\Big)^{3-\frac{d}{2}} 2^{1-2d} (-\Delta_4)^{\frac{d-5}{2}} \Theta(-\Delta_4) \\ &\qquad d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} d\mathcal{S}_4 \delta\Big(q^2 - \mathcal{S}_4\Big). \end{aligned} \tag{4.9}$$

Here Δ_4 is the Gram determinant given by the Källen function λ [21, 25],

$$\Delta_4 = \lambda(s_{12}s_{34}, s_{13}s_{24}, s_{14}s_{23}), \quad \lambda(x, y, z) = x^2 + y^2 + z^2 - 2(xy + xz + yz). \tag{4.10}$$

Also, Θ is the Heaviside step function and $d\Omega$ the differential solid hyperspherical angle element in d dimensions, given as,

$$\Omega_n = \int d\Omega_n = \frac{2\pi^{\frac{n}{2}}}{\Gamma(n/2)}. \tag{4.11}$$

Some MI appear normalised using the S_Γ term, to make the infrared singularity cancellations explicit and absorb common dimensional factors. The normalising term is given as [20, 21],

$$S_\Gamma = \left(\frac{(4\pi)^\epsilon}{16\pi^2\Gamma(1-\epsilon)}\right)^2. \tag{4.12}$$

It is also common to normalise with

$$S_{\Gamma,2} = P_2 S_\Gamma, \quad P_2 = \int d\Phi_2 = \frac{2^{-3+2\epsilon}\pi^{-1+\epsilon}\Gamma(1-\epsilon)}{\Gamma(2-2\epsilon)}\left(q^2\right)^{-\epsilon}. \tag{4.13}$$

The R_3 Integral

This MI is used only on the integration of the massless A_3^0 antenna. It is the direct integral over the three-final-state-particle phase-space. It is written as [34],

$$\begin{aligned} R_3 &= \int d\Phi_3 \\ &= \frac{1}{4}\frac{\Omega_{d-1}\Omega_{d-2}}{(2\pi)^{2d-3}} \iiint dE_1 dE_2 d\theta_{12}\,(E_1 E_2 \sin\theta_{12})^{d-3}\delta\left(q^2 - \mathcal{S}_3\right) \\ &= \frac{1}{128 q^2 \pi^3}\int d\mathcal{S}_3\,\delta\left(q^2 - \mathcal{S}_3\right). \end{aligned} \tag{4.14}$$

After analytical evaluation the integrated result, which is the one substituted in the final integrated antenna expression after IBP reduction becomes [21],

$$R_3 = 2^{-7+4\epsilon}\pi^{-3+2\epsilon}\frac{\Gamma^3(1-\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)}\left(q^2\right)^{1-2\epsilon}. \tag{4.15}$$

The R_4 Integral

The first of the tree-level NNLO MI is R_4, which corresponds to the direct integral over the four-final-state-particle phase-space. It is written as,

$$\begin{aligned} R_4 &= \int d\Phi_4 \\ &= (2\pi)^{4-3d}\left(q^2\right)^{3-\frac{d}{2}} 2^{1-2d}\int d\mathcal{S}_4 d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} \\ &\qquad\qquad (-\Delta_4)^{\frac{d-5}{2}}\Theta(-\Delta_4)\delta\left(q^2 - \mathcal{S}_4\right). \end{aligned} \tag{4.16}$$

To make the integral boundary limits integrable, we start by factorising the phase space into a two-particle phase space and a tripole phase space factor, $d\Phi_4 = d\Phi_2 d\Phi_T$ and then we map the six physical invariants s_{ij} into five dimensionless variables that range strictly between 0 and 1, making a five-dimensional hypercube. We start by defining $y_{ij} = s_{ij}/q^2$ and $y_{ijk} = s_{ijk}/q^2$. Our five dimensionless variables become [21]:

- The rescaled triple-invariant of the emitter jet, $y_{134} = \frac{s_{134}}{q^2}$;
- The longitudinal momentum fraction of the unresolved parton 3, $z_1 = \frac{p_3 \cdot \tilde{p}_2}{\tilde{p}_{134} \cdot \tilde{p}_2}$;
- The rescaled longitudinal momentum fraction of unresolved parton 4, $z_2 = (1 - z_1)t = \frac{p_4 \cdot \tilde{p}_2}{\tilde{p}_{134} \cdot \tilde{p}_2}$;
- The rescaled invariant, $y_{14} = \frac{s_{14}}{q^2} = y_{134}(1 - z_1)v$;
- The alignment variable χ, that parametrises the physical boundaries of the Gram determinant for y_{13} as, $y_{13} = (y_{13,b} - y_{13,a})\chi + y_{13,a} = \Delta y_{13}\chi + y_{13,a}$.

After applying the mapping, the MI becomes

$$
\begin{aligned}
R_4 = P_2 \frac{16^{-2}\pi^{-5+2\epsilon}}{\Gamma(1-2\epsilon)} \left(q^2\right)^{2-2\epsilon} \int_{[0,1]^5} dy_{134} dz_1 dt dv d\chi \left[y_{134}(1-y_{134})(1-z_1)\right]^{1-2\epsilon} \\
\left[z_1 t(1-t)v(1-v)\right]^{-\epsilon} \left[\chi(1-\chi)\right]^{-\frac{1}{2}-\epsilon}.
\end{aligned}
\tag{4.17}
$$

We get [9, 21],

$$
\begin{aligned}
R_4 &= P_2 \left(q^2\right)^{2-2\epsilon} \frac{2^{-11+8\epsilon}\pi^{-\frac{7}{2}+2\epsilon}\Gamma^3(1-\epsilon)}{\Gamma(3-3\epsilon)\Gamma\left(\frac{5}{2}-2\epsilon\right)} \\
&= S_{\Gamma,2} \left(q^2\right)^{2-2\epsilon} \frac{\Gamma^5(1-\epsilon)\Gamma(2-2\epsilon)}{\Gamma(3-3\epsilon)\Gamma(4-4\epsilon)} \\
&= S_\Gamma \left(q^2\right)^{2-2\epsilon} \left[\frac{1}{12} + \frac{59}{72}\epsilon + \mathcal{O}\left(\epsilon^2\right)\right].
\end{aligned}
\tag{4.18}
$$

The R_6 Integral

The R_6 MI is written as [21],

$$
\begin{aligned}
R_6 &= \int d\Phi_4 \frac{1}{s_{134}s_{234}} \\
&= (2\pi)^{4-3d} \left(q^2\right)^{3-\frac{d}{2}} 2^{1-2d} \int d\mathcal{S}_4 d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} \frac{(-\Delta_4)^{\frac{d-5}{2}}}{s_{134}s_{234}} \Theta(-\Delta_4)\delta\left(q^2 - \mathcal{S}_4\right).
\end{aligned}
\tag{4.19}
$$

Because the s_{234} is non-linear in the tripole parametrisation, we may not continue using direct integration [21]. We may, however, use the Optical Theorem as a cut of the three-loop vacuum polarisation diagram I_6 as,

$$
2\,\mathrm{Im}\, I_6 = -2P_2\,\mathrm{Re}\, A_4 + 2R_6
\tag{4.20}
$$

with I_6 and A_4 being known from the literature and written as,

$$
\begin{aligned}
I_6 = \frac{(4\pi)^{3\epsilon}}{(16\pi^2)^3} \frac{\Gamma^6(1-\epsilon)\Gamma^3(1+\epsilon)}{3\epsilon^3\Gamma^3(2-2\epsilon)} \left(-q^2\right)^{-3\epsilon} \\
\left[1 + \epsilon + \epsilon^2 + \epsilon^3(14\zeta_3 - 7) + \epsilon^4\left(-67 + 14\zeta_3 + \frac{21\pi^4}{90}\right) + \mathcal{O}\left(\epsilon^5\right)\right]
\end{aligned}
\tag{4.21}
$$

$$
\begin{aligned}
A_4 = \left(\frac{(4\pi)^\epsilon}{16\pi^2}\frac{\Gamma(1+\epsilon)\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)}\right)^2 \left(-q^2\right)^{-2\epsilon} \Bigg[& -\frac{1}{2\epsilon^2} - \frac{5}{2\epsilon} - \left(\frac{19}{2}+\frac{\pi^2}{6}\right) \\
& - \epsilon\left(\frac{65}{2}+\frac{5\pi^2}{6}-2\zeta_3\right) - \epsilon^2\left(\frac{211}{2}+\frac{19\pi^2}{6}-10\zeta_3\right) + O\left(\epsilon^3\right)\Bigg].
\end{aligned}
\tag{4.22}
$$

Note that A_4 is described in Subsubsection 4.1.2.

We then apply the Minkowski boundary prescription [21] $\left(q^2\right)^{-\epsilon} = \left(q^2\right)^{-\epsilon} e^{i\pi\epsilon}$, isolate the imaginary and real parts, and get, by using $\mathcal{A}_4$ and $\mathcal{I}_6$

$$
\begin{aligned}
R_6 &= S_\Gamma\left(q^2\right)^{-2\epsilon} \frac{\Gamma^6(1-\epsilon)\Gamma^2(1+\epsilon)}{6}\left(\frac{6\cos(2\pi\epsilon)}{\Gamma^2(1-2\epsilon)}\mathcal{A}_4 + \frac{\Gamma(1-\epsilon)\Gamma(1+\epsilon)\sin(3\pi\epsilon)}{\epsilon^3\pi\Gamma^2(2-2\epsilon)}\mathcal{I}_6\right) \\
&= S_{\Gamma,2}\left(q^2\right)^{-2\epsilon}\left[-1+\frac{\pi^2}{6}+\epsilon\left(-12+\frac{5\pi^2}{6}+9\zeta_3\right)+O\left(\epsilon^2\right)\right].
\end{aligned}
\tag{4.23}
$$

The $R_{8,a}$ Integral

The $R_{8,a}$ is the first of two MI with four denominator invariants. Its complete integral form is written as [21],

$$
\begin{aligned}
R_{8,a} &= \int d\Phi_4 \frac{1}{s_{13}s_{23}s_{14}s_{24}} \\
&= (2\pi)^{4-3d}\left(q^2\right)^{3-\frac{d}{2}} 2^{1-2d} \int dS_4 d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} \frac{(-\Delta_4)^{\frac{d-5}{2}}}{s_{13}s_{23}s_{14}s_{24}} \Theta(-\Delta_4)\,\delta\left(q^2-\mathcal{S}_4\right) \\
&= P_2\left(q^2\right)^{-2-2\epsilon}\frac{16^{-2+\epsilon}\pi^{-5+2\epsilon}}{\Gamma(1-2\epsilon)}\int_{[0,1]^5} dy_{134}dz_1dtdvd\chi \\
&\qquad \left[(1-y_{134})(1-z_1)\right]^{-1-2\epsilon}\left[tvz_1^{-\epsilon}(\Delta y_{13}\chi+y_{13,a})\right]^{-1}\left[\chi(1-\chi)\right]^{-\frac{1}{2}-\epsilon}.
\end{aligned}
\tag{4.24}
$$

This is the most complex MI integration of the set. As such, its computation will be written in detail here as most of the methods used are transferable to other MI. We start by defining the integration order. The computation will advance as,

$$
\chi \to v \to y_{134} \to t \to z_1
$$

with the inclusion of a u integral, also over the $[0,1]$ interval in between the t and z_1 integrals, which will be introduced later.

The first integral is over χ, and gives,

$$
I_\chi = \int_0^1 d\chi\left[\chi(1-\chi)\right]^{-\frac{1}{2}-\epsilon}(\Delta y_{13}\chi+y_{13,a})^{-1} = \frac{\Delta y_{13}^{-2\epsilon}}{y_{13,a}}\frac{\Gamma^2\left(\frac{1}{2}-\epsilon\right)}{\Gamma(1-2\epsilon)}{}_2F_1\left(1,\frac{1}{2}-\epsilon;1-2\epsilon;-\frac{\Delta y_{13}}{y_{13,a}}\right)
\tag{4.25}
$$

where ${}_2F_1(\ldots)$ is a hypergeometric function. We now consider the definition $y_{13,(a,b)} = y_{134}(A \mp B)$, $A = \sqrt{(1-t)(1-v)}$, $B = \sqrt{z_1tv}$, and $Z = -B/A$, and get,

$$
-\frac{\Delta y_{13}}{y_{13,a}} = \frac{4Z}{(1+Z)^2}, \quad (1+Z)^2 = \frac{y_{13,a}}{y_{134}A^2}.
\tag{4.26}
$$

Finally, by taking,

$$
{}_2F_1\left(1, b; 2b; \frac{4Z}{(1+Z)^2}\right) = (1+Z)^2 {}_2F_1\left(1, \frac{3}{2} - b; b + \frac{1}{2}; Z^2\right) \tag{4.27}
$$

and

$$
16^{\epsilon} \Delta y_{13}^{-2\epsilon} = \left[t(1-t)v(1-v)y_{134}^2 z_1\right]^{-\epsilon} \tag{4.28}
$$

we get,

$$
I_\chi = \frac{\pi\Gamma(1-2\epsilon)}{y_{134}\Gamma^2(1-\epsilon)} \left(tvy_{134}^2 z_1\right)^{-\epsilon} \left[(1-t)(1-v)\right]^{-1-\epsilon} {}_2F_1\left(1, 1+\epsilon; 1-\epsilon; \frac{z_1 tv}{(1-t)(1-v)}\right). \tag{4.29}
$$

The $R_{8,a}$ expression becomes,

$$
\begin{aligned}
R_{8,a} = P_2\left(q^2\right)^{-2-2\epsilon} \frac{16^{-2+\epsilon}\pi^{-4+2\epsilon}}{\Gamma^2(1-\epsilon)} \int_{[0,1]^4} dy_{134} dz_1 dt dv \left[y_{134}(1-y_{134})z_1(1-z_1)\right]^{-1-2\epsilon} \\
\left[t(1-t)v(1-v)\right]^{-1-\epsilon} {}_2F_1\left(1, 1+\epsilon; 1-\epsilon; \frac{z_1 tv}{(1-t)(1-v)}\right).
\end{aligned} \tag{4.30}
$$

We now continue to the v integral. To start, we restate the hypergeometric function as a series, which enables us to write,

$$
\begin{aligned}
I_v &= \int_0^1 dv \left[v(1-v)\right]^{-1-\epsilon} {}_2F_1\left(1, 1+\epsilon; 1-\epsilon; \frac{z_1 tv}{(1-t)(1-v)}\right) \\
&= \sum_{n\geq 0} \frac{(1+\epsilon)_n}{(1-\epsilon)_n} \left(\frac{z_1 t}{1-t}\right)^n \frac{\Gamma(-\epsilon-n)\Gamma(-\epsilon+n)}{\Gamma(-2\epsilon)} \\
&= -\frac{2\Gamma^2(1-\epsilon)}{\epsilon\Gamma(1-2\epsilon)} {}_2F_1\left(1, -\epsilon; 1-\epsilon; -\frac{z_1 t}{1-t}\right).
\end{aligned} \tag{4.31}
$$

Plugging this expression back in and computing the trivial y_{134} integral we get,

$$
\begin{aligned}
R_{8,a} = -P_2\left(q^2\right)^{-2-2\epsilon} \frac{2^{-7+8\epsilon}\pi^{-\frac{7}{2}+2\epsilon}}{\epsilon^2\Gamma\left(\frac{1}{2}-2\epsilon\right)} \int_{[0,1]^2} dz_1 dt\, (1-z_1)^{-1-2\epsilon} \\
\left[z_1 t(1-t)\right]^{-1-\epsilon} {}_2F_1\left(1, -\epsilon; 1-\epsilon; -\frac{z_1 t}{1-t}\right).
\end{aligned} \tag{4.32}
$$

We now restate the hypergeometric function as an Euler integral as,

$$
{}_2F_1\left(1, -\epsilon; 1-\epsilon; -\frac{z_1 t}{1-t}\right) = -\epsilon \int_0^1 du\, u^{-1-\epsilon} \left[1 + \frac{u z_1 t}{1-t}\right]^{-1} \tag{4.33}
$$

and after computing the direct t, u and z_1 integrals, in that order, we get to the final expression,

$$
\begin{aligned}
R_{8,a} = P_2\left(q^2\right)^{-2-2\epsilon} 2^{-7+4\epsilon}\pi^{-3+2\epsilon} \csc(\pi\epsilon)\Gamma(-2\epsilon) \Bigg[\frac{2^{8\epsilon}\pi}{\epsilon^2\Gamma^2\left(\frac{1}{2}-2\epsilon\right)} {}_3F_2\left(\begin{matrix} -2\epsilon, -2\epsilon, -2\epsilon \\ 1-2\epsilon, -4\epsilon \end{matrix}; 1\right) - \\
- \frac{2\Gamma^3(-\epsilon)}{\Gamma(1-4\epsilon)\Gamma(1-\epsilon)\Gamma(1+\epsilon)\Gamma(-3\epsilon)} {}_4F_3\left(\begin{matrix} 1, -\epsilon, -\epsilon, -\epsilon \\ 1-\epsilon, 1+\epsilon, -3\epsilon \end{matrix}; 1\right) \Bigg]
\end{aligned} \tag{4.34}
$$

which after a final normalisation becomes [21, 34],

$$\begin{aligned}R_{8,a} =& S_\Gamma\left(q^2\right)^{-2-2\epsilon}\left[\frac{6}{\epsilon^2}\frac{\Gamma^5(1-\epsilon)\Gamma(1-2\epsilon)}{\Gamma(1-3\epsilon)\Gamma(1-4\epsilon)}{}_4F_3\left(\begin{array}{c}1,-\epsilon,-\epsilon,-\epsilon\\1-\epsilon,1+\epsilon,-3\epsilon\end{array};1\right)-\right.\\&\left.-\frac{1}{\epsilon^4}\frac{\Gamma^3(1-\epsilon)\Gamma(1+\epsilon)\Gamma^3(1-2\epsilon)}{\Gamma^2(1-4\epsilon)}{}_3F_2\left(\begin{array}{c}-2\epsilon,-2\epsilon,-2\epsilon\\-4\epsilon,1-2\epsilon\end{array};1\right)\right]\\&= S_{\Gamma,2}\left(q^2\right)^{-2-2\epsilon}\left[\frac{5}{\epsilon^4}-\frac{20\pi^2}{3\epsilon^2}-\frac{126\zeta_3}{\epsilon}+\frac{7\pi^4}{18}+O(\epsilon)\right].\end{aligned}\tag{4.35}$$

The $R_{8,b}$ Integral

The second of the two four-denominator-invariant MI is $R_{8,b}$. It is written as [21],

$$R_{8,b} = \int d\Phi_4\,\frac{1}{s_{13}s_{134}s_{23}s_{234}}.\tag{4.36}$$

Due to the presence of s_{234}, this MI must be computed using the Optical Theorem, much like R_6 was. For this case, we are able to map $R_{8,b}$ as one of the cuts of the three-loop vacuum polarisation diagram I_8 as,

$$2\,\mathrm{Im}\,I_8 = -2P_2\,\mathrm{Re}\,A_6 + 4\,\mathrm{Re}\,V_8 + R_{8,a} + 4R_{8,b}.\tag{4.37}$$

with,

$$I_8 = S + O(\epsilon^0),\quad \mathrm{Im}\,S = -\frac{13\pi^4}{90};\tag{4.38}$$

$$V_8 = -i\int d\Phi_3\,\frac{1}{s_{12}}\,\mathrm{Box}(s_{13},s_{13},s_{12});\tag{4.39}$$

$$A_6 = \left(\frac{(4\pi)^\epsilon}{16\pi^2}\frac{\Gamma(1+\epsilon)\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)}\right)^2\left(q^2\right)^{-2-2\epsilon}\left[-\frac{1}{\epsilon^4}+\frac{5\pi^2}{6\epsilon^2}+\frac{23\zeta_3}{\epsilon}+\frac{103\pi^4}{180}+O(\epsilon)\right].\tag{4.40}$$

Note that the A_6 and V_8 integrals are described in Subsubsections 4.1.2 and 4.1.2. In those Subsubsections, the integrals are given normalised using the usual S_Γ prefactor.

Using $\mathcal{V}_8$, $\mathcal{A}_6$ and $\mathcal{R}_{8,a}$ to denote the series parts of $\mathrm{Re}\,V_8$, $P_2\,\mathrm{Re}\,A_6$ and $R_{8,a}$, respectively, we get [21],

$$\begin{aligned}R_{8,b} &= \frac{1}{2}\,\mathrm{Im}\,S + S_\Gamma\left(q^2\right)^{-2-2\epsilon}\left(\frac{1}{2}\frac{\cos(2\pi\epsilon)\Gamma^6(1-\epsilon)\Gamma^2(1-\epsilon)}{\Gamma^2(1-2\epsilon)}\mathcal{A}_6-\mathcal{V}_8-\frac{1}{4}\mathcal{R}_{8,a}\right)\\&= S_\Gamma\left(q^2\right)^{-2-2\epsilon}\left[\frac{3}{4\epsilon^4}-\frac{17\pi^2}{12\epsilon^2}-\frac{44\zeta_3}{\epsilon}-\frac{61\pi^4}{60}+O(\epsilon)\right].\end{aligned}\tag{4.41}$$

The $V_{5,a}$ Integral

$V_{5,a}$ is a one-loop bubble integrated over a three-particle phase space, written as,

$$\begin{aligned}V_{5,a} &= \mathrm{Re}\left[-i\int d\Phi_3\int\frac{d^dp}{(2\pi)^d}\frac{1}{p^2(p-k_1-k_2-k_3)^2}\right]\\&= \mathrm{Re}\left[\int d\Phi_3\,B_0\left(q^2;0,0\right)\right].\end{aligned}\tag{4.42}$$

Since $q^2 = (p_1 + p_2 + p_3)^2$, this MI factors to the three-particle phase-space integral of the $B_0(q^2; 0, 0)$ scalar one-loop function, discussed in Subsubsection 4.1.1. After parametrising the phase space and applying the methods introduced in Subsubsection 4.1.2 we get [9],

$$\begin{aligned} V_{5,a} &= S_{\Gamma,2}\left(q^2\right)^{1-2\epsilon} \cos(\pi\epsilon)\left(-\frac{1}{\epsilon}\right) \frac{\Gamma^6(1-\epsilon)\Gamma(1+\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)} \\ &= S_{\Gamma,2}\left(q^2\right)^{1-2\epsilon}\left[-\frac{1}{2\epsilon} - \frac{13}{4} + O(\epsilon)\right]. \end{aligned} \tag{4.43}$$

The $V_{5,b}$ Integral

$V_{5,b}$ is very similar in structure to $V_{5,a}$, also being the three-particle phase-space integral of a B_0 scalar one-loop function. The integral is also computed in the same manner and yields [9],

$$\begin{aligned} V_{5,b} &= \int d\Phi_3\, B_0(s_{13}; 0, 0) \\ &= S_{\Gamma,2}\left(q^2\right)^{1-2\epsilon} \cos(\pi\epsilon)\left(-\frac{1}{\epsilon}\right) \frac{\Gamma^5(1-\epsilon)\Gamma(1-2\epsilon)\Gamma(1+\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-4\epsilon)} \\ &= S_{\Gamma,2}\left(q^2\right)^{1-2\epsilon}\left[-\frac{1}{2\epsilon} - 4 + O(\epsilon)\right]. \end{aligned} \tag{4.44}$$

The V_8 Integral

The V_8 MI is the most complex of the three V-type MI. It is given as a four-denominator propagator loop and three-particle phase-space integral. That makes it be restatable as a Box function which, itself, may be restated as a D_0 function, as discussed in Subsubsection 4.1.1. It is written as [9],

$$\begin{aligned} V_8 &= \mathrm{Re}\left[-i \int d\Phi_3 \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2(p-k_1)^2(p-k_1-k_3)^2(p-k_1-k_2-k_3)^2}\right] \\ &= \mathrm{Re}\left[-i \int d\Phi_3\, \frac{1}{s_{12}} \mathrm{Box}(s_{13}, s_{23}, s_{12})\right] \\ &= \left(\frac{(4\pi)^\epsilon \mu^{-2\epsilon}}{16\pi^2}\right) \frac{\Gamma^2(1-\epsilon)\Gamma(1+\epsilon)}{\Gamma(1-2\epsilon)} \int d\Phi_3\, \frac{1}{s_{12}} \mathrm{Re}\left[D_0\left(0, 0, 0, q^2; s_{13}, s_{23}; 0, 0, 0, 0\right)\right] \end{aligned} \tag{4.45}$$

which, after normalisation and evaluation gives,

$$V_8 = S_{\Gamma,2}\left(q^2\right)^{-2-2\epsilon}\left[-\frac{5}{2\epsilon^4} + \frac{9\pi^2}{2\epsilon^2} + \frac{89\zeta_3}{\epsilon} + \frac{13\pi^4}{180} + O(\epsilon)\right]. \tag{4.46}$$

The $A_{22,\mathrm{LO}}$ Integral

$A_{22,\mathrm{LO}}$ is the first of the four A-type MI, and represents a double bubble product. These are loop momenta integrals, not phase-space integrals. This first MI is given as [9],

$$\begin{aligned}
A_{22,\mathrm{LO}} &= \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 (p - k_1 - k_2)^2 \ell^2 (\ell - k_1 - k_2)^2} \\
&= -\frac{\mu^{-4\epsilon}}{\left(16\pi^2\right)^2} \left[B_0\left(q^2; 0, 0\right)\right]^2 \\
&= S_\Gamma \left(-q^2\right)^{-2\epsilon} \frac{\Gamma^2(1+\epsilon)\Gamma^6(1-\epsilon)}{\Gamma^2(2-2\epsilon)} \left(-\frac{1}{\epsilon^2}\right) \\
&= S_\Gamma \left(-q^2\right)^{-2\epsilon} \left[-\frac{1}{\epsilon^2} - \frac{4}{\epsilon} - 12 + O(\epsilon)\right].
\end{aligned} \tag{4.47}$$

The A_3 Integral

A_3 represents a three-propagator two-loop sunset vacuum bubble, with $q^2 = s_{12}$. It is given as [9],

$$\begin{aligned}
A_3 &= \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 \ell^2 (p - \ell - k_1 - k_2)^2} \\
&= \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2} B_0\left((p - k_1 - k_2)^2; 0, 0\right) \\
&= S_\Gamma \left(-q^2\right)^{1-2\epsilon} \frac{\Gamma(1+2\epsilon)\Gamma^5(1-\epsilon)}{\Gamma(3-3\epsilon)} \left(-\frac{1}{2(1-2\epsilon)\epsilon}\right) \\
&= S_\Gamma \left(-q^2\right)^{1-2\epsilon} \left[-\frac{1}{4\epsilon} - \frac{13}{8} + O(\epsilon)\right].
\end{aligned} \tag{4.48}$$

The A_4 Integral

A_4 is a four-propagator two-loop vacuum planar double bubble. It is given as a triangle diagram containing a bubble insertion on one of its legs. That gives [9],

$$\begin{aligned}
A_4 &= \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 \ell^2 (p - k_1 - k_2)^2 (p - \ell - k_1)^2} \\
&= \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2 (p - k_1 - k_2)^2} B_0\left((p - k_1)^2; 0, 0\right) \\
&= S_\Gamma \left(-q^2\right)^{-2\epsilon} \frac{\Gamma(1-2\epsilon)\Gamma(1+\epsilon)\Gamma^4(1-\epsilon)\Gamma(1+2\epsilon)}{\Gamma(2-3\epsilon)} \left(-\frac{1}{2(1-2\epsilon)\epsilon^2}\right) \\
&= S_\Gamma \left(-q^2\right)^{-2\epsilon} \left[-\frac{1}{2\epsilon^2} - \frac{5}{2\epsilon} - \left(\frac{19}{2} + \frac{\pi^2}{6}\right) + O(\epsilon)\right].
\end{aligned} \tag{4.49}$$

The A_6 Integral

Lastly, the A_6 MI represents the two-loop two-point vertex. It is given as [9],

$$A_6 = \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 \ell^2 (p - k_1 - k_2)^2 (p - \ell)^2 (p - \ell - k_2)^2 (\ell - k_1)^2}. \tag{4.50}$$

Unlike the other A-type antennae, this one cannot be given in terms of the scalar one-loop functions B_0, C_0 or D_0. To compute this integral we have to start by expressing it as a Mellin-Barnes contour integral, then evaluate the residues, and finally expand as a Laurent series around ϵ. This process gives,

$$\begin{aligned}A_6 &= \frac{S_\Gamma\left(-q^2\right)^{-2-2\epsilon}}{(2\pi i)^2}\int_{-i\infty}^{i\infty}\int_{-i\infty}^{i\infty} dz_1 dz_2\,\frac{\Gamma(-z_1)\Gamma(-z_2)\Gamma(2+2\epsilon+z_1+z_2)\Gamma^2(-\epsilon-z_1)\Gamma^2(-\epsilon-z_2)}{\Gamma(-2\epsilon-z_1-z_2)\Gamma(2+\epsilon+z_1+z_2)}\\ &= S_\Gamma\left(-q^2\right)^{-2-2\epsilon}\left[-\frac{1}{\epsilon^4}+\frac{5\pi^2}{6\epsilon^2}+\frac{27}{\epsilon}\zeta_3+\frac{23\pi^4}{36}+O(\epsilon)\right].\end{aligned} \tag{4.51}$$

4.1.3 Exploratory Massive Master Integrals

AntCalc features an exploratory, but complete, massive Implementation of the A_3^0 route. To integrate it, two new MI are required. These two massive MI are,

$$\begin{aligned}I_1^{(m_q,0,m_q)} &= \int d\Phi_X^{(m_q,0,m_q)}\\ I_2^{(m_q,0,m_q)} &= \int d\Phi_X^{(m_q,0,m_q)}\, s_{13}\end{aligned} \tag{4.52}$$

where $\Phi_X^{(m_q,0,m_q)}$ is the massive tripole phase space and is defined as,

$$d\Phi_X^{(m_q,0,m_q)} = \frac{d\Phi_3^{(m_q,0,m_q)}(q;k_1,k_3,k_2)}{d\Phi_2^{(m_q,m_q)}(q;\tilde{k}_I,\tilde{k}_K)},\quad p\to q(k_1,m_q)+g(k_3,0)+\bar{q}(k_2,m_q) \tag{4.53}$$

and m_q is the quark mass. Note that the two- and three-particle massive phase spaces are described as in Eq. 4.9 with $k_1^2 = k_2^2 = m_q^2$, $k_3^2 = 0$, and $q^2 > 4m_q^2$.

These two integrals evaluate to [28],

$$\begin{aligned}I_1^{(m_q,0,m_q)} &= \left(q^2\right)^{1-\epsilon} r^{2-2\epsilon}2^{-2\epsilon}\pi^{-2+\epsilon}\frac{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)}{\Gamma(6-6\epsilon)}\,{}_2F_1\left(\frac{1}{2},2-2\epsilon;\frac{7}{2}-3\epsilon;r\right)\\ I_2^{(m_q,0,m_q)} &= \left(q^2\right)^{2-\epsilon} r^{3-2\epsilon}2^{1-2\epsilon}\pi^{-2+\epsilon}\frac{\Gamma(3-2\epsilon)\Gamma(4-3\epsilon)}{\Gamma(8-6\epsilon)}\,{}_2F_1\left(\frac{1}{2},3-2\epsilon;\frac{9}{2}-3\epsilon;r\right)\end{aligned} \tag{4.54}$$

where,

$$r = 1-\frac{4m_q^2}{q^2}. \tag{4.55}$$

4.2 NLO Antenna Results

The simplest non-trivial SMQCD antenna is A_3^0. As such, it may be used as an example of how AntCalc handles the physics involved in building and then integrating it over the phase space. In this chapter, we are going to go over how this antenna is mathematically computed by hand, and how AntCalc handles those same calculations on its end.

4.2.1 The Tree-Level Three-Parton Antenna A_3^0

Analytic Construction of the Unintegrated A_3^0 Antenna

The A_3^0 is the tree-level final-final quark-antiquark-gluon antenna. It represents the normalised $\gamma^*(p) \longrightarrow q(k_1)g(k_3)\bar{q}(k_2)$ process. This process, in the current model, is able to produce two distinct topologies which differ by what quark line the gluon is emitted from. Those diagrams are shown below in Figure 4.1.

(a) Gluon emitted from the quark line.

(b) Gluon emitted from the antiquark line.

Figure 4.1: Feynman diagrams depicting the two topologies described under the A_3^0 antenna, corresponding to the process $\gamma^* \longrightarrow q(k_1)g(k_3)\bar{q}(k_2)$.

The amplitude is given as follows directly from the diagrams,

$$\mathcal{M}_3^0 = e_q g_s (T^{a_3})_{i_1 i_2} \left[\bar{u}(k_1)\not{\epsilon}^*(k_3)\frac{\not{k}_1 + \not{k}_3}{s_{13}}\not{\epsilon}(p)v(k_2) - \bar{u}(k_1)\not{\epsilon}^*(p)\frac{\not{k}_2 + \not{k}_3}{s_{23}}\not{\epsilon}(k_3)v(k_2) \right] \tag{4.56}$$

where $(T^{a_3})_{i_1 i_2}$ represents the colour matrix for the quark and antiquark in positions 1 and 2 and gluon in position 3.

The amplitude is now squared as follows,

$$\begin{aligned} |\mathcal{M}_3^0|^2 &= e_q^2 g_s^2 (T^{a_3})_{i_1 i_2}(T^{a_3})_{i_1 i_2}\left(M_A M_A^\dagger + M_B M_B^\dagger - M_A M_B^\dagger - M_B M_A^\dagger \right) \\ \text{with } M_A &= \bar{u}(k_1)\not{\epsilon}^*(k_3)\frac{\not{k}_1 + \not{k}_3}{s_{13}}\not{\epsilon}(p)v(k_2) = \bar{u}(k_1)\Gamma_A v(k_2), \\ M_B &= \bar{u}(k_1)\not{\epsilon}^*(p)\frac{\not{k}_2 + \not{k}_3}{s_{23}}\not{\epsilon}(k_3)v(k_2) = \bar{u}(k_1)\Gamma_B v(k_2). \end{aligned} \tag{4.57}$$

which in total becomes

$$\begin{aligned} = &\sum_{a_3, i_1, i_2} (T^{a_3})_{i_1 i_2}(T^{a_3})_{i_1 i_2} \\ &\sum_{s_1, s_2, \lambda_g, \lambda_\gamma} \bar{u}(k_1, s_1)u(k_1, s_1)\left[\Gamma_A\bar{\Gamma}_A + \Gamma_B\bar{\Gamma}_B - \Gamma_A\bar{\Gamma}_B - \Gamma_B\bar{\Gamma}_A \right] v(k_2, s_2)\bar{v}(k_2, s_2) \end{aligned} \tag{4.58}$$

Each of the Γ terms becomes, using Dirac Algebra [ref to dirac algebra maths],

$$
\begin{aligned}
\sum_{s_1,s_2,\lambda_g,\lambda_\gamma} & \bar{u}(k_1,s_1)u(k_1,s_1)\left(\Gamma_A\bar{\Gamma}_A\right)v(k_2,s_2)\bar{v}(k_2,s_2) = \\
&= \frac{1}{s_{13}^2}\sum_{\lambda_g,\lambda_\gamma}\varepsilon_\mu^*\varepsilon_\nu\varepsilon_\rho\varepsilon_\sigma^*\,\mathrm{Tr}\left(\not{k}_1\gamma^\mu(\not{k}_1+\not{k}_3)\gamma^\rho\not{k}_2\gamma^\sigma(\not{k}_1+\not{k}_3)\gamma^\nu\right) \\
&= \frac{1}{s_{13}^2}g_{\mu\nu}g_{\rho\sigma}\,\mathrm{Tr}\left(\not{k}_1\gamma^\mu(\not{k}_1+\not{k}_3)\gamma^\rho\not{k}_2\gamma^\sigma(\not{k}_1+\not{k}_3)\gamma^\nu\right) \\
&= \frac{1}{s_{13}^2}\,\mathrm{Tr}\left(\not{k}_1\gamma^\mu(\not{k}_1+\not{k}_3)\gamma^\rho\not{k}_2\gamma_\rho(\not{k}_1+\not{k}_3)\gamma_\mu\right) \\
&= \left(\frac{2-d}{s_{13}}\right)^2\mathrm{Tr}(\not{k}_1\not{P}\not{k}_2\not{P}),\quad \not{P}=(\not{k}_1+\not{k}_3) \\
&= 4\left(\frac{2-d}{s_{13}}\right)^2\left(2(k_1\cdot P)(k_2\cdot P)-(k_1\cdot k_2)P^2\right) \\
&= 2(d-2)^2\frac{s_{23}}{s_{13}}
\end{aligned} \tag{4.59}
$$

$$
\begin{aligned}
\sum_{s_1,s_2,\lambda_g,\lambda_\gamma} & \bar{u}(k_1,s_1)u(k_1,s_1)\left(\Gamma_B\bar{\Gamma}_B\right)v(k_2,s_2)\bar{v}(k_2,s_2) = \\
&= 2(d-2)^2\frac{s_{13}}{s_{23}}
\end{aligned} \tag{4.60}
$$

$$
\begin{aligned}
\sum_{s_1,s_2,\lambda_g,\lambda_\gamma} & \bar{u}(k_1,s_1)u(k_1,s_1)\left(\Gamma_A\bar{\Gamma}_B\right)v(k_2,s_2)\bar{v}(k_2,s_2) = \\
&= 2(d-2)\frac{(d-4)s_{13}s_{23}-2s_{12}^2-2s_{12}(s_{13}+s_{23})}{s_{13}s_{23}}
\end{aligned} \tag{4.61}
$$

$$
\begin{aligned}
\sum_{s_1,s_2,\lambda_g,\lambda_\gamma} & \bar{u}(k_1,s_1)u(k_1,s_1)\left(\Gamma_B\bar{\Gamma}_A\right)v(k_2,s_2)\bar{v}(k_2,s_2) = \\
&= 2(d-2)\frac{(d-4)s_{13}s_{23}-2s_{12}^2-2s_{12}(s_{13}+s_{23})}{s_{13}s_{23}}
\end{aligned} \tag{4.62}
$$

and computing the colour matrix sum as

$$
\sum_{a_3,i_1,i_2}(T^{a_3})_{i_1i_2}(T^{a_3})_{i_1i_2}=\frac{1}{2}(N^2-1) \tag{4.63}
$$

we are finally able to reach the final result, as follows, using dimensional regularisation as $d\to 4-2\epsilon$ [ref to dimreg],

$$
\begin{aligned}
|\mathcal{M}_3^0|^2 &= e_q^2 g_s^2 (N^2-1)\left(2(d-2)^2\left(\frac{s_{13}}{s_{23}}+\frac{s_{23}}{s_{13}}\right)-4(d-2)\frac{(d-4)s_{13}s_{23}-2s_{12}^2-2s_{12}(s_{13}+s_{23})}{s_{13}s_{23}}\right)\\
&= 2(d-2)e_q^2 g_s^2 (N^2-1)\left((d-2)\left(\frac{s_{13}}{s_{23}}+\frac{s_{23}}{s_{13}}\right)-2\frac{(d-4)s_{13}s_{23}-2s_{12}^2-2s_{12}(s_{13}+s_{23})}{s_{13}s_{23}}\right)\\
&= \frac{8}{s_{13}s_{23}}\left(2q^2 s_{12}+s_{13}^2+s_{23}^2-2\epsilon\left(q^2 s_{12}+s_{13}^2+s_{23}^2+s_{13}s_{23}\right)+\epsilon^2\left(s_{13}+s_{23}\right)^2\right)
\end{aligned}
\tag{4.64}
$$

The A_3^0 antenna also requires the $\mathcal{M}_2^0$ amplitude. This amplitude corresponds to the $\gamma^* \to q\bar{q}$ Born process. The amplitude is

$$
\mathcal{M}_2^0 = e_q \delta_{i_1 i_2} \bar{u}(k_1) \not{\epsilon}(p) v(k_2). \tag{4.65}
$$

Applying the same spin, polarization and colour sums as above, and considering that, for a two final-state particles system $s_{12} = q^2$, we get

$$
\begin{aligned}
|\mathcal{M}_2^0|^2 = 4e_q^2(d-2)N s_{12} &= 4e_q^2(d-2)Nq^2\\
&= 8e_q^2(1-\epsilon)Nq^2
\end{aligned}
\tag{4.66}
$$

The antenna is now finally ready to be built. That is accomplished using the following formula,

$$
A_3^0 = \frac{1}{2g_s^2 C_F}\frac{|\mathcal{M}_3^0|^2}{|\mathcal{M}_2^0|^2}. \tag{4.67}
$$

By taking $C_F = (N^2-1)/(2N)$, the formula yields, using Eqs. 4.64 and 4.66,

$$
\begin{aligned}
A_3^0 &= \frac{2q^2 s_{12}+s_{13}^2+s_{23}^2}{q^2 s_{13}s_{23}} - \epsilon\frac{(s_{13}+s_{23})^2}{q^2 s_{13}s_{23}}\\
&= \frac{1}{q^2}\left[\left(\frac{s_{13}}{s_{23}}+\frac{s_{23}}{s_{13}}+2\frac{s_{12}q^2}{s_{13}s_{23}}\right)-\epsilon\left(2+\frac{s_{13}}{s_{23}}+\frac{s_{23}}{s_{13}}\right)\right]
\end{aligned}
\tag{4.68}
$$

Phase-Space Integration through R_3

We now integrate the A_3^0 antenna over the appropriate (3 final-state particle, in this case) phase-space. This integral is computed in d dimensions and over the phase-space measure in terms of the Mandelstam invariants (s_{13}, q^2, etc.). The integral takes the form,

$$
\begin{aligned}
\mathcal{A}_3^0 &= C(\epsilon,1)\int d\Phi_{ijk}\, A_3^0\\
&= \mathcal{N}(\epsilon,1)\,(4\pi)^{3-2d}\left(q^2\right)^{1-\frac{d}{2}}\int ds_{12}ds_{13}ds_{23}d\Omega\,(s_{12}s_{13}s_{23})^{\frac{d-4}{2}}\,\delta\left(q^2-s_{12}-s_{13}-s_{23}\right)\,A_3^0.
\end{aligned}
\tag{4.69}
$$

with $d\Phi_{ijk} = d\Phi_3/\Phi_2$ and with the derivation of the $d\Phi_3$ measure being present in [ref to this derivation]. $C(\epsilon, 1)$ is the integral normalisation which is here given as

$$C(\epsilon, 1) = 8\pi^2 (4\pi)^{-\epsilon} e^{\epsilon\gamma_E}. \tag{4.70}$$

This integral is quite complex and spans across all A_3^0 terms. Accordingly, we are not going to compute this integral directly: instead we are going to use IBP reduction to reduce the expression to a linear combination of master integrals, compute those master integrals and then use the result to compute the integrated antenna. This process is explained in [ref to IBP section].

After applying IBP identities and reducing the integral to a linear combination of this master integral[2], we reach the following expression,

$$\mathcal{A}_3^0 = \frac{C(\epsilon, 1)}{\Phi_2} \frac{2(2 - 6\epsilon + 5\epsilon^2 - 2\epsilon^3)}{q^2\epsilon^2} R_3 \tag{4.71}$$

This antenna has a single master: R_3 [ref to master integrals section/appendix]. This master integral is the integral over phase-space itself,

$$R_3 = \int d\Phi_3 = \frac{2^{-7+4\epsilon}\pi^{-3+2\epsilon}\Gamma^3(1-\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)} \left(q^2\right)^{1-2\epsilon}. \tag{4.72}$$

After using the expression in Eq. 4.72 explicitly in Eq. 4.71becomes

$$\begin{aligned} \mathcal{A}_3^0 &= \frac{C(\epsilon, 1)}{\Phi_2} \left(q^2\right)^{-2\epsilon} (4\pi)^{-3+2\epsilon} (2 - 6\epsilon + 5\epsilon^2 - 2\epsilon^3) \frac{\Gamma^3(1-\epsilon)}{\epsilon^2\Gamma(3-3\epsilon)\Gamma(2-2\epsilon)} \\ &= \left(q^2\right)^{-\epsilon} e^{\epsilon\gamma_E} (1-2\epsilon)\big(2 + \epsilon(\epsilon+2)\big) \frac{\Gamma^2(-\epsilon)}{\Gamma(3-3\epsilon)} \end{aligned} \tag{4.73}$$

and, after expanding the integrated expression as a Laurent series in ϵ, we finally arrive at the infrared pole structure,

$$\begin{aligned} \mathcal{A}_3^0 = \left(q^2\right)^{-\epsilon} \Bigg[&\frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + \frac{57 - 7\pi^2}{12} + \epsilon\left(\frac{84 - 21\pi^2 - 8\psi^{(2)}(1) + 108\psi^{(2)}(3)}{24}\right) \\ &+ \epsilon^2\left(\frac{35640 - 3990\pi^2 - 71\pi^4 - 720\psi^{(2)}(1) + 9720\psi^{(2)}(3)}{1440}\right)\Bigg] \end{aligned}$$

$$= \left(q^2\right)^{-\epsilon} \left[\frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + \frac{19}{4} - \frac{7\pi^2}{12} + \epsilon\left(\frac{109}{8} - \frac{7\pi^2}{8} - \frac{25\zeta_3}{3}\right) + \epsilon^2\left(\frac{639}{16} - \frac{133\pi^2}{48} - \frac{71\pi^4}{1440} - \frac{25\zeta_3}{2}\right)\right] \tag{4.74}$$

where the usual polygamma identities, $\psi^{(2)}(1) = -2\zeta_3$ and $\psi^{(2)}(3) = -2\zeta_3 + \frac{9}{4}$, were employed.

Implementation in *AntCalc*

Using AntCalc, building the antenna is very simple, since the function `BuildAntenna[type, numFinalParticles, numLoops]` already encodes the entire logical chain and is able to return all intermediate steps using the appropriate options.

First, by running the function naively we get the final result in Eq. 4.68,

[2] This process is shown completely in [ref to ibp reduction appendix].

```
In[2]:= BuildAntenna[A,3,0]//Collect[#,Epsilon]&
Out[2]:=
```

$$\frac{2s_{12}}{q^2 s_{13}} + \frac{2s_{12}}{q^2 s_{23}} + \frac{2s_{12}^2}{q^2 s_{13} s_{23}} + \frac{s_{13}}{q^2 s_{23}} + \frac{s_{23}}{q^2 s_{13}} + \epsilon\left(\frac{-2}{q^2} - \frac{s_{13}}{q^2 s_{23}} - \frac{s_{23}}{q^2 s_{13}}\right)$$

We can also use the `ReturnRecord` option to access all intermediate steps, and check that AntCalc goes through the same steps as the by-hand derivation in Sec. 4.2.1. We can see that with,

```
In[3]:=
BuildAntenna[A,3,0,ReturnRecord->True]["IntermediateSteps"]
Out[3]:=
   Amplitude:
```

$$T^{a_3}_{i_1 i_2}\left[\frac{\bar{u}(k_1)\,\not{\epsilon}^*(k_3)\,(\not{k}_1+\not{k}_3)\,\not{\epsilon}(p)\,v(k_2)}{(k_1+k_3)^2} - \frac{\bar{u}(k_1)\,\not{\epsilon}(p)\,(\not{k}_2+\not{k}_3)\,\not{\epsilon}^*(k_3)\,v(k_2)}{(k_2+k_3)^2}\right]$$

```
   Interference:
```

$$\frac{4(\epsilon-1)(N^2-1)\Big(-2s_{12}^2 - 2s_{12}(s_{13}+s_{23}) + (\epsilon-1)s_{13}^2 + 2\epsilon\, s_{13}s_{23} + (\epsilon-1)s_{23}^2\Big)}{s_{13}s_{23}}$$

```
   Result:
```

$$-\frac{2\epsilon}{q^2} + \frac{2s_{12}^2}{q^2 s_{13}s_{23}} + \frac{2s_{12}}{q^2 s_{13}} + \frac{2s_{12}}{q^2 s_{23}} + \frac{s_{13}}{q^2 s_{23}} - \frac{\epsilon\, s_{13}}{q^2 s_{23}} + \frac{s_{23}}{q^2 s_{13}} - \frac{\epsilon\, s_{23}}{q^2 s_{13}}$$

where `Amplitude`, `Interference` and `Result` respectively refer to the expressions in Eqs. 4.56, 4.64 and 4.68.

Integrating antennae using AntCalc is also simple. Every step in 4.2.1 is handled by a single function: `IntegrateAntenna[antenna]`.

Before going over how this function is used, a small clarification needs to be made: the `IntegrateAntenna` function is not compatible with the default output of the `BuildAntenna` function described earlier. This is so that the built antenna is immediately useful and easily readable. As described in Chapter 3, however, the IBP reduction steps are handled by the LiteRed2 package, which does not expect Mandelstam invariants, but rather, momenta. Accordingly, to use `BuildAntenna`'s output in `IntegrateAntenna` we must toggle the relevant flag: `BuildAntenna[A,3,0,IntegrableForm->True]`. This will return a large association with all required information for integrating the antenna.

We may now proceed to the integration. The whole process goes as follows,

```
In[4]:= intA30 = BuildAntenna[A,3,0,IntegrableForm->True];
In[5]:= IntegrateAntenna[intA30,ExpansionOrder->2]
Out[5]:=
```

$$\frac{19}{4} + \epsilon^{-2} + \frac{3}{2\epsilon} - \frac{7\pi^2}{12} + \epsilon^2\left(\frac{639}{16} - \frac{133\pi^2}{48} - \frac{71\pi^4}{1440} - \frac{25\zeta_3}{2}\right) + \epsilon\left(\frac{109}{8} - \frac{7\pi^2}{8} - \frac{25\zeta_3}{3}\right)$$

Much like with `BuildAntenna`, `IntegrateAntenna` also allows us to inspect its intermediate steps. Here, however, the available options show the master integral linear combination expression both with R_3 and with it substituted by the appropriate expression, that is, the equivalent to Eqs. 4.71 and 4.73 before normalisation. The code and results are,

```
In[6]:= resA30 = IntegrateAntenna[intA30,ExpansionOrder->2,
ReturnRecord->True];
In[7]:= resA30["MasterCombination"]//Simplify//Collect[#,R3]&
Out[7]:=
```

$$\frac{-4(d-3)\left(4+d(\epsilon-2)-4\epsilon\right)}{(d-4)^2q^2}R_3$$

```
In[8]:= resA30["MasterSubstitutedExpression"]//Simplify
Out[8]:=
```

$$-\left(q^2\right)^{-2\epsilon}\frac{2^{-5+4\epsilon}(d-3)\left(4+d(\epsilon-2)-4\epsilon\right)\pi^{-3+2\epsilon}\Gamma^3(1-\epsilon)}{(d-4)^2\Gamma(3-3\epsilon)\Gamma(2-2\epsilon)}$$

The results here are given as a combination of d and ϵ. This is a result of the way the function processes each step. Since these are snapshots of what is happening inside the function itself, they show the expressions before dimension regularisation is applied.

Finally, AntCalc also enables us to forgo the work we did building and the integrating the antenna with the use of the `BuildAndIntegrateAntenna[type, numFinalParticles,numLoops]` function. It goes over the whole process, as we did, and outputs the ϵ Laurent series,

```
In[9]:= BuildAndIntegrateAntenna[A,3,0,ExpansionOrder->2]
Out[9]:=
```

$$\frac{19}{4}+\epsilon^{-2}+\frac{3}{2\epsilon}-\frac{7\pi^2}{12}+\epsilon^2\left(\frac{639}{16}-\frac{133\pi^2}{48}-\frac{71\pi^4}{1440}-\frac{25\zeta_3}{2}\right)+\epsilon\left(\frac{109}{8}-\frac{7\pi^2}{8}-\frac{25\zeta_3}{3}\right)$$

The agreement between the analytic derivation and the corresponding *AntCalc* outputs validates both the construction and the integration workflow for A_3^0. Its local soft and collinear behaviour is validated separately in the following subsection.

4.2.2 Local Validation of A_3^0

To be usable as a subtraction term, an antenna function's unintegrated form must also reproduce the singular behaviour of the corresponding matrix element locally in phase space. In particular, A_3^0 must reproduce the universal eikonal factor when the gluon becomes soft and the Altarelli-Parisi splitting function in the relevant collinear limits. These two quantities provide local validation of the results obtained analytically and using *AntCalc*.

Soft Limit and the Eikonal Factor

When a gluon becomes soft, gauge theory amplitudes factorise universally to the amplitude without the soft gluon times a universal soft current as [5],

$$|\mathcal{M}_{n+1}(...,i,j,k,...)|^2 \xrightarrow{j\to 0} -g_s^2\sum_{i,k}T_i\cdot T_k\mathcal{S}_{ik}^{(j)}|\mathcal{M}_n(...,i,k,...)|^2 \tag{4.75}$$

where T_i are colour-charge operators and,

$$\mathcal{S}_{ik}^{(j)}=\frac{2s_{ik}}{s_{ij}s_{jk}} \tag{4.76}$$

is the eikonal factor.

Since this property is universal, we can use it to validate our results. For the A_3^0 antenna, the eikonal factor may be computed as,

$$\mathcal{S}_{12}^{(3)} = \lim_{k_3 \to 0} A_3^0(1_q, 3_g, 2_{\bar{q}}). \tag{4.77}$$

After building the antenna using *AntCalc*, the limit may be computed using *Wolfram Mathematica* as follows,

```
In[2]:= A30 = BuildAntenna[A,3,0];

softSubs = {s13 -> lambda * s13, s23 -> lambda * s23};
A30Soft = A30 /.  softSubs;

A30SoftSeries = Series[A30Soft, {lambda, 0, 0}];

eikonalFactor = SeriesCoefficient[A30SoftSeries, -2] /.  q2 -> s12
// Simplify;
Out[2]:=
```

$$\frac{2s_{12}}{s_{13}s_{23}}$$

This result agrees with the expected eikonal factor.

Collinear Limit and the Altarelli–Parisi Splitting Function

In the same manner, when two partons i and j become collinear, the amplitude factorises to [5, 35],

$$|\mathcal{M}_{n+1}(..., i, j, ...)|^2 \xrightarrow{i||j} \frac{8\pi\alpha_s}{s_{ij}} P_{IJ\to K}(z)|\mathcal{M}_n(..., K, ...)|^2 \tag{4.78}$$

where K is the parent parton in which i and j recombine, z is the momentum fraction carried by one of the daughter partons, and $P_{IJ\to K}$ is the splitting function. This function is universal but channel-dependent and, since it carries the spin and helicity structure of the collinear splitting, it depends on the partons involved. For A_3^0, where a gluon may become collinear with a quark, the function appears as,

$$P_{qg\to q}(z) = C_F \frac{1+z^2}{1-z}. \tag{4.79}$$

Noting that the building process already normalises the fundamental Casimir C_F, using *AntCalc* in *Wolfram Mathematica*, the AP splitting function may be computed as,

```
In[3]:= A30 = BuildAntenna[A,3,0];

collinearSubs = {s13 -> lambda, s12 -> z q2, s23 -> (1 - z) q2,
Epsilon -> 0};
A30Collinear = A30 /.  collinearSubs;

A30CollinearSeries = Series[A30Collinear, {lambda, 0, 0}];

apSplittingFunc = SeriesCoefficient[A30CollinearSeries, -1]
```

```
// Simplify;
Out[3]:=
```

$$\frac{1+z^2}{1-z}$$

This result agrees with the expected colour-stripped $qg \to q$ AP kernel.

4.2.3 The Virtual A_2^1 Contribution

To compute A_2^1, we may follow the same steps. This is a loop-antenna, however, which means that its build result has been PaVe reduced. It is given in terms of the B_0 and C_0 scalar one-loop function. The unintegrated antenna is computed, using *AntCalc*, as,

```
In[2]:= BuildAntenna[A,2,1]
Out[2]:=
```

$$-\frac{1}{2}\Big[(3+2\epsilon)B_0\Big(q^2;0,0\Big)\Big] - q^2 C_0\Big(0,0,q^2;0,0,0\Big)$$

The integrated final result is, then [9],

```
In[3]:= BuildAndIntegrateAntenna[A,2,1,ExpansionOrder->2]
Out[3]:=
```

$$-\frac{1}{\epsilon^2} - \frac{3}{2\epsilon} - 4 + \frac{7\pi^2}{12} + \epsilon\left(-8 + \frac{7\pi^2}{8} + \frac{7\zeta_3}{3}\right) + \epsilon^2\left(-16 + \frac{7\pi^2}{3} - \frac{73\pi^4}{1440} + \frac{7\zeta_3}{2}\right)$$

Note that, exclusively for this antenna, the same PaVe scalar functions appear in both the build- and integration-side reductions. No phase-space IBP reduction is required.

4.3 NNLO Antenna Results

4.3.1 Two-Parton Contributions: the A_2^2 Set

The A_2^2 route is computed similarly to A_2^1. This route, however, yields four different antenna components and is the longest-running calculation supported by *AntCalc* in its current version. The unintegrated antennae are computed using,

```
In[2]:= BuildAntenna[A,2,2]
Out[2]:=
```

$$\left\{\begin{array}{l}\dfrac{11\Big[(3+2\epsilon)B_0(s_{12};0,0)+2s_{12}C_0(0,0,s_{12};0,0,0)\Big]}{12\epsilon}+\ldots,\\[2ex] 24\epsilon^{14}\Big(391\pi^{12}-1149918\pi^8\zeta_3-62009136\pi^4\zeta_3^2-823011840\zeta_3^3\Big)+\ldots,\\[2ex] -\dfrac{1}{6}\dfrac{(3+2\epsilon)B_0(s_{12};0,0)+2s_{12}C_0(0,0,s_{12};0,0,0)}{\epsilon}+\ldots,\\[2ex] 80(13-42\epsilon)\dfrac{\zeta_3}{1920\epsilon\Gamma^2(2-2\epsilon)}+\ldots\end{array}\right\}$$

The expressions have been truncated here. These four components represent, respectively, the leading, subleading, quark-loop and self-interference terms, or, A_2^2, $\tilde{A}_2^2$, $\hat{A}_2^2$ and $\check{A}_2^2$, in that order. Their complete unintegrated forms are lengthy and are, therefore collected in Appendix [ref to appendix]. We may then obtain the integrated expressions as,

```
In[3]:= BuildAndIntegrateAntenna[A,2,2,ExpansionOrder->0]
Out[3]:=
```

$$\left\{\frac{1}{4\epsilon^4}+\frac{17}{8\epsilon^3}+\frac{433-72\pi^2}{144\epsilon^2}+\frac{121350-44820\pi^2+15120\zeta_3}{25920\epsilon}\right.$$
$$+\frac{-45415-64590\pi^2+4734\pi^4+37440\zeta_3}{25920},$$
$$-\frac{1}{4\epsilon^4}-\frac{3}{4\epsilon^3}+\frac{-1476+312\pi^2}{576\epsilon^2}+\frac{-3978+864\pi^2+1536\zeta_3}{576\epsilon}$$
$$+\frac{-10359+2850\pi^2-118\pi^4+4176\zeta_3}{576},$$
$$-\frac{1}{4\epsilon^3}-\frac{1}{9\epsilon^2}+\frac{65+9\pi^2}{216\epsilon}+\frac{4085-546\pi^2+72\zeta_2}{1296},$$
$$\frac{1}{4\epsilon^4}+\frac{3}{4\epsilon^3}+\frac{1230-20\pi^2}{480\epsilon^2}+\frac{-60\pi^2+560(6+\zeta_3)}{480\epsilon}$$
$$\left.+\frac{-205\pi^2-7\pi^4-240(-36+7\zeta_3)}{480}\right\}$$

The integrated Laurent series agree with the established $\mathcal{A}_2^2$ set results [20].

4.3.2 Three-Parton Contributions: the A_3^1 Set

Computing the A_3^1 antenna set is very similar to the work done for the A_2^2 set, with the big differentiator being that this set does not include a self-interference term. The unintegrated antennae are computed using,

```
In[2]:= BuildAntenna[A,3,1]
Out[2]:=
```

$$\left\{\epsilon(1+3\epsilon)s_{13}s_{23}+\ldots,\right.$$
$$\frac{2s_{12}\Big[(-1+\epsilon)s_{13}+\epsilon s_{23}\Big]}{2(-1+\epsilon)q^2s_{13}}D_0\Big(0,0,0;q^2,s_{12},s_{23};0,0,0,0\Big)+\ldots,$$
$$\left.\frac{2}{3}\frac{s_{12}^2}{\epsilon q^2 s_{13}s_{23}}+\ldots\right\}$$

The expressions have been truncated here. These three components represent, respectively, the leading, subleading and quark-loop terms, or, A_3^1, $\tilde{A}_3^1$, and $\hat{A}_3^1$, in that order. Their complete unintegrated forms are lengthy and are, therefore collected in Appendix [ref to appendix]. We may then obtain the integrated expressions as,

```
In[3]:= BuildAndIntegrateAntenna[A,3,1,ExpansionOrder->0]
Out[3]:=
```

$$\left\{\frac{1}{4\epsilon^4}+\frac{31}{12\epsilon^3}+\frac{-9540-660\pi^2}{1440\epsilon^2}+\frac{38820-3520\pi^2+11040\zeta_3}{1440\epsilon}\right.$$
$$+\frac{-156930-12240\pi^2+123\pi^4+55120\zeta_3}{1440}+O(\epsilon),$$
$$\frac{-15+4\pi^2}{24\epsilon^2}+\frac{-19+\pi^2+28\zeta_3}{4\epsilon}+\frac{1215\pi^2+56\pi^4+540(-35+18\zeta_3)}{760},$$
$$\left.\frac{1}{3\epsilon^3}-\frac{1}{2\epsilon^2}+\frac{114-14\pi^2}{72\epsilon}+\frac{327-21\pi^2+200\zeta_3}{72}+O(\epsilon)\right\}$$

The integrated Laurent series agree with the established $\mathcal{A}_3^1$ set results [9, 20].

4.3.3 Four-Parton Contributions

The A_4^0 Antenna Set

The double real-radiation antenna set is A_4^0, and it has two components, A_4^0 and $\tilde{A}_4^0$. It represents the $\gamma^* \to qgg\bar{q}$ process.

Building the unintegrated antenna follows the same usual process,

```
In[2]:= BuildAntenna[A,4,0]
Out[2]:=
```

$$\left\{-\frac{3s_{12}}{2(-1+\epsilon)\left(q^2\right)^3}+\ldots,\right.$$
$$\left.-\frac{\epsilon}{(-1+\epsilon)q^2 s_{13}}+\ldots\right\}$$

and so does computing the integrated antenna functions,

```
In[3]:= BuildAndIntegrateAntenna[A,4,0,ExpansionOrder->0]
Out[3]:=
```

$$\left\{\frac{3}{4\epsilon^4}+\frac{65}{24\epsilon^3}-\frac{1}{\epsilon^2}\left(\frac{217}{18}-\frac{13\pi^2}{12}\right)+\frac{1}{\epsilon}\left(\frac{43224}{864}-\frac{589\pi^2}{144}-\frac{71\zeta_3}{4}\right)\right.$$
$$+\frac{1076717}{5184}-\frac{7955\pi^2}{432}+\frac{373\pi^4}{1440}-\frac{1327\zeta_3}{18}+O(\epsilon),$$
$$\frac{1}{\epsilon^4}+\frac{3}{\epsilon^3}+\frac{1}{\epsilon^2}\left(13-\frac{3\pi^2}{2}\right)+\frac{1}{\epsilon}\left(\frac{845}{16}-\frac{9\pi^2}{2}-\frac{80\zeta_3}{3}\right)$$
$$\left.+\frac{6921}{32}-\frac{473\pi^2}{24}+\frac{17\pi^4}{72}-80\zeta_3+O(\epsilon)\right\}$$

where for each case, the lists are $\{A_4^0, \tilde{A}_4^0\}$ and $\left\{\mathcal{A}_4^0, \tilde{\mathcal{A}}_4^0\right\}$, respectively.

The complete unintegrated expressions are collected in Appendix [ref to appendix]. The integrated results agree with the established $\mathcal{A}_4^0$ antenna results [9].

The B_4^0 and C_4^0 Antennae

Finally, the $\gamma^* \to q\bar{q}q\bar{q}$ process is represented by the B_4^0 and C_4^0 antennae. The B_4^0 antenna describes non-identical quark flavours, while C_4^0 gives the identical-flavour contribution.

The antenna building process is completed as usual using *AntCalc*,

```
In[2]:= BuildAntenna[B,4,0]
Out[2]:=
```

$$-\frac{\epsilon s_{12}}{(-1+\epsilon)q^2 s_{134}^2} + \ldots$$

```
In[3]:= BuildAntenna[C,4,0]
Out[3]:=
```

$$-\frac{1}{2}\frac{\epsilon s_{12}}{(-1+\epsilon)q^2 s_{123}s_{134}} + \ldots$$

The integration process follows in the same way,

```
In[4]:= BuildAndIntegrateAntenna[B,4,0,ExpansionOrder->0]
Out[4]:=
```

$$-\frac{1}{12\epsilon^3} - \frac{7}{18\epsilon^2} + \frac{1}{\epsilon}\left(-\frac{407}{216} + \frac{11\pi^2}{72}\right) - \frac{11753}{1296} + \frac{77\pi^2}{108} + \frac{67\zeta_3}{18} + O(\epsilon)$$

```
In[5]:= BuildAndIntegrateAntenna[C,4,0,ExpansionOrder->0]
Out[5]:=
```

$$\frac{1}{\epsilon}\left(-\frac{13}{32} + \frac{\pi^2}{16} - \frac{\zeta_3}{4}\right) - \frac{339}{64} + \frac{17\pi^2}{48} - \frac{\pi^4}{45} + \frac{21\zeta_3}{8} + O(\epsilon)$$

The complete unintegrated expressions for both antennae are collected in Appendix [ref to appendix]. The integrated results of both antennae agree with the established $\mathcal{B}_4^0$ and $\mathcal{C}_4^0$ results [9].

4.4 Summary of the Massless Antenna Set

The massless results presented in this chapter demonstrate that, using *AntCalc*, we are able to build and integrate all quark-antiquark final-final standard-model (SMQCD) antenna functions through NNLO. The selected case studies cover tree-level NLO and NNLO, one-, and two-loop contributions, showcasing the programme's ability to handle these different cases. Due to the size of the unintegrated antennae, some have been truncated. The complete results are shown in Appendix [ref to appendix]. The resulting integrated antennae agree with the established results.

The more detailed A_3^0 explanation shows a method of local validation of the unintegrated result based on computing the eikonal factor and the AP splitting function using the antenna. This method might also be used with other antenna functions whenever the corresponding unresolved limit is present. A global validation is presented in Chapter 5, where we compute the massless hadronic R-ratio using all necessary integrated antennae.

Having established the massless sector, the following section considers the exploratory massive extension of the A_3^0 route.

4.5 Exploratory Massive Extension: A_3^0

The massive A_3^0 route is currently the only massive route supported in *AntCalc*. Its inner workflow is similar to that of the massless A_3^0 route with the added complexity of the quarks being massive.

Kinematically, instead of considering only the centre of mass energy as a scale, the quark mass must also be considered. Hence, q^2 and m_q^2 are scales, and they combine to create the relevant threshold variable r,

$$r = 1 - \frac{4m_q^2}{q^2}. \tag{4.80}$$

The phase space has $k_1^2 = k_2^2 = m_q^2$ and $k_3^2 = 0$. The cut families also shift the massless cuts by a mass term as,

$$\begin{aligned} k_{1,2}^2 &\to m_q^2 - k_{1,2}^2, \\ k_3^2 &\to k_3^2. \end{aligned} \tag{4.81}$$

This also implies using massive Mandelstam invariants which, in this case, with k_3 presenting the massless gluon's momentum, become,

$$s_{ij} = (q - k_k)^2 - m_i^2 - m_j^2, \quad \{i, j, k\} = \{1, 2, 3\}. \tag{4.82}$$

LiteRed2, then, has to be updated with these new families before recognising the topology. It will then generate two master integrals. In the current *AntCalc* implementation, the IBP reduction step produces two MI which do not exactly match the $I_1^{(m_q,0,m_q)}$ and $I_2^{(m_q,0,m_q)}$ integrals described in Subsection 4.1.3. Instead, it produces J_1 and J_2, shown below using *LiteRed2*'s $j[...]$ notation, where the complete basis, named MX30Basis123, uses the denominators $\left\{m_q^2 - k_1^2, m_q^2 - k_2^2, k_3^2, -s_{13}, -s_{23}\right\}$ in that order,

$$\begin{aligned} J_1^{(m_q,0,m_q)} &= j[\text{MX30Basis123}, 1, 1, 1, 0, 0] = \mathcal{N}_{\text{LiteRed2}} \int \text{Cut}\left[\frac{1}{D_1 D_2 D_3}\right], \\ J_2^{(m_q,0,m_q)} &= j[\text{MX30Basis123}, 2, 1, 1, 0, 0] = \mathcal{N}_{\text{LiteRed2}} \int \text{Cut}\left[\frac{1}{D_1^2 D_2 D_3}\right]. \end{aligned} \tag{4.83}$$

Here, $\mathcal{N}_{\text{LiteRed2}}$ represents the normalisation implicit in the cut-integral convention used for the family in *LiteRed2*. These two MI do not directly equal the two literature masters, but they can be mapped to them. We need here to introduce a conversion factor of $-1/4$ between the literature and *AntCalc*'s conventions. Using it, we can complete the mapping as,

$$\begin{aligned} I_1^{(m_q,0,m_q)} &= -\frac{1}{4} J_1^{(m_q,0,m_q)}, \\ I_2^{(m_q,0,m_q)} &= -\frac{1}{4}\left(a J_1^{(m_q,0,m_q)} + b J_2^{(m_q,0,m_q)}\right). \end{aligned} \tag{4.84}$$

with,

$$\begin{aligned} a &= \frac{(1 - 2\epsilon)m_q^2 + (1 - \epsilon)q^2}{3(1 - \epsilon)}, \\ b &= \frac{m_q^2\left(4m_q^2 - q^2\right)}{3(1 - \epsilon)}. \end{aligned} \tag{4.85}$$

4.5.1 Build and Integrate Results

To build and integrate the massive A_3^0 in *AntCalc*, we use the same commands as for the massless A_3^0 route, as discussed in Subsection 4.2.1, with the added option `quarkMass -> mQ`, where `mQ` is the quark mass, so far written here as m_q. The building process is completed as,

```
In[2]:= BuildAntenna[A,3,0,quarkMass->mQ]
Out[2]:=
```

$$\left[-4(-2+\epsilon)m_q^4\left(s_{13}^2+s_{23}^2\right)\right]+\ldots$$

The integration process follows in the same usual way, noting that much like in all other integrated cases, the default output is given with $q^2 \to 1$, making here $r \to 1-4m_q^2$,

```
In[3]:= BuildAndIntegrateAntenna[A,3,0,quarkMass->mQ,
ExpansionOrder->0]
Out[3]:=
```

$$\frac{1}{2100m_q^2\left(1+2m_q^2\right)^2\pi^2}\left[\frac{1}{\epsilon}\left(70m_q^2\left(-1+2m_q^2+8m_q^4\right){}_2F_1\left(\frac{1}{2},2;\frac{7}{2};1-4m_q^2\right)+\ldots\right)\right.$$
$$\left.+\left(10\left(-1+2m_q^2+8m_q^4\right){}_2F_1\left(\frac{1}{2},2;\frac{7}{2};1-4m_q^2\right)+\ldots\right)\right]+O(\epsilon)$$

These expressions agree with the established literature results [28]. The full unintegrated and integrated antenna expressions were truncated in this section. They are shown completely in Appendix [ref to appendix].

Chapter 5

Global Validation: The Massless Hadronic R-Ratio

5.1 R-Ratio Definition

The hadronic R-ratio is usually defined as the ratio of total cross-section for e^+e^- annihilation into hadrons to the point-like cross-section for $e^+e^- \to \mu^+\mu^-$, that is, it is the ratio of the cross-sections of the topologies where the electron-positron pair annihilates into hadrons and into a muon-antimuon pair. It is mathematically described as follows,

$$\begin{aligned} R_{\text{hadr}} &= \frac{\sigma(e^+e^- \to \text{hadrons})}{\sigma(e^+e^- \to \mu^+\mu^-)} \\ &= \frac{3q^2}{4\pi\alpha_{\text{em}}^2}\,\sigma(e^+e^- \to \text{hadrons}). \end{aligned} \tag{5.1}$$

Looking closely, the $\sigma(e^+e^- \to \text{hadrons})$ term is what QCD predicts for the $\gamma^* \to \text{hadrons}$, computed via the optical theorem [21]. What makes the R-ratio a clean QCD observable, and a good benchmark for *AntCalc* is that the ratio features terms for higher perturbation orders as a perturbative series in α_s,

$$R_{\text{hadr}} = \left(c_{\text{LO}} + \frac{\alpha_s}{\pi} c_{\text{NLO}} + \left(\frac{\alpha_s}{\pi}\right)^2 c_{\text{NNLO}} + O\left(\alpha_s^3\right)\right) \tag{5.2}$$

here the perturbative coefficients, c_{LO}, c_{NLO} and c_{NNLO}, respectively receive contributions from the multiparticle final states at the relative orders, and respective antennae, as described in Chapter 2. For example, A_3^0 as described in Chapter 4 contributes to c_{NLO}. Moreover, the leading order term $R^{(0)}$ simply counts the number of quark colours and flavours weighted by their electric charges,

$$R^{(0)} = N_c \sum_q e_q^2 \tag{5.3}$$

from $R_{\text{hadr}} = R^{(0)}\big(1 + O(\alpha_s)\big) = R^{(0)} R(\alpha_s)$.

5.2 Global Validation with *AntCalc*

As discussed, the hadronic R-ratio is a very useful quantity to compute using the integrated antenna functions we computed using *AntCalc*. As such, the programme provides an R-ratio calculator. It is the longest running function in the whole package. Its result gives,

```
In[2]:= BuildRRatio[SMQCD]
Out[2]:=
```

$$1+\left(\frac{\alpha_s}{2\pi}\right)\frac{3}{2}C_F+\left(\frac{\alpha_s}{2\pi}\right)^2 C_F\left[N\left(\frac{243}{16}-11\zeta_3\right)+\frac{3}{16N}+N_f\left(-\frac{11}{4}+2\zeta_3\right)\right]$$

As expected, the results for each perturbative order are finite. This validates the mutual normalisation and assembly of the integrated antennae: at each perturbative order, their infrared poles cancel in the inclusive sum, leaving a finite coefficient.

5.2.1 Assembly from Integrated Antennae

Using *AntCalc*, we are also able to compute the R-ratio step-by-step, without having to resort to the automated function by just computing the intermediate steps ourselves. The R-ratio is assembled using the $\mathcal{T}$-objects discussed in Chapter 2. We simply sum the normalised objects order-by-order as,

$$\begin{aligned} R=\frac{R_{\text{hadr}}}{R^{(0)}}&=R_{\text{LO}}+R_{\text{NLO}}+R_{\text{NNLO}} \\ &=r_{\text{LO}}+\frac{\alpha_s}{2\pi}r_{\text{NLO}}+\left(\frac{\alpha_s}{2\pi}\right)^2 r_{\text{NNLO}} \\ &=1+\frac{\alpha_s}{2\pi}\frac{\mathcal{T}^4_{q\bar{q}}+\mathcal{T}^4_{q\bar{q}g}}{\mathcal{T}^2_{q\bar{q}}}+\left(\frac{\alpha_s}{2\pi}\right)^2\frac{\mathcal{T}^6_{q\bar{q}}+\mathcal{T}^6_{q\bar{q}g}+\mathcal{T}^6_{q\bar{q}q\bar{q}}+\mathcal{T}^6_{q\bar{q}q'\bar{q}'}+\mathcal{T}^6_{q\bar{q}gg}}{\mathcal{T}^2_{q\bar{q}}}+O\left(\alpha_s^3\right). \end{aligned} \tag{5.4}$$

Expanding the $\mathcal{T}$-objects, we are able to describe the R-ratio formula using the integrated antenna functions as,

$$\begin{aligned} r_{\text{LO}}&=1 \\ r_{\text{NLO}}&=C_F\left(\mathcal{A}^1_2+\mathcal{A}^0_3\right) \\ r_{\text{NNLO}}&=C_F\Bigg[N\left(\mathcal{A}^1_2\mathcal{A}^0_3+\mathcal{A}^1_3+\mathcal{A}^0_4+\mathcal{A}^2_2\right)-\frac{1}{N}\left(\mathcal{A}^1_2\mathcal{A}^0_3+\tilde{\mathcal{A}}^1_3+\tilde{\mathcal{A}}^0_4+C^0_4-\tilde{\mathcal{A}}^2_2\right) \\ &\qquad\qquad+N_f\left(\hat{\mathcal{A}}^1_3+\hat{\mathcal{A}}^2_2\right)+C_F\breve{\mathcal{A}}^2_2\Bigg]. \end{aligned} \tag{5.5}$$

Assembling these pieces together, using the integrated antenna results from *AntCalc*, we finally get,

$$R=1+\left(\frac{\alpha_s}{2\pi}\right)\frac{3}{2}C_F+\left(\frac{\alpha_s}{2\pi}\right)^2 C_F\left[N\left(\frac{243}{16}-11\zeta_3\right)+\frac{3}{16N}+N_f\left(-\frac{11}{4}+2\zeta_3\right)\right] \tag{5.6}$$

which is the exact expression for the massless hadronic R-ratio at NNLO [20] and is also exactly the expression we get by running `BuildRRatio[SMQCD]`.

Chapter 6

Future Work

In its current version, *AntCalc* is implemented entirely in *Wolfram Mathematica*. This is the right language to build a trial-stage programme of its nature due to its forgiving mathematical environment and its general adoption throughout the field. But this choice of language comes at a cost: efficiency. Though convenient, *Mathematica* is slow and memory-hungry. In its current state, running the full `BuildRRatio` call takes three to four hours and, on machines with less RAM, may cause the calculation to crash halfway through. It is then apparent that the next step in *AntCalc*'s development is taking it mostly out of the *Mathematica* environment and making it a truly modular, multi-language programme that exploits only the best characteristics of each tool, including *Mathematica*, keeping the current high-level workflow. The current plan for this step is to make the orchestrator a *Python* programme and to store the profiles in plain, machine-readable files, such as those in the `JSON` format. Then, retain *FeynArts* and *FeynCalc* for diagram generation and amplitude manipulation, respectively, but hand the algebraic manipulations required to obtain the antenna, or its components, over to *FORM*. Then, lastly, make IBP reduction faster and more stable by replacing *LiteRed2* with *Kira*, written in *C++*, keeping *Mathematica* as an algebraic simplifier.

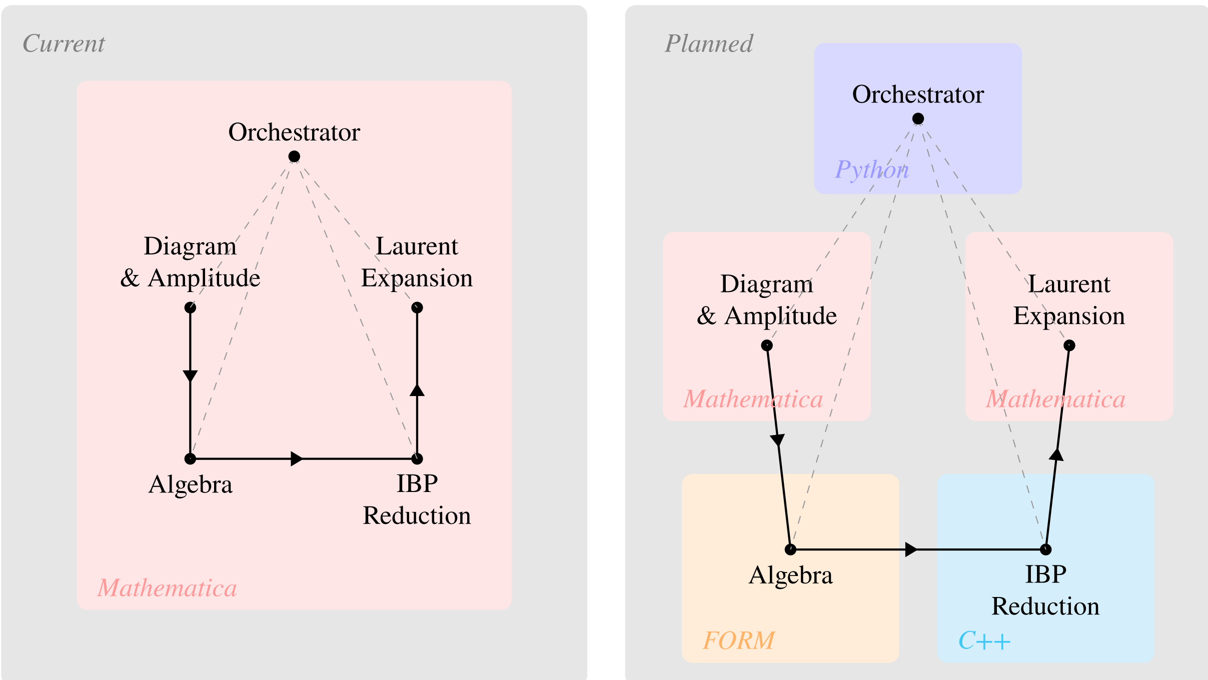

Figure 6.1: Diagram showing the current and planned workflow steps relative to their programming languages.

This step will then allow us to keep expanding *AntCalc*'s profile library without fear of hitting a computational bottleneck. The first point of expansion would be to the rest of the massive quark-antiquark final-final antenna functions, especially A_4^0, B_4^0 and C_4^0 at first, and then A_3^1 and A_2^2. After these are completed, expanding the scope to other relevant antenna types, particularly the quark-gluon (D and E) and the gluon-gluon (F, G and H) antennae, both in the massless and in the massive regimes becomes the logical next step. Lastly, extensions to kinematics other than final-final, namely initial-initial and initial-final configurations, and higher perturbative orders, particularly N^{3}LO, antennae would also be relevant.

Connections to other established tools would also be pertinent. A connection to a Feynman diagram generation tool such as *MadGraph* and one to a parton-level event generator such as *NNLOJet* could allow us to generate more complex antennae, from more complex topologies and also enable the numerical validation of the results.

Chapter 7

Conclusion

Conclusion goes here.

Appendix A

Master Integral Derivations

Appendix A goes here.

Appendix B

Integrated Antenna Trees

Appendix B goes here.

Appendix C

IBP Derivation

Appendix C goes here.

Appendix D

Future Work Details

Appendix D goes here.

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